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BACHELOR THESIS

Physiological and geodata analyses of methane reduction potential in ruminants

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ABSTRACT

Methane is the second most abundant greenhouse gases (GHG) (1935 ppb) after CO_2 (423 ppm) and is responsible for around 30% (gross) of anthropogenic global temperature rise. Because of its high global warming potential (GWP) especially on a short time period (GWP20 = 79.7 and GWP100 = 27 for non-fossil CH_4) is methane emission an excellent target for mitigation to keep global temperature rise at least below the 2 °C target of the Paris Agreement. Right now methane is still accumulating in the atmosphere through the emission of mainly three sectors on the global scale. While the emission from the energy and the waste sector could be reduced with today's technics quit easily, the agriculture sector is classified as "hard-to-abate". Here, this bachelor thesis investigates, to solve this, one possible mitigation strategy the "immunization of cattle" with a holistic approach containing a biochemistry and a geography part.

The biochemical site aims to identify a specific and potent vaccine that targets only the anaerobic, methanogenic archaea in the rumen of cattle. The goal is to shift the microbiome from methanogenic to more non-methanogenic microbes, such as acetogenic bacteria, so that the digestive efficiency of cattle is not compromised. Here, the genus *Methanobrevibacter*, which makes up for 60% of rumen archaea, was microbiological investigated and tried to transform for the first time, primarily focusing on two species (*Methanobrevibacter thaueri* CW^T and *Methanobrevibacter boviskoreani* AbM4). It was determined that the species grow in DSM 1523 and can be subcultured, are sensitivity to puromycin at least at concentrations of 20 µg/ml, and DMSO was found to be unsuitable as a solvent because its oxidative effects also inhibit growth. Transformation using a suicide plasmid via interdomain conjugation with *Escherichia coli* showed promising growth in three preparation lines, but verification via whole-genome sequencing is still pending.

The geographical site tries to evaluate the potential role of the investigated mitigation strategy for slow down climate change. For this purpose, the changed methane emission from enteric fermentation processes in cattle for diffrent success scenarios of microbiome shift for three countries (Germany, New Zealand, USA) and its impact on national CH_4 emission. The calculation is based on the 2006 published guidelines by the IPCC. The geodata analyses showed for a shift of at least 30% a relevant impact to achieve international emission targets. Given the current state of vaccine development, this method cannot play a significant role in reducing emissions over the next decade. However, in the long term, the technology could offer a solution that allows for a sustainable cattle population size that is socially and politically acceptable, even though a reduction will likely still be necessary.

Future research should further develop genetic tools to intensify the investigation of methanogenic archaea. Additionally, it should be assessed whether immunization of cattle could affect emissions beyond enteric methane, either amplifying or offsetting the overall emission reduction potential.

CONTENTS

LIST OF FIGURES

LIST OF TABLES

LIST OF ABBREVIATIONS

CO_2eq.	CO_2 equivalent.
hpt	hypoxanthine phosphoribosyltransferase.
8-Aza	8-Azahypoxanthin.
AA	aminoacids.
AB	antibiotic.
bp	base pairs.
CFU	colony forming unit.
CI	confidence intervall.
CRT	common reporting table.
Cyc-Se.	cycloserin.
DAP	diaminopimelic acid.
DC	dairy cattle.
DE	digestibility of food.
DEU	Germany.
DMSO	dimethylsulfoxid.
DTT	dithiothreitol.
EF	emission factor.
EnterFe	enteric fermentation.
eq.	equivalent.
E. coli	*Escherichia coli*.
E. coli DH5α	*Escherichia coli* DH5α.
E. coli WM6026	*Escherichia coli* WM6026.
E. coli WM6029	*Escherichia coli* WM6029.
EtOH	ethanol.
gDNA	genomic DNA.
GE	gross energy intake.

GHG	greenhouse gases.
GMP	global methane pledge.
GWP	global warming potential.
IPCC	intergovermental panel on climate change.
LULUCF	land use and land use change and forestry.
M. boviskoreani	*Methanobrevibacter boviskoreani* AbM4.
M.	*Methanobrevibacter*.
MIC	minimum inhibitory concentration.
M. millerae	*Methanobrevibacter millerae* ZA-10^T.
M. thaueri	*Methanobrevibacter thaueri* CW^T.
M. wolinii	*Methanobrevibacter wolinii* SH^T.
NDC	non-dairy cattle.
NZL	New Zealand.
OD600	optical density at 600 nm.
OECD	organisation for economic co-operation and development.
PB	planetary boundaries.
Puro	puromycin.
rRNA	ribosomal RNA.
rRNA	ribosomal RNA.
Sqr.	square.
ss-DNA	singlestranded-DNA.
S. epidermidis	*Staphylococcus epidermidis*.
UNFCCC	united nations framework convention on climate change.

USA	United States of America.
Ym	methane conversionfactor.

1 INTRODUCTION

The pressing situation of the planets "health" is a scientific consensus (IPCC, 2022; Planetary Boundaries Science (PBScience), 2025). Already we can see the negative impact in our day to day live that the taken projecteries over the last decades are to be proven true, for example a temperature rise in Europe of 2.5 ℃ towards preindustrial times can be measured as well as a higher frequency of compound events (C3S/ECMWF & WMO, 2026; Ionita & Nagavciuc, 2026). So it is highly necessary to develop strategies for adaption to this changing environment, but more so for mitigation to stabilize the climate and to keep at least the 2 ℃ global-goal of UNFCCC (2015).
One of the main driver for this development is the anthropogenic climate change. Climate change is one of nine processes that are relevant for containing a stable system on earth, to grant condition, in which human mankind can knowingly thrive Planetary Boundaries Science (PBScience) (2025). Like six other of them climate change exceeded its planetary boundaries (PB) already (ibid.).

The relevant value for climate change is the anthropogenic radiative forcing, which not only exceeded the PB (+1 W/m^2) 2025 but the boundary for the zone of an increasing risk (+1.5 W/m^2) as well with a value of 2.97 W/m^2 (Irfan et al., 2024; Planetary Boundaries Science (PBScience), 2025, p. 13). One strong anthropogenic influence of this value is the emission of greenhouse gases (GHG) (Team, 2025). Very prominent is CO_2 as being the most abundant (423 ppm in the atmosphere 2025) and biggest responsibility for the temperature rise of over 1.5 ℃ (more than 50%, see fig. 1.1) and change in radiative forcing (Planetary Boundaries Science (PBScience), 2025).

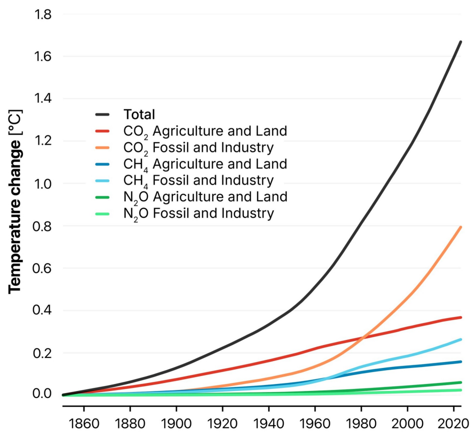

Figure 1.1: The share of the three main GHG (CO_2, CH_4, N_2O) on the anthropogenic temperature rise since 1850. Each GHG is subdivided in two groups depending on the emission source ("Agriculture and Land" or "Fossil and Industry"). Source: (Planetary Boundaries Science (PBScience), 2025), licensed under CC BY 4.0

The second most abundant GHG and with the biggest relative increase in atmospheric concentration (around 265% from 720 ppb 1750 to 1935 ppb in 2026) is methane (Appelhans et

al., 2025; Planetary Boundaries Science (PBScience), 2025; Thoning et al., 2022). Methane is also responsible (see fig. 1.1) for around 30% (gross) of the anthropogenic temperature rise (Appelhans et al., 2025, p. 9).
If compared to CO_2 some features makes it highly intresting as a potential target for mitigation. The stronger impact of methane on radiative forcing and its relatively short lifetime in the atmosphere of 12.4 years (15-40% of CO_2 is still in the atmosphere after 1000 years) (Appelhans et al., 2025) lead to global warming potential (GWP) values that strongly decline over longer time horizonts (GWP20: 79.7 and GWP100: 27 for non-fossil CH_4 (Intergovernmental Panel On Climate Change (Ipcc), 2023, Table 7.15)).
This means, as a mitigation that addresses CH_4 it is enough to reduce the methane emission and within two decades a notable impact will be achieved, while for carbon dioxide strategies must also be focus the removal of CO_2 from the atmosphere. Because of that there are already international initiatives to reduce methane emissions such as the global methane pledge (GMP), which was initiated in 2021 and signed by 160 states including Germany (DEU), New Zealand (NZL) and United States of America (USA) (CCAC, n.d.).The participants commit to emitting 30% less than they did in 2020 and 40% less than in 2010, without any penalties (Appelhans et al., 2025).

Right now methane is still accumulating in the atmosphere (Saunois et al., 2025; Thoning et al., 2022). Whereas the three biggest sources globally for anthropogenic CH_4 are the agriculture- (approx 40%), energy- (36%) and waste-sector (22%) (Appelhans et al., 2025; Mundra & Lockley, 2024). The emission derived from the energy and waste sector can be reduced with today's technical knowledge and options quit easily (Appelhans et al., 2025; Intergovernmental Panel On Climate Change (IPCC), 2022). Emissions from agriculture, on the other hand, are counted to "hard-to-abate", because we are lacking methane reducing technologies and strategies that are socially and economically accepted (Appelhans et al., 2025; Edelenbosch et al., 2024; Intergovernmental Panel On Climate Change (IPCC), 2022; Thoning et al., 2022).
This is due to the fact that methane is not produced specifically through any particular economic activity but arises naturally from biological processes, primarily through enteric fermentation, especially in cattle (77%), and manure management, together they make up for 31% of global methane emission (Appelhans et al., 2025; Smith et al., 2021).

The only option for reducing methane emissions for this right now is to reduce the number of live stock dramatically, which is contrary to the global rise of demand for livestock products and globally rising numbers of livestock (Appelhans et al., 2025; Edelenbosch et al., 2024; FAO, 2009). This bachelor thesis wants to attack this lack of options by further investigating a mitigation strategy for the largest single source of methane emissions in the agricultural sector: "immunization of cattle" against its methanogenic microbiome in the rumen. For this purpose, an interdisciplinary approach was chosen with a biochemical and geographical part.

This interdisciplinary approach allows uniting two scales of investigation in one work. The biochemistry site, on one hand, which develops new approaches to attack methane production on cellular or organism level. On the other hand, the geographical perspective, which places the results in the context of multifactorial climate change and can assess the extent to which the biochemical approach can play a role to mitigate.

In the following each site will have its own brief introduction.

1.1 Biochemestry

To attack methane emission from cattle it is necessary to understand where it occurs in the digestive system. Cattle as a ruminant has a rumen, or large "fore-stomach", which enables them to break down and digest long fibers of carbohydrates such as plant cells via microbial fermentation (EPA, 2024; Moss et al., 2000). These fermentation processes carried out by a wide variety of microbes produce H_2 and CO_2, which must be removed as quickly as possible due to negative feedback inhibition of fermentation. Put simply, this can be achieved by strictly anaerobic microbes. Two relevant groups are acetogenic microbes, which produce short-chain hydrocarbons, or methanogenic archaea, which produce CH_4 (Boadi et al., 2004; Garnett & Eckard, 2024).

In the rumen, methanogenic archaea are often the dominant group, as they can metabolize H_2 and CO_2 more efficiently (Moss et al., 2000). This also means that completely eliminating methanogenic archaea would not be advisable since, according to Morvan et al. (1994), H_2 would accumulate, making digestion in cattle less efficient.
However, the goal of the immunization strategy is to shift the microbiome toward a higher proportion of acetogenic bacteria by training the cattle's immune system to recognize and attack methanogenic archaea (Malyugina et al., 2025). This would result, on the one hand, in less methane production and, on the other hand, in the production of more short-chain hydrocar-

bons as acetate or butyrate that can still be digested and absorbed by the cattle. In theory, this would also lead to more efficient digestion (Boadi et al., 2004; Malyugina et al., 2025; Moss et al., 2000).

Already, there were some experiments trying to immunize cattle against methanogenic archaea with an inactivated vaccine. But the success was low and inconsistent, mainly due to high surface variability of methanogens (Malyugina et al., 2025; Wright, 2004).To solve this problem a robust and specific antigen has to be found for the dominant methanogen species. The problem here, however, is that while the composition of anaerobic, methanogenic archaea in the rumen is now somewhat better understood (four main groups and the dominant genus is *Methanobrevibacter* (*M.*), which accounts for approximately 60% (Baca-González et al., 2020, p. 3)), the individual species remain poorly understood and characterized, even though some of them have already been isolated (Baca-González et al., 2020).

Here, the strains *Methanobrevibacter thaueri* CW^T (*M. thaueri*) and *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*) isolated from the rumen of cattle will be further investigated (Lee et al., 2013; Miller & Lin, 2002). For this purpose, the investigation will have two sites.

First, general analyses and cultivation characterization as cell-count per ml or checking if they grow on solid media.
Second, trying to implement for the first time a genetic tool for *Methanobrevibacter*. Following the approach of Klein et al. (2025) for markerless mutagenesis of *Methanothermobacter marburgensis*, the plan is to transform *Methanobrevibacter* via interdomain conjugation with *Escherichia coli* (*E. coli*). Therby a suicide plasmid was used, with four main functional sites to allow the selection of transformed *M.* and also the counterselection (see fig. 1.2):

- *E. coli* casette with an ORI and a selection via chloramphenicol resistance plus the *traJ* as relevant gen for the conjugation
- Promoter region containing the *pac* gene, which serves as a selection marker for the transformed *M.* with puromycin (Puro)
- Homology arms, where the segments of the genomic DNA (gDNA) from the transforming *M.* are located. The homology regions flank the gene to be deleted. Through single homologous recombination events, the plasmid is first integrated and then removed

again via counterselection, whereby the gene flanked by the homology arms is also excised (Klein et al., 2025).

- The hypoxanthine phosphoribosyltransferase (*hpt*) gene ensures sensitivity to 8-Azahypoxanthin (8-Aza)

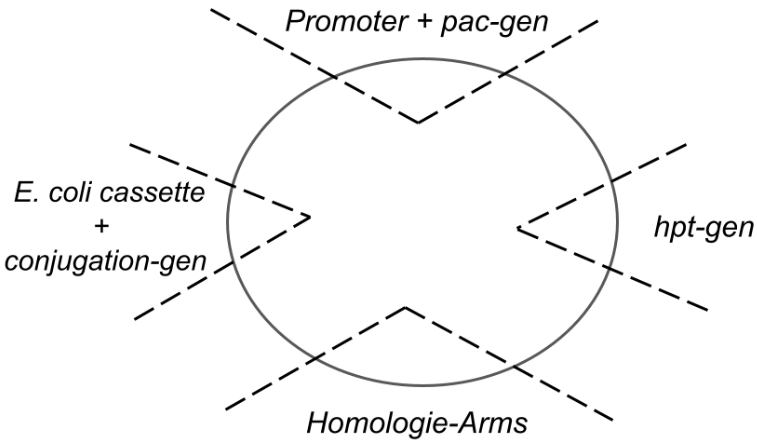

Figure 1.2: Sketch of the general structure of the suicide plasmids for the transformation of *Methanobrevibacter thaueri* CW^{T} (*M. thaueri*) and *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*) via conjugation using *E. coli*. The plasmid contains the *E. coli* cassette with ORI and *catp* as a selection marker using chloramphenicol, as well as the conjugation gene *traJ*. In addition, all plasmids contain the *hpt* gene to enhance counterselection and the two homology arms that flank the genomic *hpt* gene of the corresponding organism. The final important region is the one containing the *pac* gene and its associated promoter, which serves as a selection marker.

,

In the following experiments, regions on the homology arms are selected that flank the *hpt* of the respective organism, resulting in a mutant lacking genomic *hpt*. This makes it possible to target any other gene in the genome by selecting the appropriate homology regions and removing them via markerless counterselection (Klein et al., 2025).

1.2 Geography

Based on a closer examination of three countries (DEU, NZL, USA), the mitigation strategy “immunization of cattle” is evaluated. First, it has to be checked if the global methane emission patterns earlier in the introduction fit on the national scale and cattle is a valid target for mitigation.

Second, the calculation of methane emission is based on the guideline developed by the intergovermental panel on climate change (IPCC) in the year 2006 (IPCC, 2006a) and its refinement in 2019 (IPCC, 2019). This guideline standardized the international calculation of national GHG emission for all emission sources.

To calculate methane emissions resulting from enteric fermentation processes in cattle (IPCC, 2006b, 2019), cattle are typically divided into two subcategories: “dairy” and “non-dairy” cattle. Emissions are first calculated separately for these two categories and then summed. There are three different calculation methods (Tier 1 through 3), which differ in their complexity and the amount of data required.

The Tier 1 approach is quite straightforward. The number of animals in the corresponding animal category is multiplied by the appropriate emission factor (EF), which is provided as a default value by the IPCC and specifies how much methane is emitted per animal (IPCC, 2006b, Eq. 10.19).
In Tier 2, this EF is now calculated for each animal category and country based on specific local activity data. Here, the gross energy intake (GE) per animal is calculated from energy consumption and the average digestibility of food (DE). The EF is then obtained by multiplying the GE by the methane conversionfactor (Ym), which can be taken as the default value from the IPCC or has been scientifically determined for the respective country and animal category (IPCC, 2006b, Eq. 10.21).
Tier 3 involves even more complex calculations and uses detailed, mechanistic models or direct measurements at the farm or animal level (e.g., individual feeding data, respiratory gas measurements of individual animals) (IPCC, 2006a).

The mitigation strategy "immunization of cattle" influences clearly two values of the Tier 2 approach. First the Ym, whereas a shift in the rumen microbiome towards non-methanogenic microbes leads to a lower energy conversion into methane. Second the DE, as described in section 1.1, the microbiome shift is expected to lead to increased digestive efficiency, since less energy is converted into methane, which cannot be utilized, and more into short-chain hydrocarbons that can be absorbed by the cattle.

The performed calculation of methane emission followed the Tier 2 approach and this thought as can be seen further in section 2.2 and section 3.2.2.

2 DATA AND MATERIALS

2.1 Biochemistry

2.1.1 Chemicals

All chemicals are from Carl Roth GmbH + Co. KG unless explicitly labeled otherwise.

2.2 Geography

2.2.1 Study area

The geodata analyses of methane reduction potential in rumineszenz will focus on the impact of the mitigation strategy immunization of cattle. Two emission sectors of agriculture discussed in the IPCC (2019) could be effected of this. On the one hand methane resulting resulting from enteric fermentation and on the other hand from manure management. However, in this bachelor thesis the manure management sector will not be investigated, primarily due to as-yet-unclear links to a possible microbiome shift, as well as time constraints.

All calculations are based on the methodology described in IPCC (2019) for calculating CH_4 emission from enteric fermentation in cattle.

Furthermore, is the calculation done for only three countries: Germany (DEU), New Zealand (NZL) and United States of America (USA). These three countries were picked, because of three main reasons.
First, the project partners for the biochemical research project to develop an immunization method of cattle against methanogen archaea are located in those very countries.
Second, all three countries belong of the ANNEX-I states. This means that, due to their larger economies and higher levels of consumption, they emit a significant amount of GHGs, including CH_4, and therefore bear a greater responsibility with regard to climate change and its mitigation.
Third, although all three are categorized as ANNEX-I states, there are some notable difference between these countries, as e.g. in their size in terms of area, population, and economy (the US. and Germany as leading economic nations of the world) or in their predominant way to raise animals, particularly cattle (e.g. in New Zealand (NZL) pasteur grazing is dominant in Germany loose housing (for the Environment, 2026, p. 155; Niebuhr et al., 2026, Table 510)).

The time period considered in this bachelor's thesis is determined, on the one hand, by the data series from the published CRT (1990-2024 for DEU and NZL plus 1990-2022 for the USA) and, on the other hand, by the reference data for the evaluation of the mitigation strategy. For the latter, the decade 2010–2019 is relevant, which is used in Saunois et al. (2025), well as the year 2030 as the year for the emission targets of the GMP.

2.2.2 CRT Data

Common reporting table (CRT) is the data source for the calculations performed. The CRTs were downloaded from the UNFCCC (2026).
For Germany and New Zealand, the published data range from 1990 to 2024, and for the United States, from 1990 to 2022. Where is one CRT published per year and the values represent thereby the entire year.

For the analysis of methane emissions from enteric fermentation processes involving cattle, the "Table3" and "Table3.A" sheets of the .xlsx files were used. These show only the values for the year it was published for.
The "TableS3" sheet serves as the basis for analyzing methane emissions from the various sectors in each country as well as total emissions. This sheet lists not only the values for the current year but also all past years, so the CRT for 2024 and 2022 for the U.S. also includes all values for the remaining years.

Germany and New Zealand have defined two subcategories for cattle according to IPCC (2019): "dairy cattle" and "non-dairy cattle." The United States, on the other hand, chose to define its own subcategories for cattle, so that "dairy cattle" and "non-dairy cattle" were further subdivided. Section section 3.2.1 explains how the data was processed to ensure that all three countries are comparable with one another.

The used variables from the CRT table are given in following units for each animal category:

- CH_4 emission in kt CH_4
- Population size in 1000s
- GE in MJ head^{-1} year^{-1}
- Ym in %
- DE in %

2.2.3 *Reference data*

In this bachelor thesis two different scaled data for methane emission are taken into account as a reference to evaluate the possible impact of a mitigation strategy against methane emissions derived from cattle enteric fermentation processes.
On the global scale Saunois et al. (2025, fig. 7) calculated the average yearly growth rate of methane 20.9 Tg CH_4 for the decade 2009-2010. Here they also used the tier 1 and tier 2 calculation guidelines of the IPCC (2006a).
The global methane pledge (GMP) calls for a methane emission reduction of each country until 2030. The aim is hereby to emitted 30% less over all sectors than in 2020 and 40% less than in 2010. Germany and the USA signed this pledge in the year 2021 (Appelhans et al., 2025). Because of that this serves as a reference on a country scale.

In this thesis the focus is on the strategy of immunization of cattle and not possible others, as e.g. food additives. These are used only selectively to compare what makes immunization unique and how it measures up against them.

2.3 *Software*

All analyses and processing of the data as well as the visualization took place in R (version 4.6.1). The main package used for processing and analyses was "tidyverse" and "purrr". The visualization was performed via "ggplot". For more information on that, check the R scripts.

3 METHODS

3.1 Biochemestry

Die Rotverfärbung wird durch das resazurin im Medium verursacht

Base composi:

Table 3.1: Composition of the starting solution for the preparation of DSM 1523. The quantities listed are for 1 L. The final pH should be between 6.8 and 7.0. Solutions that have already been premixed are shown in bold.

ingredient	quantity
Solution A	10.00 ml
Trace element solution SL-10	1.00 ml
Brain heart infusion (BD Bacto)	6.00 g
Proteose peptone (BD Difco)	6.00 g
Yeast extract (OXOID)	2.00 g
Na-acetate	1.00 g
Na-formate	2.00 g
Sodium resazurin (0.025% w/v)	4.00 ml
$NaHCO_3$	4.00 g
Milli-Q H_2O	985.00 ml

Composi sol A:

Table 3.2: Composition for the **solution A** as a part of the DSM 1523 and DSM 119. The quantities listed are for 250 ml.

ingredient	quantity
KH_2PO_4	12.50 g
$MgSO_4$ x 7 H_2O	10.00 g
NaCl	10.00 g
NH_4Cl	10.00 g
Milli-Q H_2O	250.00 ml

Composi sol B:

Table 3.3: Composition for the **solution A** as a part of the DSM 1523. The quantities listed are for 275 ml. Solutions that have already been premixed are shown in bold.

ingredient	quantity
$CaCl_2$	0.94 g
Seven vitamins solution	25 ml
Milli-Q H_2O	250.00 ml

Composi Trace:

Composi Seven vitamins:

Table 3.4: Composition for the **trace element solution SL-10** as a part of the DSM 1523 and DSM 119. The quantities listed are for 1 l.

ingredient	quantity
HCl (12%)	10.00 ml
$FeCl_2$ x 4 H_2O	1.50 g
$ZnCl_2$	70.00 mg
$MnCl_2$ x 4 H_2O	100.00 mg
H_3BO_3	6.00 mg
$CoCl_2$ x 6 H_2O	190.00 mg
$CuCl_2$ x 2 H_2O	2.00 mg
$NiCl_2$ x H_2O	24.00 mg
Na_2MoO_4 x 2 H_2O	36.00 mg
Milli-Q H_2O	990.00 ml

Table 3.5: Composition for the **seven vitamins solution** as a part of the DSM 1523. The quantities listed are for 1 l.

ingredient	quantity
Vitamin B_{12}	100.00 mg
p-Aminobenzoic acid	80.00 mg
D-(+)-biotin	20.00 mg
Nicotinic acid	200.00 mg
Calcium pantothenate	100.00 mg
Pyridoxine hydrochloride	300.00 mg
Thiamine-HCl x 2 H_2O	200.00 mg
Milli-Q H_2O	1.00 l

3.2 Geography

The general methodology for the processing, analyses and visualization of the CRT data in R was to build a function and then apply these with the interested variables and data from the specific countries.
In the following part, the general procedure is described in the part below. For a more detailed overview, please check the R scripts.

3.2.1 Processing CRT-Data

Cattle enteric fermentation

For Germany and New Zealand all given data for cattle and its subcattegories "dairy cattle" and "non-dairy cattle" were imported from the "Table3.A" sheet of th CRT tables for each year, and processed so, that for each year of each country a table was created with one column with the year, one column with the cattle type and the following columns are for the measured or calculated values. Afterwards, all one year tables of one country were merged, so that one country has one table, where the values for the cattle categories from 1990 to the last published year are present.

For the U.S. there are no direct "Dairy cattle" and "Non-dairy cattle" categories, but more detailed one. So after the import of the same value columns for each cattle category as described before, these more detailed subcategories have to be merged in the same 2 as for Germany and New Zealand. For this purpose, the official definition and classifications from the U.S. (EPA, 2024, Chapter 5, p. 6) were used. DE, GE and Ym were merged as a weighted mean with the populations as weight. The methane emission were summed.

Methane emission for each sector

For sector-specific emissions, only the “TableS3” sheet was used for the most recent CRT (for the U.S., the year 2022; for Germany and New Zealand, the year 2024). Here, all sectors and their subcategories were imported for all years, as were the total emissions. After processing the data, only the methane emission values for the sectors in general were retained, as well as the total methane emissions for each country.

3.2.2 Calculation of methane emissions from enteric fermentation processes in cattle

The total methane emission for cattle and its subcategories dairy and non-dairy was not directly extracted from the published common reporting table (CRT) for the countries, but calculated from the indicators methane conversionfactor (Ym), gross energy intake (GE) and the population number of the animal categories. Therefore, the CH_4 emissions were calculated for dairy and non-dairy cattle following the eq. (1). The methane emission for cattle than is the sum of dairy and non-dairy.

$$CH_{4,T} = \frac{GE \cdot (\frac{Ym}{100}) \cdot 365}{55.65} \cdot \frac{N_T}{10^6} \tag{1}$$

where:

- $CH_{4,T}$ = yearly methane emission of a animal category [kt $\mathrm{CH_4}$ year^{-1}]
- GE = gross energy intake [MJ head^{-1} day^{-1}]
- Ym = methane conversion factor [%]
- 55.65 = energy content of CH_4 [MJ kg^{-1}]
- N_T = population-size of animal category
- 10^6 = conversion-factor kg to kt

3.2.3 Share of cattle methane emissions from enteric fermentation

The share of enteric fermentation was claculated for its fraction on the total amount of anthropogen produced CH_4 in a country and on the methane emissions resulting from the agriculture sector. For this purpose the published data in the CRT was used for each country.

3.2.4 Calculation of saved methane

To calculate the potential methane savings resulting from the mitigation strategy under investigation (vaccination of cattle), various success scenarios are defined: 10, 20, 30, 40, and 50% shifts in the rumen microbiome of cattle from methanogens toward non-methane-producing microbes.
These will have an impact on two indicators used in IPCC (2019). Ym, which represents the proportion of energy or feed consumed by cattle that is converted into methane by that microbiome, and DE, assuming that the reduced amount of feed energy converted into methane now benefits the animal during digestion, meaning that the percentage of feed digestibility for the animal increases.

The values of the resulting Ym (Ym_{Red}) and DE (DE_{Red}) are calculated as in eq. (2) and eq. (3).

$$Ym_{Red} = Ym \cdot \left(1 - \frac{Red}{100}\right) \tag{2}$$

where:

- Ym_{Red} = adjusted Ym for the investigated reduction scenario [%]
- Ym = original methane conversion factor from CRT [%]
- Red = investigated reduction scenario [%]

$$DE_{Red} = DE + \left(Ym \cdot \frac{Red}{100}\right) \tag{3}$$

where:

- DE_{Red} = adjusted DE for the investigated reduction scenario [%]
- DE = Digestibility of feed as a fraction of gross energy intake [%]
- Red = investigated reduction scenario [%]

The eq. (1) presents the relation of Ym and methane emission of cattle already. DE, on the other hand, is named in three different equations in the calculation of IPCC (2019), shown in the eq. (4) through (6) with little adjustments. It has his impact on produced methane through his role in calculate the GE of the animal, which has a linear relation towards methane emission (see eq. (1)).

$$GE = \frac{\left(\frac{NE_x}{REM}\right) + \left(\frac{NE_y}{REG}\right)}{\left(\frac{DE}{100}\right)} \tag{4}$$

where:

- GE = gross energy intake [MJ head^{-1} day^{-1}]
- NE_x = sum of the net energy required for maintenance (NE_m), for the activity (NE_a), for lactation (NE_l), for the work (NE_work) and for the pregnancy (NE_p) of the animal [MJ head^{-1} day^{-1}]
- REM = ratio of net energy available in a diet for maintenance (Equation (5))
- NE_y = sum of the net energy required for growth and for wool production [MJ head^{-1} day^{-1}]
- REG = ratio of net energy available for growth in a diet (Equation (6))
- DE = Digestibility of feed as a fraction of gross energy intake [%]

$$REM = \left[1.123 - (4.092 \cdot 10^{-3} \cdot DE) + (1.126 \cdot 10^{-5} \cdot DE^2) - \left(\frac{25.4}{DE}\right)\right] \tag{5}$$

where:

- REM = ratio of net energy available in a diet for maintenance
- DE = Digestibility of feed as a fraction of gross energy intake [%]

$$REM = \left[1.164 - (5.16 \cdot 10^{-3} \cdot DE) + (1.308 \cdot 10^{-5} \cdot DE^2) - \left(\frac{37.4}{DE}\right)\right] \tag{6}$$

where:

- REG = ratio of net energy available for growth in a diet
- DE = Digestibility of feed as a fraction of gross energy intake [%]

To calculate the new methane emission for cattle corresponding to the reduction scenario now, GE has to be recalculated with the adjusted DE. Afterwards eq. (1) can be used again to calculate the adjusted CH_4 emission with the adjusted GE (GE_{Red}) and Ym_{Red}.
For calculation the GE_{Red} DE_{Red} can not just simply be put in the eq. (4), because of the values for NE_x and NE_y. Although the values for the individual net energy intake categories (see eq. (4)) should not change, these are not given in the published CRTs of the countries. To solve this problem, two things are assumed:

1. The net energy required for wool production an irrelevant categorie is for the investigation of cattle.

2. Most of the animal in the population of dairy and non-dairy cattle are fully grown or close to it. Here, too, the IPCC only classifies calves "dairy" or "non-dairy" once they started eating solid food. Because of that the net energy needed for growth will overlooked and set to 0.

So, the eq. (4) can be slimmed down and rearranged to calculate a fix value for NE_x for the corresponding year, because the part with NE_y turns zero (see eq. (7)).

$$NE_x = GE \cdot REM \cdot \left(\frac{DE}{100}\right) \tag{7}$$

$$GE = \frac{\left(\frac{NE_x}{REM}\right)}{\left(\frac{DE}{100}\right)} \tag{8}$$

where:

- NE_x = sum of the net energy required for maintenance (NE_m), for the activity (NE_a), for lactation (NE_l), for the work (NE_work) and for the pregnancy (NE_p) of the animal [MJ head^{-1} day^{-1}]
- GE = gross energy intake [MJ head^{-1} day^{-1}] (eq. (4))
- REM = ratio of net energy available in a diet for maintenance (Equation (5))
- DE = Digestibility of feed as a fraction of gross energy intake [%]

Now, GE_{Red} can be calculated for a given year using the fixed values for NE_x and DE_{Red}, all due to the assumption that NE_y is 0 (see eq. (8)). The calculated gross energy intake is then substituted into eq. (1) along with the corresponding Ym_{Red} value. This yields the methane emissions ($CH_{4,R}$) for a cattle category under a reduction scenario for a specific year.

The saved methane emissions ($CH_{4,S}$) for a specific reduction scenario are calculated by subtracting the original value for methane emissions from the value adjusted for the scenario (see eq. (9))

$$CH_{4,S} = CH_{4,T} - CH_{4,R} \tag{9}$$

where:

- $CH_{4,S}$ = saved CH_4 through reduction scenario [kt CH_4 year^{-1}]
- $CH_{4,T}$ = yearly methane emission of a animal category [kt CH_4 year^{-1}]
- $CH_{4,R}$ = yearly methane emission of a animal category for a corresponding reduction scenario [kt CH_4 year^{-1}]

3.2.5 Prediction of CH_4 emission until 2030

Two steps are required to calculate total methane emissions, as well as methane emissions for three reduction scenarios (10, 30, and 50) through the year 2030.

First, total methane emissions without emissions from EnterFe processes in cattle are calculated and then extrapolated.

Second, to calculate the methane emission from enteric fermentation (EnterFe) in cattle, the values for DE, Ym, Population size and NE_x were predicted for each subcategorie ("dairy" and "non-dairy cattle") until 2030. For that, NE_x had to be first calculated (see eq. (7)). From the predicted values for these indicators the cattle EnterFe CH_4 emission could be calculated (via eq. (8) and (1)).
Also the the methane emission values for three different reduction scenarios (10, 30 and 50) were calculated with these values using the eq. (2), (3), (8) and (1).
Afterwards, the emissions of the subcategories were summed to the CH_4 emission for cattle resulting from enteric fermentation processes.

Third, the predicted emission values for total without cattle enteric fermentation were added with the calculated emission values for cattle enteric fermentation to achieve the total methane emission of a country.

The forecast was made using a linear regression, with the value to be predicted depending on the respective year. However, the regression was not based on the entire time period, but rather on three different time intervals (the last 5, 10, and 15 years). For each linear model, both the confidence intervall (CI) with a confidence level of 95% and the total emissions were calculated. In addition, the R^2 value was determined for each model.

The extrapolation extends from 2025 to 2030 for New Zealand and Germany, and from 2023 to 2030 for USA.

4 RESULTS

4.1 Biochemistry

All following experiments were done with *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*) and *Methanobrevibacter thaueri* CW^T (*M. thaueri*), in some *Methanobrevibacter wolinii* SH^T (*M. wolinii*) and *Methanobrevibacter millerae* ZA-10^T (*M. millerae*) were investigated as well.

4.1.1 Cultivation

Media

For cultivation two liquid, complex media were tested, the DSM 1523 and the DSM 119. Both media are suitable for the cultivation of *M. thaueri* and *M. boviskoreani*, although some specific behavior could be detected.

The *M. thaueri* grew more slowly and not as dense as the *M. boviskoreani*. Also the *M. thaueri* started to lyse rapidly in a few days after full growth. This was observed much more prominently in DSM 1523 than in 119.

M. boviskoreani on the other hand did not show the described lysing behavior but precipitated in both media. These precipitates were not completely dissolvable through shaking. *Here I also could implement the Microscopy picture of the precipitate?*

Substrate

Besides the general cultivation with a H_2/CO_2 headspace, it was tested if *M. thaueri* and *M. boviskoreani* could use ethanol (EtOH) as an alternative source of reducing potential for growth, as a liquid option to hydrogen. For the tested concentration of EtOH (10 mM, 20 mM, 50 mM, 100 mM, 200 mM) no visual growth could be detected. In einer literatur steht für typstamm bovis er wachse auf EtOh, und für Format auch -> das könnte man reintheoretisch noch testen

4.1.2 Cell count

Cell counting was performed with a phase-contrast microscope at a 400 times magnification and a Neubauer improved counting chamber, only the *M. wolinii* with an optical density at 600 nm (OD600) of 0.387 was counted with a Buerker counting chamber. In the Buerker, squares were counted, in the Neubauer improved, the group squares were counted, both have a volume of 4 · 10^{-6} ml, as can be seen in GmbH (2026). der part hiervor vmtl eher in die methoden

The cell-number per ml for the cultures was then determined using the following calculation:

$$\textit{cellnumber} \cdot ml^{-1} = \frac{\textit{ø cellnumber} \cdot \textit{dilution factor}}{\textit{Volume of the counted Square}} \tag{10}$$

Table B1 through B6 in the appendix contain the original counted cellnumbers per filled chamber and square. In addition, fig. A2 in the appendix shows an example of a microscopy image used for cell counting, in this case for *M. boviskoreani*. The black dots are thereby the archaea cells. It is also easy to see how some of them link together in a row or cluster slightly.

To calculate the cell count per ml, the arithmetic mean was determined for each culture (tab. B7) and substituted into the equation 10 along with the corresponding dilution factor (see ibid.). The results of the calculation were recorded in table 4.1. The calculation was performed as an example using *M. wolinii* (OD600 = 0.387):

$$\textit{cellnumber} \cdot ml^{-1} = \frac{216.375 \textit{ cells} \cdot 20}{4 \cdot 10^{-6}\, ml} = 1,081,875,000 = 10.819 \cdot 10^{8}$$

Table 4.1: Cellnumber per ml of different *Methanobrevibacter* (*M.*) culture for corresponding optical density at 600 nm (OD600). For counting the cells a phase-contrast microscope was used and a Buerker or Neubauer improved counting chamber at a 400x magnification. The cell count per ml was calculated using the equation 10, and the values were taken from the table B7.

culture	OD600	cellnumber · ml^{-1}
M. wolinii	0.387	10.819 · 10^{8}
M. wolinii	0.396	10.672 · 10^{8}
M. thaueri	0.235	5.603 · 10^{8}
M. thaueri	0.116	3.644 · 10^{8}
M. millerae	0.378	8.880 · 10^{8}
M. boviskoreani	0.287	9.272 · 10^{8}

A clear trend can be observed for the cellnumber per ml of the different *Methanobrevibacter* (*M.*) culture throughout the table 4.1:

Higher OD600 values correspond to a higher cell count per ml. At the lowest measured OD600 of 0.116 (*M. thaueri*), 3.644 · 10^{8} cells per ml was determined, and at the highest OD600 of 0.396 (*M. wolinii*), 10.672 · 10^{8} cells per ml was determined.

Nevertheless, differences between the organisms are also apparent. For example, *M. boviskoreani* (OD600 = 0.287) has a higher cell count per mL (9.272 · 10^{8}) than *M. millerae* (OD600 = 0.378 and celnumber per ml = 8.880 · 10^{8}) despite having a lower OD600 of nearly

0.1. Furthermore, both *M. wolinii* cultures and *M. millerae* have similar OD600 values between 0.378 and 0.396, but those of *M. wolinii* are around $2 \cdot 10^8$ cells per ml higher (10.819 or $10.672 \cdot 10^8$ to $8.880 \cdot 10^8$ cells per ml).
The *M. wolinii* cultures are quite close to each other in terms of cell count per ml, with the higher value being determined for the culture with the lower OD600.
Besides *M. wolinii*, two cultures of *M. thaueri* were also examined to determine cell count. Here, one culture has an OD600 of 0.235 that is slightly more than twice as high as the other one with 0.116, but a cell count per ml that is slightly less than twice as high ($5.603 \cdot 10^8$ to $3.644 \cdot 10^8$ cells per ml).

4.1.3 Plating

Spread plating

100 µl (nochmal im Laborbuch nachschauen) of a full-grown culture were spread plated as duplicates on a DSM 1523 solid media (1.5% agar) under anaerobic condition in a glove-box. In figure 4.1 the plating results are shown.
Various spots can be seen on all 4 plates. Most of these are gas bubbles, as indicated by the smaller size and transparency of the spots (see red rectangles in Figure 4.1). In contrast, a few spots are opaque, larger, round, slightly milky, and yellow (see green squares ibid.). These are grown colonies. Das die einen Kolonien sind und die anderen Gasblasen ist streng genommen ja bereits eine Interpretation, finde es aber in dem Kontext recht nervig und wenig förderlich für die Verständlichkeit, das nur vage zu bezeichnen als milchig und weniger milchig. Wie siehst du das Max?

No mature colonies of *M. thaueri* were detected on the two plates. *M. boviskoreani*, on the other hand, grew on both plates. Individual colonies were picked from the plates, propagated in liquid DSM 1523, and verified as *M. boviskoreani* via 16S-Sequencing.

Pour plating

The pour plating was not successful for 200 µl of a full-grown *M. thaueri* culture with DSM 119 (1% agar) as you can see in fig. A3. This experiment was also performed as duplicate.

Roll Plating

200 µl and 400 µl of full-grown *M. thaueri*, *M. wolinii* and *M. boviskoreani* culture have been roll plated in DSM 1523 (1.5% agar). The headspace was regulary charged up for the 200 µl

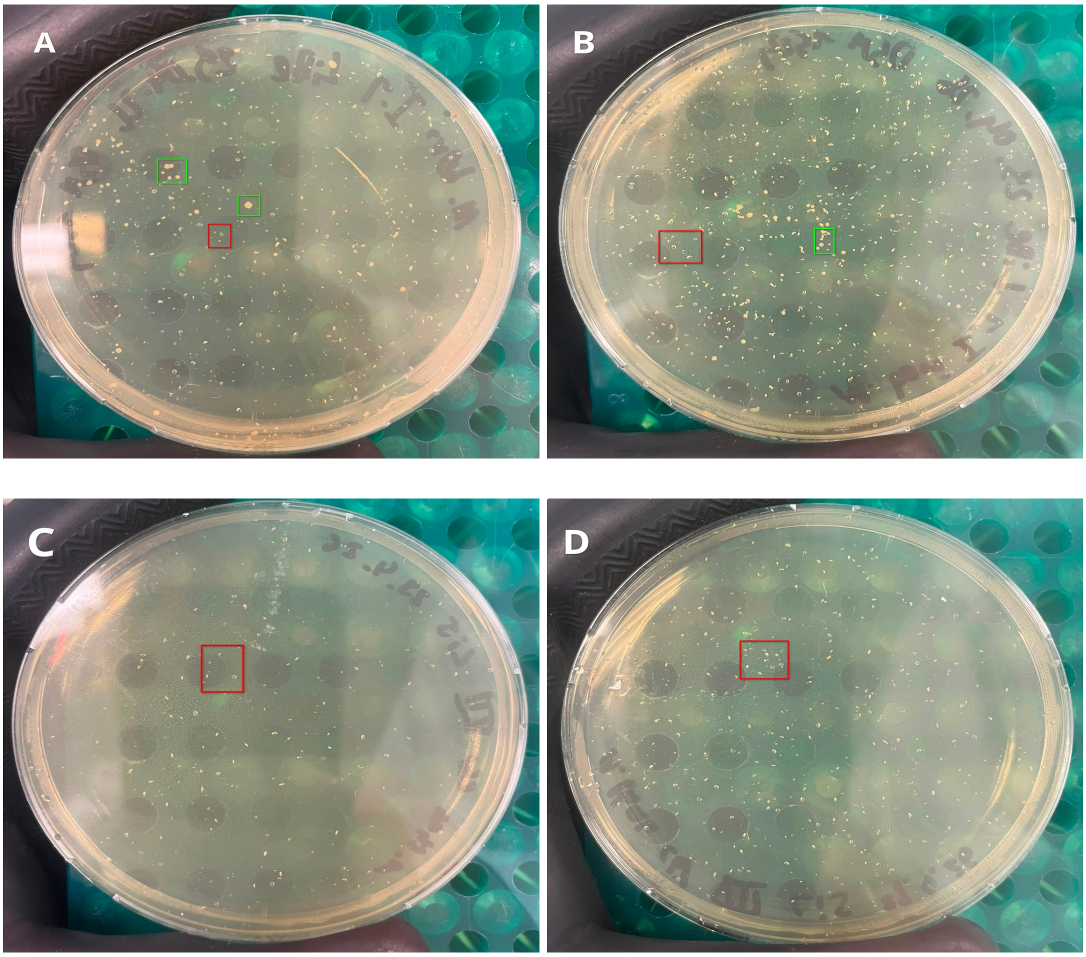

Figure 4.1: Spread plating results of *Methanobrevibacter thaueri* CW^T (*M. thaueri*) and *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*). 100 µl of a full-grown culture was plated on DSM 1523 (1.5% agar) in duplicates. Red rectangles highlight gas bubble spots, green rectangles grown colonies. In image A and B *M. boviskoreani* was plated and in image C and D *M. thaueri*.

preparations over a month, for the 400 µl not.

For the 400 µl there were no visible colonies for *M. thaueri* and just a lot of very, very little dots for *M. boviskoreani* and *M. wolinii*, that could have grown into visible colonies. The 200 µl, on the other hand, showed for all three organism growth (see fig. 4.2).

In all three bottles colony forming unit (CFU) were formed (see fig. 4.2). *M. thaueri* showed notable less colonies (see table 4.2: 15 for 2432 of *M. wolinii* or 3536 of *M. millerae*), but they were larger than the little dots for the other cultures(fig. 4.2). Another difference between the bottle of *M. wolinii* (fig. 4.2 B) and the bottles of *M. thaueri* (ibid. C, D) and *M. boviskoreani* (ibid. A) is the agar structure. At *M. wolinii* it is not as smooth and a bit more clumpy. In diskussion auf unterschiedliche technick beim rollen verweisen -> schlussfolgern wie am besten zu performen ist

However, besides the growing spots inside the bottles, shown in figure 4.2, also the H_2/CO_2

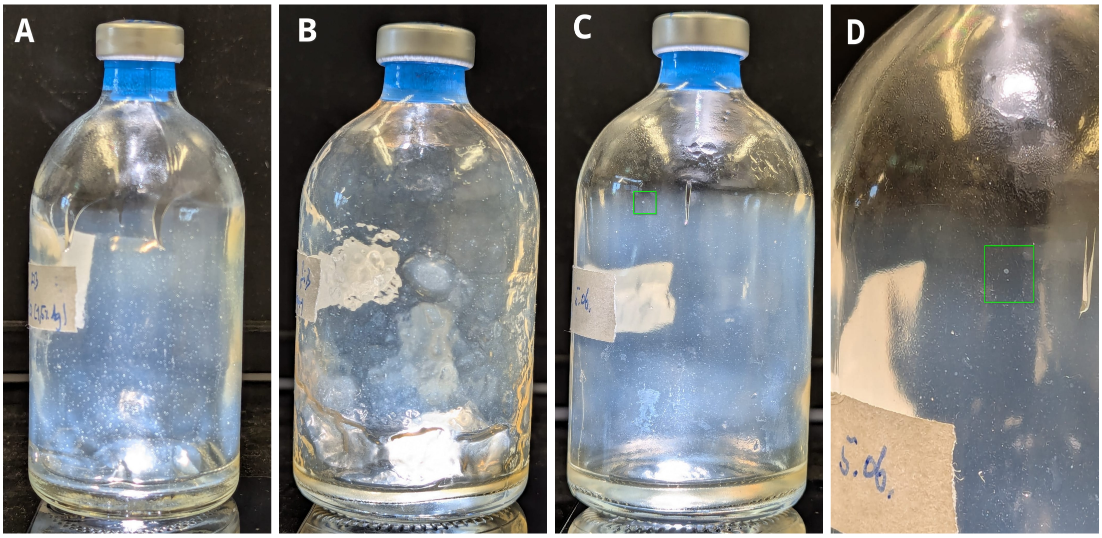

Figure 4.2: Roll plating of *Methanobrevibacter thaueri* CW^T (*M. thaueri*), *Methanobrevibacter wolinii* SH^T (*M. wolinii*) and *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*) in 10ml DSM 1523 (1.5% agar). 200 µl of a full-grown culture were used. The green rectangle marks the same colony in C and D. A: *M. boviskoreani*, B: *M. wolinii*, C: *M. thaueri*, D: *M. thaueri* zoomed in on colony

headspace changed. So approximately every two to four days the gas pressure in the bottle dropped from about 1.1 bar to below 0.8 bar and hat to be refilled.
Verification via 16S-Sequencing, as is done with spread plating, is still pending.

Even though the OD600 of the cultures used was not measured prior to plating, the plating efficiency was estimated. This was done on the assumption that the cultures that appeared visually full-grown had an OD600 approximately equal to those listed in tab. 4.1, with the higher value for cell count per mL always being used.

$$\textit{Plating efficiency [ppm]} = \frac{\textit{counted CFU}}{\textit{number of used cells}} \cdot 10^6 \tag{11}$$

The calculation was based on equation 11 and was performed as follows, for example:

$$\textit{Plating efficiency} = \frac{2432}{0.2\ ml \cdot 10.819 \cdot 10^8\ cells \cdot ml^{-1}} \cdot 10^6 = 11.24\ ppm$$

Table 4.2: Counted colony forming unit (CFU) and estimated plating efficiency of different *Methanobrevibacter* (*M.*) culture for roll plating technic. 200 µl of a full-grown culture was used for plating. For *M. wolinii* and *M. boviskoreani* was counted a sixteenth of the bottle. Always the higher number of cells per ml from table 4.1 were used as a reference for the cell count in the initial solution. Plating efficiency than was calculated following eq. (11).

culture	counted CFU	plating efficiency [ppm]
M. wolinii	2432	11.24
M. thaueri	15	0.13
M. boviskoreani	3536	19.07

4.2 lists the CFU counts per bottle as well as the corresponding plating efficiency. For *M. wolinii* and *M. boviskoreani*, the efficiency is in the lower two-digit ppm range (between 10 and 20%). The efficiency for *M. thaueri* is notable lower, at 0.13 ppm.

4.1.4 Antibiotic sensitivity

Puromycin

Growing experiment for *M. thaueri* and *M. boviskoreani* was performed in DSM 1523 media with various concentration of puromycin (Puro): 0 µg/ml, 1 µg/ml, 2 µg/ml, 5 µg/ml, 10 µg/ml and 20 µg/ml. The experiment was done in triplicates (Cltr 1, Cltr 2 and Cltr 3), with each antibiotic (AB) concentration series inoculated with one full-grown pre-culture. For plotting negative measured OD600 values were set to 0.001. The growth curves are plotted in fig. 4.3 (*M. boviskoreani*)and 4.4 (*M. thaueri*)

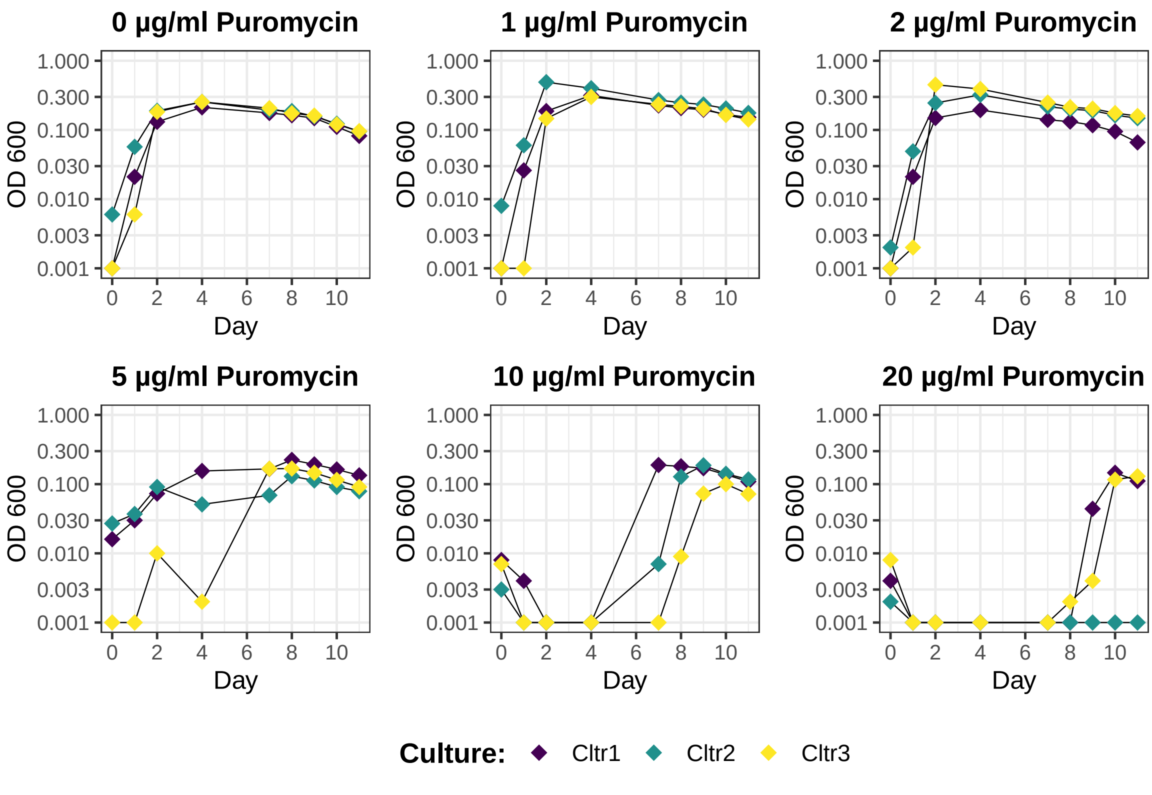

Figure 4.3: Growthcurve in DSM 1523 for *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*) for various concentration of puromycin (Puro): 0 µg/ml, 1 µg/ml, 2 µg/ml, 5 µg/ml, 10 µg/ml and 20 µg/ml. Experiment was performed as triplicates(Cltr. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The optical density at 600 nm (OD600) of the pre-culture were 0.110 for Cltr 1, 0.091 for Cltr 2 and 0.034 for Cltr 3. The OD600 was plotted on a logarithmic scale against time. OD600 values less than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

Fig. 4.3 shows that *M. boviskoreani* exhibits a very similar growth pattern for Puro concentrations of 0, 1, and 2 µg/ml. Exponential growth occurs within the first two days. This is followed by a delayed growth phase, such that the growth maximum is reached between days two and four at an OD600 of around 0.3. Until the final measurement day (day 10), the OD decreases continuously to an OD600 value of around 0.1. One difference that can still be noted here, however, is that the variation between the triplicates at 0 µg/ml is smaller than that observed with Puro. In the latter case, culture 3 does not begin exponential growth until one day later as the other triplicates.

At 5 µg/ml, the growth curves for Cltr 1 and Cltr 2 are generally similar to those just described. However, the two cultures do not grow as quickly. Cltr 3 shows delayed growth and exhibits greater fluctuations in the OD600 values. Thus, all three triplicates here reach their maximum growth only after 8 days. Nevertheless, the maximum OD600 values here are also slightly lower than in the previous growth curves. Thereafter, the OD600 values decrease continuously.

At 10 and 20 µg/ml Puro concentrations, exponential growth begins with a notable delay. At 5 µg/ml, this occurs between 4 and 7 days. The maximum is reached between 7 and 10 days, with maximum OD600 values ranging from approximately 0.1 to 0.25. At 20 µg/ml, one replicate shows no growth at all by the end of the measurement (Cltr 2). The other two replicates do not begin exponential growth until day 8 or 9. The maximum is reached on day 10 at approximately 0.1. By day 11, a slight decrease in the OD600 values can already be observed.

The growth curves of *M. thaueri* for Puro concentrations of 0 and 1 µg/ml reach their OD600 maximum after 3 days, similar to *M. boviskoreani* (see Figure 4.4). However, the maximum does not exceed 0.3 and is closer to 0.2 than to 0.3.
After that, the OD600 decreases rapidly again, so that at 0 µg/ml, Cltr 1 and 2 already have an OD below 0.003 after 9 days, and Cltr 3 is also just barely above 0.01 after 10 days. For 1 µg/ml, the final measured OD values range between 0.001 and 0.05.

For 2 µg/ml puromycin, Figure 4.4 shows growth for all three triplicates, reaching its maximum on day 3. However, the increase in OD600 values is significantly lower. Thus, the maximum for Cltr 2 is below 0.03, and for the others it is around 0.1. This is followed by another decrease in the values, so that the final OD600 range between 0.001 and 0.03.

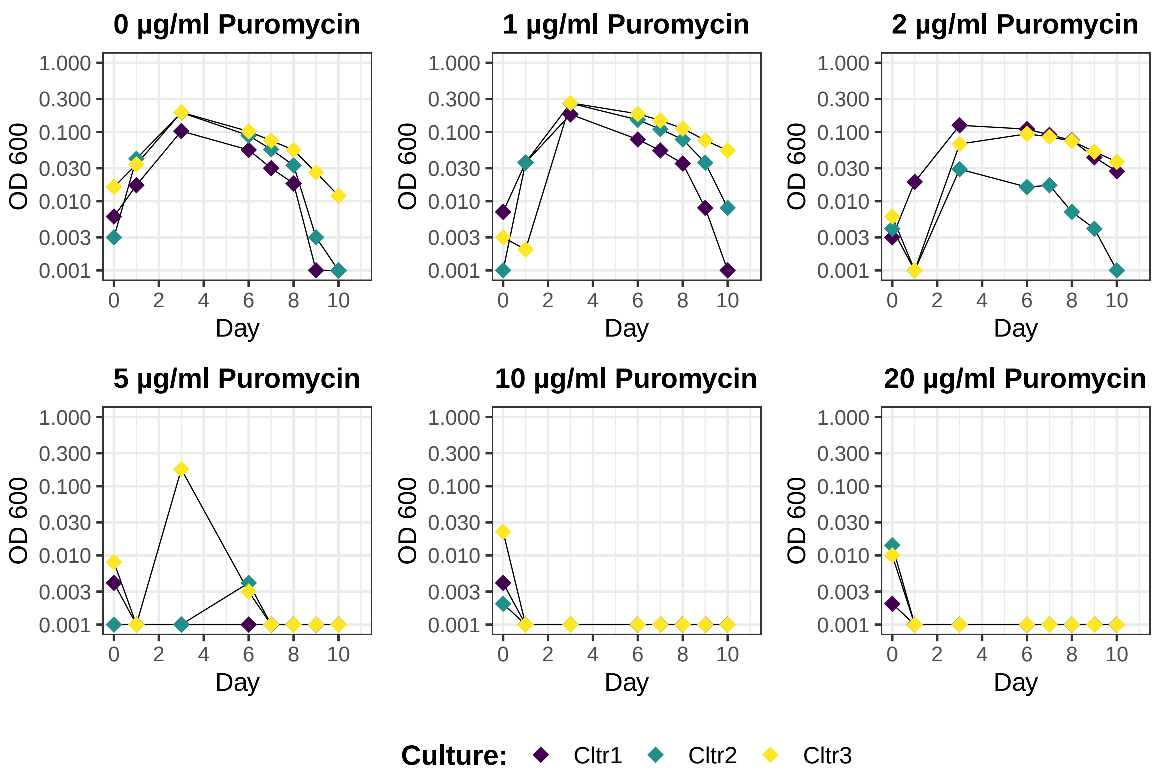

Figure 4.4: Growthcurve in DSM 1523 for *Methanobrevibacter thaueri* CW^T (*M. thaueri*) for various concentration of puromycin (Puro): 0 µg/ml, 1 µg/ml, 2 µg/ml, 5 µg/ml, 10 µg/ml and 20 µg/ml. Experiment was performed as triplicates(Cltr. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The optical density at 600 nm (OD600) of the pre-culture were 0.154 for Cltr 1, 0.173 for Cltr 2 and 0.120 for Cltr 3. The OD600 was plotted on a logarithmic scale against time. OD600 values less than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

For the other measured Puro concentrations, in contrast to *M. boviskoreani*, no growth above an OD600of 0.003 was observed with one exception (Cltr 3 at 5 µg/ml) (see Fig. 4.3 and 4.4). For Cltr. 3 at a Puro concentration of 5 µg/ml, an OD600 of more than 0.1 was measured on day 3, but this value dropped back to 0.003 by day 6.

Cycloserin

In duplicates it was tested if 200 µg/ml cycloserin (Cyc-Se.) would inhibit the growth of *M. thaueri* in DSM 1523. In all four bottles (two negativ and two with 200 µg/ml Cyc-Se.) visual growth could be detected after 1 day.

4.1.5 Nucleotide-baseanalogica sensitivity test

Similar to the AB sensitivity test, a growth experiment was conducted for the base analogs 8-Azahypoxanthin (8-Aza) for *M. thaueri*, *M. boviskoreani*, *M. millerae*. One difference is that the base analogs do not dissolve in H_2O, so they were dissolved in dimethylsulfoxid (DMSO). Furthermore, different concentrations were selected: 0 µg/ml, 100 µg/ml, 250 µg/ml, 500 µg/ml, 750 µg/ml, 1000 µg/ml, and one with 1 ml DMSO. The OD600 was measured and plotted on a logarithmic scale against time (see Figures 4.5 through 4.6).

8-Azahypoxanthin

4.5 shows a growth curve for the positive control (0 µg/ml 8-Aza) of *M. boviskoreani* that is similar to that of the positive control in Fig. 4.3. Thus, exponential growth occurs within the first two days. On days 3 through 4, the OD600 reaches a maximum of around 0.3, followed by a continuous decline with a final OD600 of just under 0.1 on day 13.

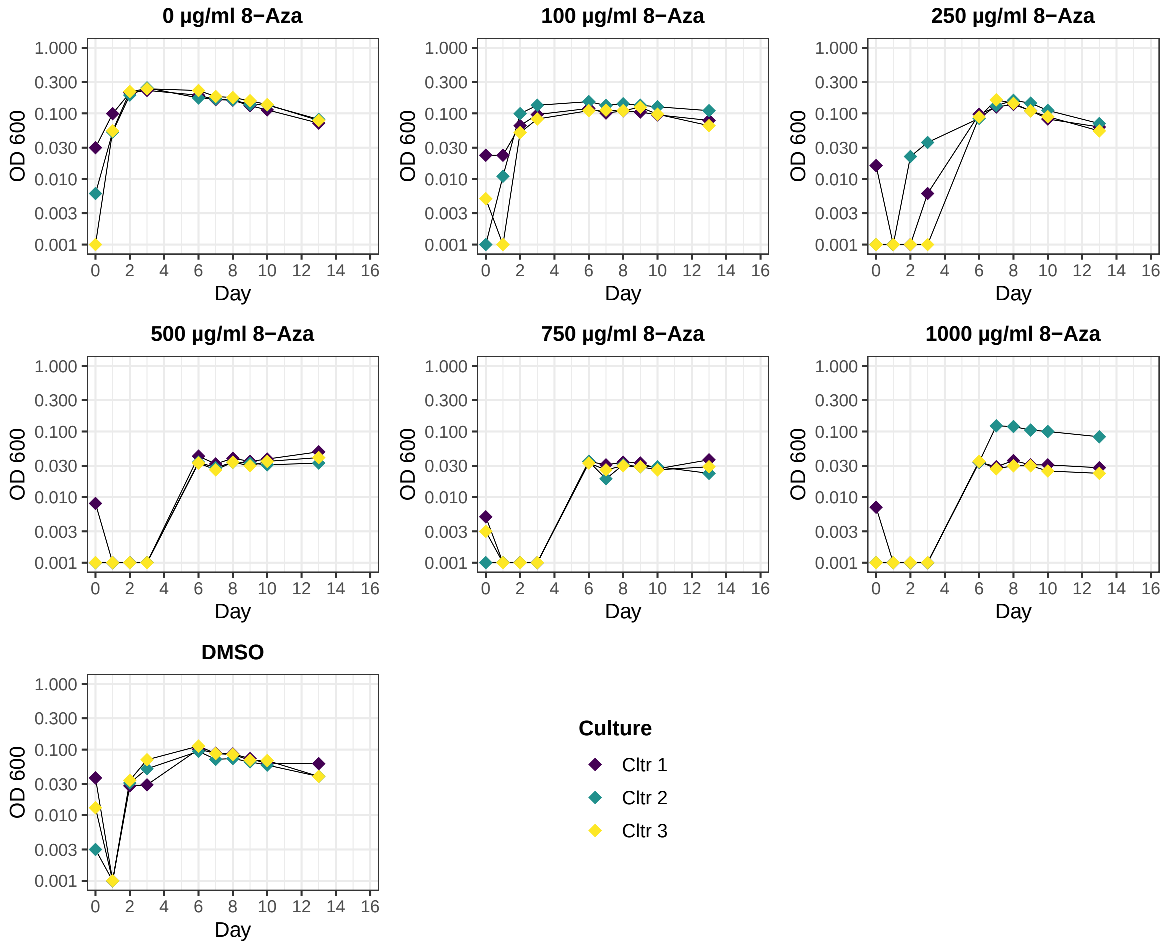

Figure 4.5: Growthcurve in DSM 1523 for *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*) for various concentration of 8-Azahypoxanthin (8-Aza) (0 µg/ml, 100 µg/ml, 250 µg/ml, 500 µg/ml, 750 µg/ml, 1000 µg/ml) and one bottle with 1 ml DMSO as solvent of 8-Aza. Experiment was performed as triplicates(Cltr. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The optical density at 600 nm (OD600) of the pre-culture were 0.096 for Cltr 1, 0.105 for Cltr 2 and 0.152 for Cltr 3. The OD600 was plotted on a logarithmic scale against time. OD600 values lass than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

The solvent control (DSMO), in which no base analogs were added but only 1 ml of the solvent

DMSO, showed a different growth curve. Initially, none of the triplicates begin to grow immediately. Instead, the OD600 values first decrease. On day 3, slight growth begins, reaching a maximum of 0.1 on days 4–6. Afterward, the values decline to approximately 0.03 on day 13.

At the lower concentrations of 8-Aza (100 and 250 µg/ml), *M. boviskoreani* in fig. 4.5 shows a growth curve shape similar to that of the corresponding positive control. However, at 100 µg/ml 8-Aza, the maximum OD600 value is reached within 1 to 3 days, but it is approximately 0.1 instead of just under 0.3. 0.1 is also the maximum OD600 value for the 250 µg/ml concentration. Here, growth does not begin until 2 to 4 days later, so that the maximum is not reached until 7 days later.

The three remaining 8-Aza concentration solutions (500, 750, and 1000 µg/ml) exhibit a very similar growth curve. An OD600 below 0.01 was measured for *M. boviskoreani* during the first 4 to 6 days. Then there was an increase to an OD600 of 0.03, which remained constant until the final measurement day (day 13). An exception is Culture 2 in the 1000 µg/ml 8-Aza culture. This culture grows from Day 3 to Day 5 to an OD600 of 0.1, which then decreases slightly until the end of the measurement series.

Figure 4.6 visualizes the growth curves for *M. thaueri*. The first thing that stands out in comparison to the other archaea (fig. 4.5 and 4.7) is that the baseline OD600 is higher, at approximately 0.03.
Except the positive control (0 µg/ml 8-Aza), no significant increase in the OD600 values was observed. Instead, there was a fairly constant fluctuation around the initial value of 0.03 until the last measurement day (Day 16). An exception here is the triplicate Cltr 3 in the DMSO sample. Here, an OD600 of 0.3 was measured on Day 2, but it fell back to 0.03 by the next measurement day (Day 4). At the end of the measurement series, this triplicate also ended at approximately 0.07 instead of 0.03 like the other triplicates.

In the positive control, slight growth of *M. thaueri* was detected. However, the growth curve differs quite noticeably from that of the positive control for *M. thaueri* in fig. 4.4. In addition, there is considerable heterogeneity among the individual triplicates.
For example, Cltr 1 begins to grow slightly from the first day and reaches its maximum OD600 between 0.1 and 0.3 on day 9 (see fig. 4.6). By the end of the measurement, the OD600 decreases continuously to 0.1.

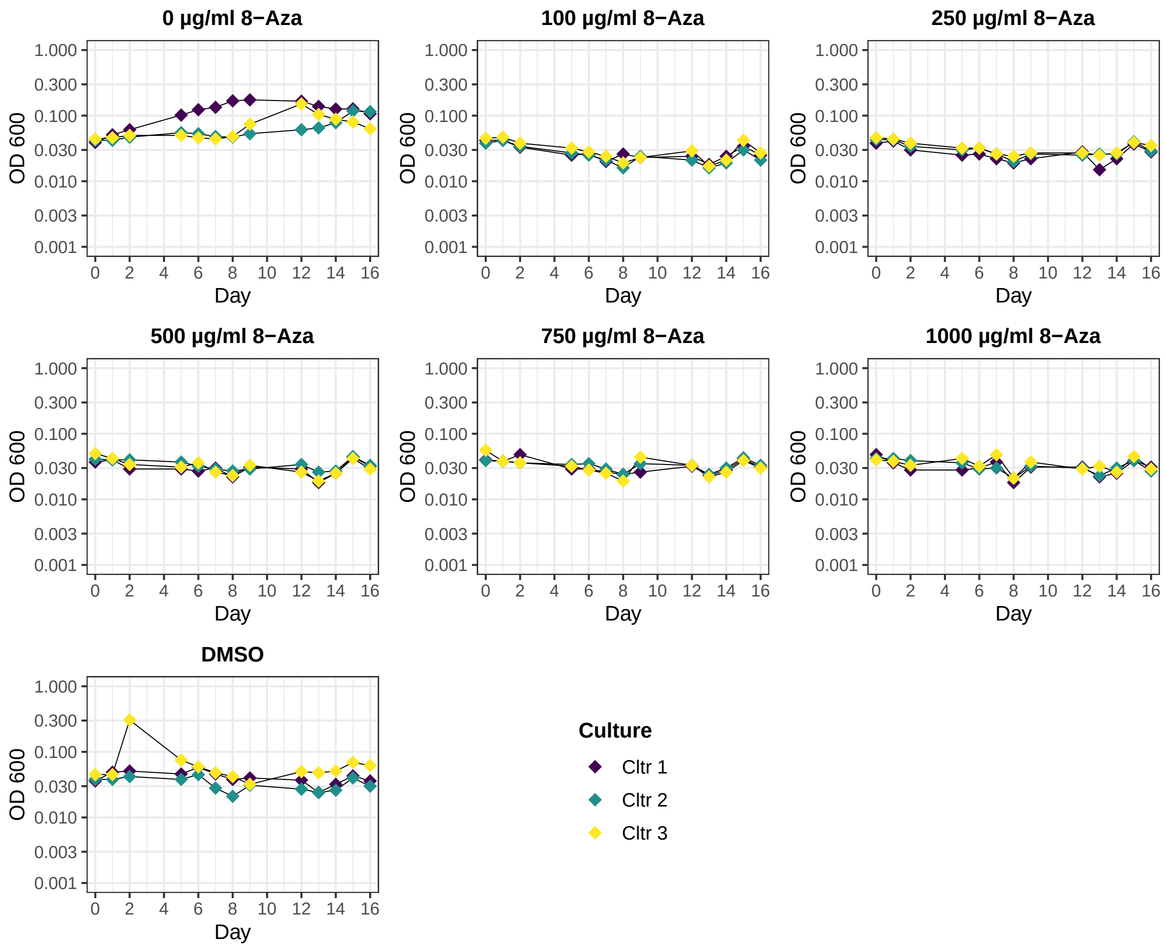

Figure 4.6: Growthcurve in DSM 1523 for *Methanobrevibacter thaueri* CW^T (*M. thaueri*) for various concentration of 8-Azahypoxanthin (8-Aza) (0 µg/ml, 100 µg/ml, 250 µg/ml, 500 µg/ml, 750 µg/ml, 1000 µg/ml) and one bottle with 1 ml DMSO as solvent of 8-Aza. Experiment was performed as triplicates(Cltr. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The optical density at 600 nm (OD600) of the pre-culture were 0.125 for Cltr 1, 0.148 for Cltr 2 and 0.210 for Cltr 3. The OD600 was plotted on a logarithmic scale against time. OD600 values lass than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

Cltr 3 begins to grow on Day 8, reaching a maximum OD600 of approximately 0.2 on Day 12. Between Days 13 and 14, the last triplicate then begins to show slight growth, reaching an OD600 of just over 0.1 on Day 16.

The growth curves of *M. millerae* for the positive control (0 µg/ml 8-Aza) and the solvent control (DMSO) have a similar shape for all triplicates, with the measured OD600 being reached on day three, followed by a continuous decline until the end of the measurement series (day 14). However, the curves differ in their OD600 values. The positive control reaches a maximum value of 0.3, while the solvent control reaches just under 0.1. This also results in a lower final OD600 measurement for DMSO (below 0.03) and for the positive control (just under 0.1).

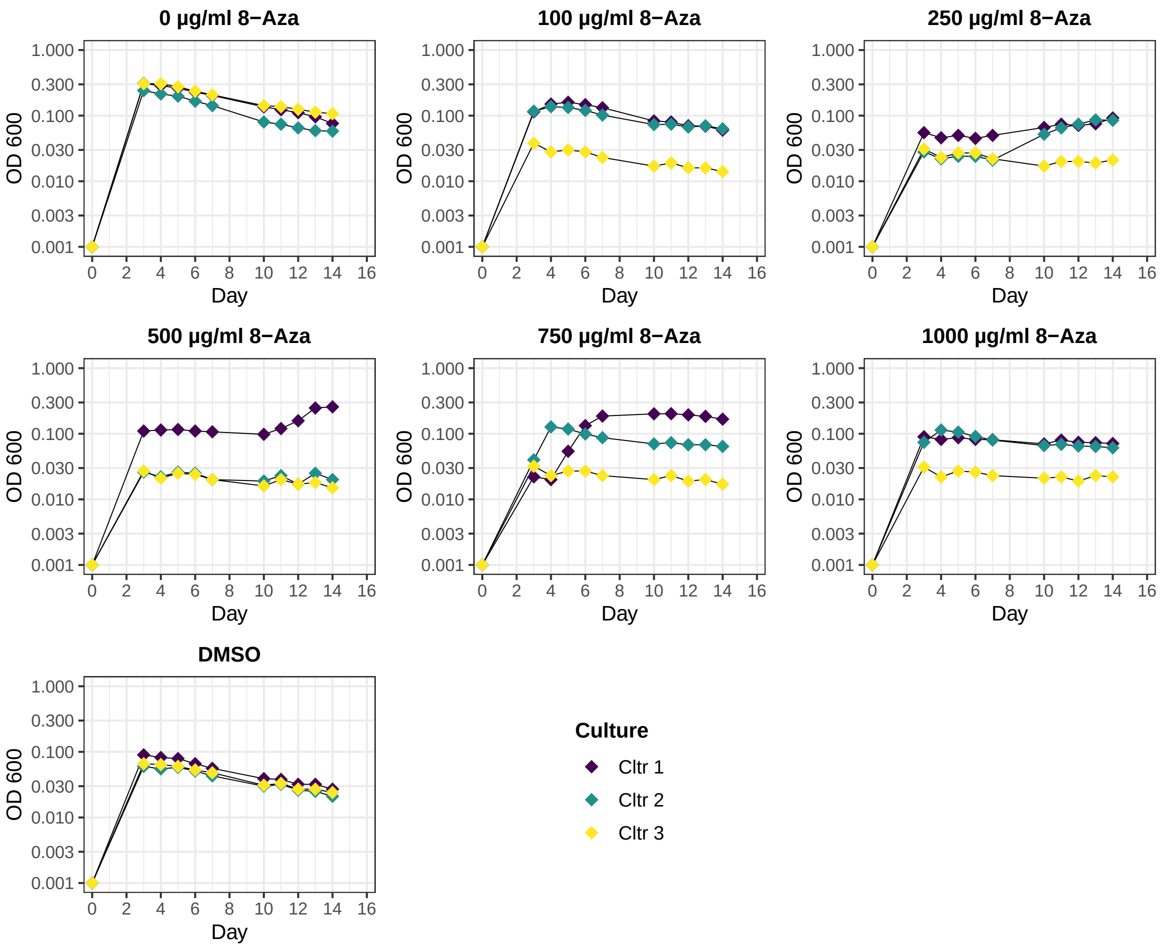

Figure 4.7: Growthcurve in DSM 1523 for *Methanobrevibacter millerae* ZA-10^T (*M. millerae*) for various concentration of 8-Azahypoxanthin (8-Aza) (0 µg/ml, 100 µg/ml, 250 µg/ml, 500 µg/ml, 750 µg/ml, 1000 µg/ml) and one bottle with 1 ml DMSO as solvent of 8-Aza. Experiment was performed as triplicates(Cltr. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The optical density at 600 nm (OD600) of the pre-culture were 0.366 for Cltr 1, 0.339 for Cltr 2 and 0.281 for Cltr 3. The OD600 was plotted on a logarithmic scale against time. OD600 values lass than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

Clear trends for the other samples are generally difficult to discern at first glance. However, most triplicates do not reach an OD600 higher than 0.1.

The 100 µg/ml 8-Aza generally follows the curve shape of the positive and solvent controls, although triplicates Cltr 1 and 2 reach their maximum OD600 value on Day 4 with a value above 0.1, while Cltr 3 does not exceed 0.03.
For the 250 µg/ml concentration, the triplicates have a constant OD600 of approximately 0.03, with Cltr 1 and 2 showing slight growth starting on day 7, reaching approximately 0.1 by the end of the measurement. The triplicates for Ctr. 2 and 3 at the 500 µg/ml concentration show a constant OD600 value of 0.03 starting on Day 3. For Ctr. 1, an OD600 of 0.1 was measured between Days 2 and 10, which increases to just under 0.3 by Day 13.

For the 750 µg/ml and 1000 µg/ml 8-Aza solutions, replicate Ctrl 3 exhibits a constant value of 0.03 starting on day 2. At 1000 µg/ml, the other two triplicates show a constant OD600 of just under 0.1 starting on day 2, while at 750 µg/ml, only Cltr 2 shows this value starting on day 2. The Cltr 1 triplicate in the 750 µg/ml solution shows a value of 0.03 on days 2 and 3, then increases to an OD600 between 0.2 and 0.3 by day 7. This value decreases slightly by the end of the measurement series.

DMSO

In general, all preparation with DMSO showed the tendency to be more vulnerable to turn red when taking a sample for photometric measurement. In addition, The decolorization also took longer than in the positive control or other experimental conditions without DMSO, sometimes over an hour.

This interaction of DMSO with the DSM 1523 media was investigated more closely in the *M. thaueri* measurement series. Here, one triplicate with DSM 1523 was prepared with just 1 ml DMSO, but without inoculation (DMSO –, see fig. 4.8).

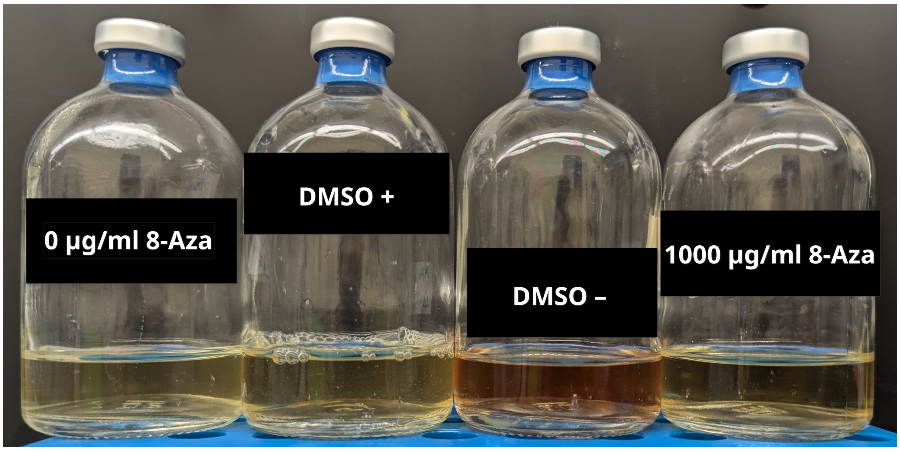

Figure 4.8: Color differences with and without DMSO after 6 days of incubation. Image of various preparations for *M. thaueri* from the same triplicate (Cltr. 3). The positive control (0 µg/ml 8-Azahypoxanthin (8-Aza)) contains only the organism (*M. thaueri*) in DSM 1523, 1000 µg/ml 8-Aza *M. thaueri* and the named base analog, DMSO + *M. thaueri* and 1 ml DMSO; DMSO – also contains 1 ml DMSO but was not inoculated.

After the first few days of incubation, a difference in color was observable among the cultures. Specifically, the DMSO bottles exhibited a red color, in contrast to the yellow color of the other samples in the 8-Azahypoxanthin (8-Aza) measurement series for *M. thaueri*. This is illustrated in fig. 4.8 using selected samples from the triplicate Cltr. 3 after 6 days of incubation. The OD600 was also measured for DMSO, but showed no difference from the baseline OD600 values of the other samples (approximately 0.03 to 0.04).

4.1.6 Plasmid designing

***pac*-gene**

The *pac*-gene in the initial plasmid (*pac_old*), that should have been taken for the transformation via conjugation of the archaea, showed a different aminoacids (AA) sequence to the original sequence (*pac_org*) from *Streptomyces alboniger* (see (a) in fig. 4.9). They differ in five AA on the C-terminal end (see (a) in fig. 4.9).

(a)

MTEYKPTVRLATRDDVPRAVRTLAAAFADYPATRHTVDPDRHIERVTELQ
ELFLTRVGLDIGKVWVADDGAAVAVWTTPESVEAGAVFAEIGPRMAELSG
SRLAAQQQMEGLLAPHRPKEPAWFLATVGVSPDHQGKGLGSAVVLPGVE
AAERAGVPAFLETSAPRNLPFYERLGFTVTADVEVPEGPRTWCMTRKPGA
$\}$ *pac_org*

MTEYKPTVRLATRDDVPRAVRTLAAAFADYPATRHTVDPDRHIERVTELQ
ELFLTRVGLDIGKVWVADDGAAVAVWTTPESVEAGAVFAEIGPRMAELSG
SRLAAQQQMEGLLAPHRPKEPAWFLATVGVSPDHQGKGLGSAVVLPGVE
AAERAGVPAFLETSAPRNLPFYERLGFTVTADVECPKDRATWCMTRKPGA
$\}$ *pac_old*

(b) **(c)**

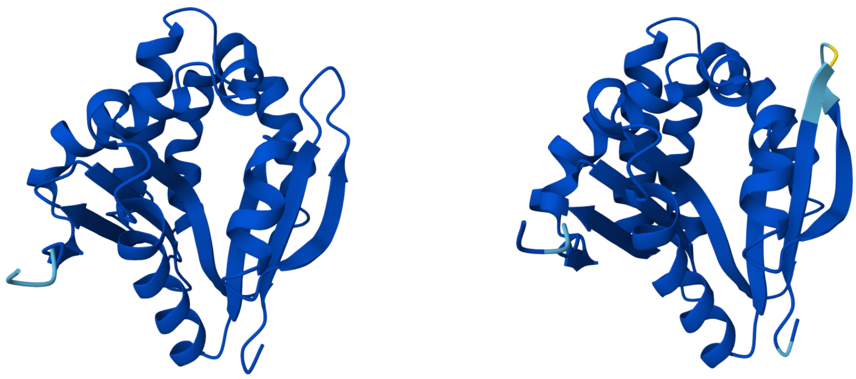

Figure 4.9: Aminoacids (AA) sequence of two *pac* genes and their 3-D structures. (a) shows the AA sequence of the original sequence from *Streptomyces alboniger* (*pac_org*) as well as from the *pac* gene that was located in the plasmid originally intended for transformation (*pac_old*). The differing section is highlighted in red. (b) The projected 3D structure for *pac_org* calculated by AlphaFold. (c) The projected 3D structure for *pac_old* calculated by AlphaFold. The color of the structure in AlphaFold shows the confidence of the predicted structure: blue, very confident, light blue less and yellow not very confident.

These mutation lid in AlphaFold to different 3-D structures. In (b) of fig. 4.9 the original *pac* has an antiparallel β-sheet with a big loop on the right side of its structure. The *pac_old* (see (c) in fig. 4.9) shows also an antiparallel β-sheet at this position, but it is longer and has a shorter loop. Interestingly is, in its prediction for this region is not so confident as before (see colorchange between (b) and (c) in fig. 4.9).

Plasmid

In SnapGene, four plasmid maps were created for *M. thaueri* and two for *M. boviskoreani*. The constructs followed the general scheme shown in fig. 1.2 and are listed in table B8. For the plasmids, the original *pac* sequence (*pac_opt*) was codon-optimized for the respective archaea.

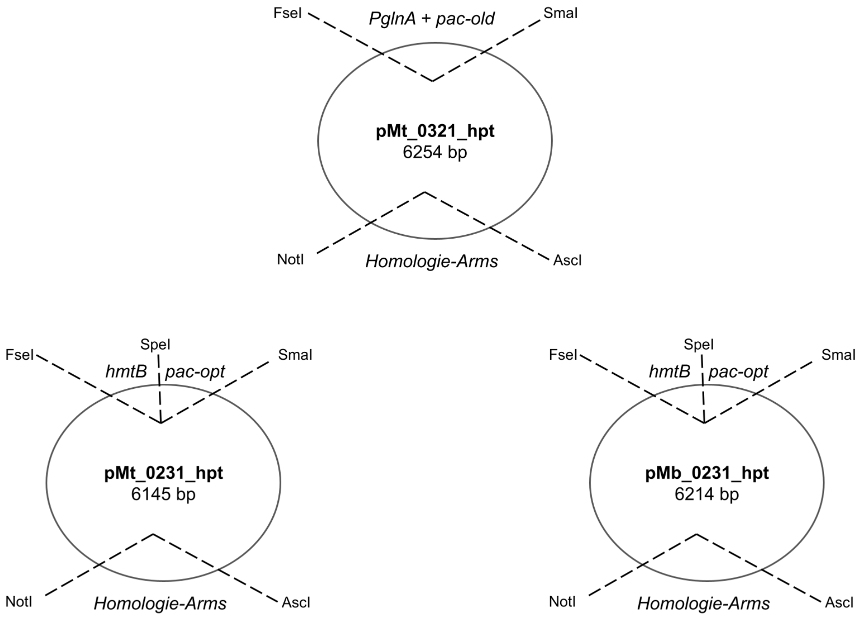

Figure 4.10: Suicide plasmid maps for transformation by conjugation for *Methanobrevibacter thaueri* CW^{T} (*M. thaueri*) and *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*). pMt_0321_hpt (6254 bp) is the parent plasmid used to construct the other two, pMt_0231_hpt (6145 bp) and pMb_0231_hpt (6214 bp). The maps show only the gene regions (between the dashed lines) that have been modified, as well as the restricitionsite sites (at the end of the dashed lines) used or the newly created restricitionsite sites. The new produced plasmids were verified via nanopore sequencing.

Two plasmids were constructed, as shown in fig. 4.10: pMt_0231_hpt for *M. thaueri* and pMb_0231_hpt for *M. boviskoreani*. Compared to the plasmid originally used for the construction, a new restricitionsite (SpeI) was inserted between the *pac* and its promoter (see fig. 4.10). In addition, the plasmid *pac_old*, which contained the alternative AA sequence, was converted to the codon-optimized version with the original AA sequence (*pac_opt*), and the promoter was changed to hmtB.

Both plasmids were verified by sequencing.

4.1.7 Transformation via conjugation

E. coli cultivation test in DSM 1523

Cultivation of *E. coli* DH5α was tested under similar settings in DSM 1523 as for *Methanobrevibacter*. Three drops for inoculation was enough to get a dense culture the next day.

Preparation for transformation

Max soll ich die Tabelle mit den Ansätzen doch lieber in den Anhang hauen, weil sehr groß oder möchtest d die hier im results part haben, da die Konjugation der interessanteste Versuch ist?

The DAP-auxotrophic *E. coli* WM6026 and *E. coli* WM6029 were successfully transformed with the plasmids pMt_0231_hpt, pMm_0231_hpt and pMb_0231_hpt. For the conjugation each plasmid combination with *E. coli* WM6026 and *E. coli* WM6029 were cultivated in 20 ml LB media with 0.1 mM DAP and 30 µg/ml chloramphenicol. For all *Methanobrevibacter* species except *M. boviskoreani** the media was inoculated with 1 ml *E. coli* WM6026 and WM6029 pre-culture and cultivated for 4.5 h. For *M. boviskoreani*, it was inoculated with 1.5 ml and the *E. coli* were cultivated for 3.75 h. The final OD600 is shown in table 4.3.

For the growth of *E. coli* WM6026 and WM6029 is quit heterogenous. So have the strains with 1 ml inoculation final OD600 values between 1.02 and 1.59. The maximum OD600 (1.79) was measured for WM6026 with 1.5 ml inoculation and a 45 min shorter cultivation. But the minimum was measured for this preparation as well with 0.60 for WM6029.

For the conjugation 28 preparations were prepared (see table 4.3). They differ in combination of *E. coli* strain and *Methanobrevibacter* species (cultures with an asterisk showed first visual growth on the day they were used for the conjugation) and in the recovery-time (time until puromycin (Puro) was added, see t_P in table 4.3) as well as in the time until then they were transferred in new selection media (see t_{1_t} and t_{2_t} in table 4.3).

Of the 28 preparations 3 preparation lines showed visual growth could be detected after the transfer into new media (highlighted red in table 4.3). Although most of them showed in the recovery media after some time dense growth, but usually a bit untypically kind for an only *Methanobrevibacter* culture. All three preparations belong to fresh grown *Methanobrevibacter* culture (*M. boviskoreani** and *M. millerae**) and the *M. boviskoreani** also showed its typical precipitates. In addition, all had as conjugation strain the *E. coli* WM6026.

Furthermore, two belong to the preparation that have a short recovery time of 4.5 to 5 h (5.6A and 6.6A), and one with a long recovery time 19.5 h (5.6B).

Of these three preparation lines the 1st and 2nd transfer of 5.6B, the 1st transfer of 5.6 A, and th 1st and 2nd transfer as well as the recovery of 6.6A were used for further investigation. All

Table 4.3: Preparation of the conjugation for the transformation of *Methanobrevibacter thaueri* CW^{T} (*M. thaueri*), *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*), and *Methanobrevibacter millerae* ZA-10^{T} (*M. millerae*). Two DAP-auxotrophic *Escherichia coli* (*E. coli*) strains were used for conjugation, WM6026 and WM6029. The final OD600 was measured as a 1:10 dilution with LB-media before the Plating with *Methanobrevibacter*. The *E. coli* for *M. thaueri* contained the plasmid pMt_0231_hpt, for *M. boviskoreani* the plasmid pMb_0231_hpt, and for *M. millerae* the plasmid pMm_0231_hpt. Cultures marked with an asterisk showed visible growth for the first time on the day they were used for conjugation. Preparations from the same *Methanobrevibacter* pre-culture share the same first number (No.). Preparations from the same plate are distinguished by the letters A and B. Before being transferred to a 10 ml DSM 1523 medium containing cycloserin (Cyc-Se.) (200 µg/ml) and no DAP, the strains *Methanobrevibacter* and *E. coli* were incubated overnight on agar plates containing 10 m $MgSO_4$, 5 mM $CaCl_2$, and 0.1 mM DAP overnight in a jar (headspace with H_2/CO_2 mix). t_P is the time that elapsed during recovery before puromycin (final concentration of 20 µg/ml) was added. The time from this point until the first transfer of the culture to a new DSM 1523 medium with the same concentrations of Cyc-Se. and Puro was measured and denoted as t_{1_t}. t_{2_t} then indicates the number of days that elapsed until the second transfer. The samples marked in red are those that showed visible growth in one of the transferred cultures and were collected for sequencing.

No.	*Methanobrevibacter*	*E. coli* (OD600)	t_P [h]	t_{1_t} [h]	t_{2_t} [d]
1.6A	*M. thaueri*	WM6026 (1.02)	5	17	4
1.6B			19.5	2.5	
1.9A		WM6029 (1.30)	5	17	
1.9B			19.5	2.5	
2.6A	*M. boviskoreani*	WM6026 (1.22)	5	17	4
2.6B			19.5	2.5	
3.6A			5	17	
3.6B			19.5	2.5	
2.9A		WM6029 (1.15)	5	17	
2.9B			19.5	2.5	
3.9A			5	17	
3.9B			19.5	2.5	
4.6A	*M. millerae**	WM6026 (1.24)	5	17	4
4.6B			19.5	2.5	
5.6A			5	17	
5.6B			19.5	2.5	
4.9A		WM6029 (1.59)	5	17	
4.9B			19.5	2.5	
5.9A			5	17	
5.9B			19.5	2.5	
6.6A	*M. boviskoreani**	WM6026 (1.79)	4.5	0.5	4
6.6B			21	0.5	3
7.6A			4.5	0.5	4
7.6B			21	0.5	3
6.9A		WM6029 (0.60)	4.5	0.5	4
6.9B			21	0.5	3
7.9A			4.5	0.5	4
7.9B			21	0.5	3

six were transferred after 16-20 days again, and then a week later again. The 1st and 2nd transfer of 6.6A were transferred a third time after 6 days.
For all a similar behavior could be detected. The 2nd and 1st transfer of 6.6A grew very well while the others grew but not as dense.

The genomic DNA (gDNA) was for all six cultures isolated and a 16S-Sequencing for the bacterial and archaeal 16S-rRNA was performed. The archaea 16S results and the negative control of the used primer set proof the corresponding *Methanobrevibacter* are in the cultures. The analyses of the bacterial 16S showed no sign of *E. coli* WM6026, but a contamination with *Staphylococcus epidermidis* (*S. epidermidis*). A negative check of the original used primer revealed also a contamination of them with *S. epidermidis*. To check the culture again, two primer sets for the PCR of the bacterial 16S-rRNA were used, that had a negative control test without any bands in the gel image.
The results of this PCR are shown in fig. 4.11 (see "bac. 16S fresh" and "bac. 16S extra" lanes). All six cultures show in their gDNA a clear band at approx. 1000 bp, which were extracted and verified as *S. epidermidis* via sequencing. The band thereby is pretty thick and tends to smear.

Figure 4.11 shows also the result for the PCR with a primer set (FW-screen-BBpMr and RV-screen-BBpMr) for a part of the suicide plasmid. All six cultures (lane 1 to 6 of the "Plasmid" lanes) show a clear band at 750 bp and is a bit less thick as the 1000 bp band of the bacterial 16S-rRNA.
For the lane 1,3 and 4, which belong to the three *M. millerae** (1st (4) and 2nd (1) transfer of 5.6B plus 1st (3) transfer of 5.6A), there is a second smaller band slightly below 500 bp. Furthermore, is the band at 750 bp a bit smaller than the band of *M. boviskoreani** cultures.
On NCBI with the "Primer Blast" it was also checked if the used primer set could hybridize with the gDNA of *M. millerae*, *M. boviskoreani*, *E. coli* and *S. epidermidis* and produce an PCR product of 4000 bp or less. No hit was found for the default parameters for specificity.

A full genome sequencing of the culture is still pending due to the contamination problems.

4.2 Geography

All methane emission analyses focused on three countries: Germany (DEU), New Zealand (NZL) and United States of America (USA). In addition, only the anthropogenic emission

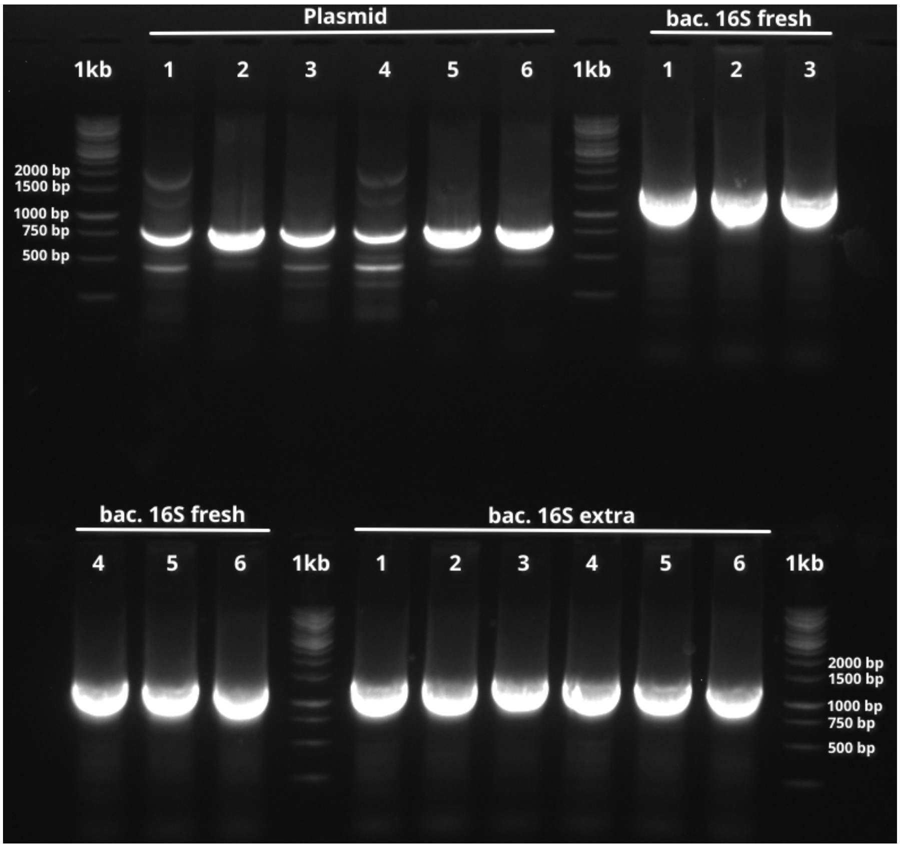

Figure 4.11: Gel image for different PCR products of growing cultures of *M. boviskoreani* and *M. millerae* after conjugation. For the lanes labeled 1, gDNA from culture 5.6B (1st transfer) was used; for 2, from 6.6A (1st transfer); for 3, from 5.6A (2nd transfer); for 4, from 5.6B (2nd transfer); for 5, 6.6A (2nd transfer); for 6, 6.6A (recovery). For a closer look at the cultures see table 4.3. “Plasmid” displays the results of the PCR for a portion of the plasmid used in the conjugation. The expected size would be 838 bp. "bac. 16S fresh" and "bac. 16S extra" show both the PCR product for the bacterial 16S-rRNA. Thereby were different primers used, but both had a clear negativ control before. For "bac. 16S fresh" the primers were freshly aliquoted. For "bac. 16S extra" primers different from those used in the first 16S sequencing were used. As ladder the 1 kb ladder from ThermoFischer was apllied ("1kb" lanes).

will be investigated with a special focus on CH_4 derived from enteric fermentation (EnterFe) processes in cattle.

4.2.1 National methane emission since 1990

Figure 4.12 shows the development of the total methane emission of a year as well as of the specific sectors from 1990 to 2020s for each country.

In general, the first thing that stands out is that the time series for the U.S. ends as early as 2022, while those for the other two countries do not end until 2024. Furthermore, the scale of methane emissions shows that the U.S. generally produces notably more CH_4. The maximum total emissions are 35 Tg CH_4, while for DEU the maximum is 5 Tg and for NZL it is 1.5 Tg.

For Germany, a clear downward trend is evident in total CH_4 emissions (see fig. 4.12). Emis-

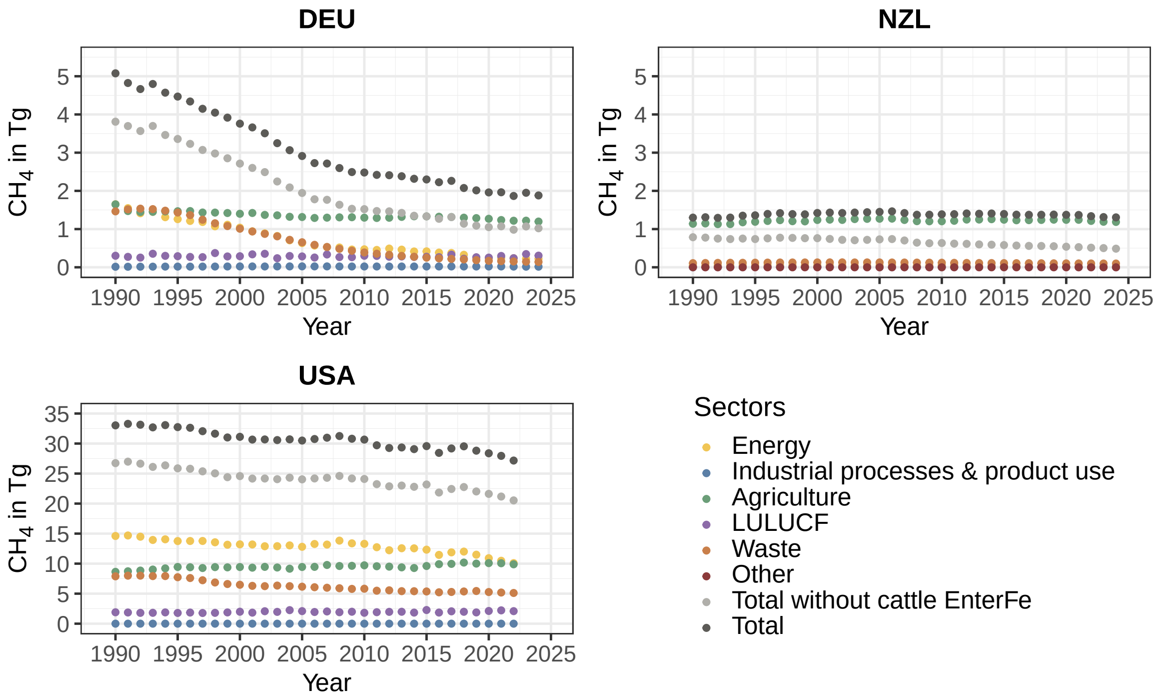

Figure 4.12: Time series of the anthropogenic methane emission of Germany (DEU), New Zealand (NZL) and United States of America (USA). The different relevant emission sectors are shown with their total sum (Total). Furthermore, the total emission minus the CH_4 produced in enteric fermentation (EnterFe) processes of cattle is plotted as well. Methane is plotted as Tg CH_4 of a year against the time.

sions fell around 3 Tg CH_4 in the 34 year long time series (1990 5 Tg to 2 Tg in 2024), representing a decrease of approximately 60%. Two distinct phases of this trend can be identified: first, from 1990 to 2006, with a sharp decline of about 2.4 Tg CH_4 (approximately 2.6 Tg CH_4 in 2006), followed by a phase with a more moderate decline of around 0.6 Tg to 2 Tg of anthropogenically produced methane in 2024.

Total emissions excluding those from enteric fermentation (EnterFe) in cattle follow a similar trend, with the curve's value being around 1 Tg CH_4 lower (fig. 4.12 DEU).

Looking at the sectoral emissions of DEU, it becomes apparent that in 1990 there were three equally large CH_4-producing sectors: “Energy,” “Waste,” and “Agriculture,” each accounting for approximately 1.5 Tg CH_4 (see fig. 4.12). However, these sectors have developed differently. The "Energy" and the "Waste" sector have followed quite similar trends. Following a sharp decline between 1990 and 2006 - with the "Waste" sector not beginning to change until 1994 - from 1.5 to 0.5 Tg, a moderate decline then sets in, until approximately 0.025 Tg of CH_4 is emitted in 2024.

Agriculture is also changing, though only very slightly compared to the other two sectors: from 1.6 Tg of methane in 1990 to approximately 1.25 Tg CH_4 in 2024, with the largest single change occurring between 1990 and 1991, amounting to a difference of about 0.1 Tg. In 2015, total emissions without cattle EnterFe are exactly as high as those of the agricultural sector and fall below that level thereafter.

In addition to these three sectors, two others are shown in fig. 4.12 (“Industrial Processes & Product Use” and "LULUCF" (land use and land use change and forestry)).The “Industrial Processes & Product Use” sector is a negligibly small source of methane emissions. Its CH_4 emissions have remained consistently well below 0.1 Tg over the 35-year period. With CH_4 emissions consistently ranging between 0.3 and 0.4 Tg, the "LULUCF" sector was the fourth-largest single methane emission sector in Germany during the 1990s, but it surpassed the waste sector in 2016 and has been the second-largest CH_4-emitting sector in DEU after agriculture since 2019.

For the USA the distance between total methane emissions and total emissions excluding cattle EnterFe remains fairly constant at around 10 to 15 Tg CH_4 (see fig. 4.12). In addition, a clear, steady decline can be observed for the total methane emission from 1990 (approx. 33 Tg CH_4) to 2022 (approx. 27 Tg CH_4), although the curve does not follow the trend linearly but rather fluctuates slightly. Over this 32-year period, the U.S. has reduced its methane emissions across all sectors by about 6 Tg CH_4, which corresponds to a reduction of 18%.

As in Germany, the three most significant methane-producing sectors in the U.S. are “Energy,” “Agriculture,” and “Waste” in 1990. However, here the "Energy" sector is the largest contributor, accounting for nearly half of the emissions in 1990 with 15 Tg CH_4, followed by the waste and agriculture sectors with approximately 8 Tg CH_4 each.
The "Energy" and "Waste" sector show a downward trend over the time series (energy: from 15 to 10 Tg CH_4 and waste: from 8 to 5 Tg CH_4), while emissions from agriculture rise slightly over time, from 8.5 Tg to 10 Tg CH_4. Thus, in 2022, agriculture and energy - each at 10 Tg CH_4 - are the largest methane emission sectors and together account for more than two-thirds of total methane emissions in the U.S. in 2022.

Two other sectors contribute to U.S. methane emissions: the “Industrial Processes & Product Use” sector, which is relatively small and therefore negligible, and the “LULUCF” sector, with a constant range of 0.2 to 0.25 Tg CH_4.

Compared to the other two countries, New Zealand presents a different picture (see fig. 4.12). First, total CH_4 emissions in 2024 are nearly the same as those in 1990, namely about 1.4 Tg CH_4. Furthermore, a slight increase in emissions to 1.5 Tg CH_4 can be observed through 2006, followed by a slight decline.
Second, “Agriculture” is the absolute dominant sector in terms of methane emissions and accounts for nearly 90% of CH_4 emissions from NZL (see fig. 4.12). All other sectors account for significantly less than 0.1 Tg CH_4 per year, with the exception of the waste sector, whose methane emissions are slightly above 0.1 Tg CH_4.
Third, the difference between total methane emissions and total emissions excluding cattle EnterFe increases by about 60% (0.5 Tg CH_4 in 1990 and 0.8 Tg in 2024).

4.2.2 Enteric fermentation processes of cattle

Methane emissions

Figure 4.13 shows the trend in methane emissions from enteric fermentation in cattle, as well as their subcategories “dairy cattle” and “non-dairy cattle” for each country. The values for the subcategories are calculated according to eq. (1) and based on the required data from the reported CRT (visualized in fig. A4 through A6). Total emissions from cattle are therefore simply the sum of these values.

Cattle emissions for Germany (DEU) in fig. 4.13 decline from 1990 to 2024 (from 1.25 Tg CH_4 to approximately 0.8 Tg), with a steeper decline in the first year of the time series, followed by a more gradual decline.
For dairy and non-dairy cattle, this initial decline is also evident in fig. 4.13, after which a slight decline can also be observed (dairy cattle: 0.75 Tg CH_4 in 1990 to 0.5 Tg in 2024; non-dairy cattle: 0.55 in 1990 to 0.35 in 2024). Dairy cattle are an exception: their emissions drop to 0.5 Tg CH_4 by 2006, then rise slightly again through 2015, and subsequently decline once more, returning to 0.5 Tg by 2014.

Dairy cattle consistently produce slightly more CH_4 per year than non-dairy cattle (0.1–0.2 Tg CH_4, see fig. 4.13). This contrasts with the population ratio shown in fig. A4, where the dairy cattle population size generally accounts for about half of the non-dairy cattle over the years.
Further differences in the indicators are evident for Ym, where dairy cattle had a significantly higher value of 7% in 1990 compared to 6.25, which, however, declines sharply to 6.1% by 2024, while the value for non-dairy cattle rises to 6.5%. GE changes similar, the value for the

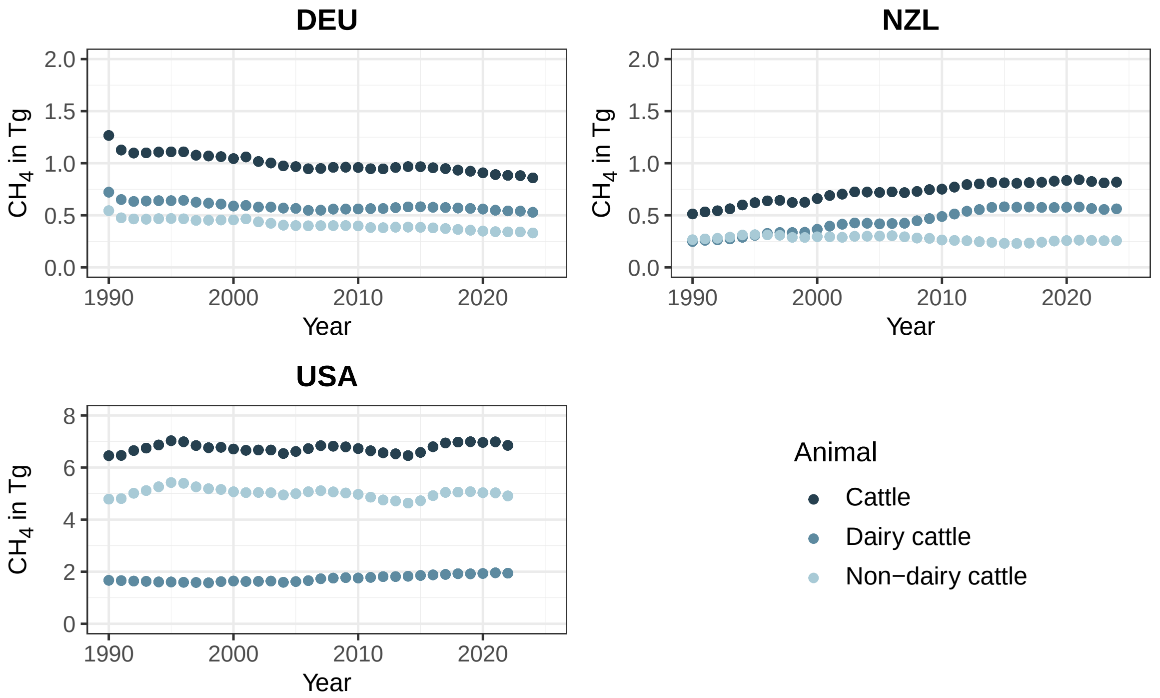

Figure 4.13: Time series of methane emission derived from enteric fermentation processes in cattle and its subcategories dairy and non-dairy for the countries: Germany (DEU), New Zealand (NZL) and United States of America (USA). Methane emission are measured in emitted Tg CH_4. Emissions are alculated for dairy and non-dairy cattle with the indicators in fig. A4:A6 and the eq. (1). The cattle emission is than the sum of its subcategories.

non-dairy cattle remains constant at 100 MJ head^{-1} day^{-1}, while the value for the dairy cattle population rises from 250 to 380 MJ head^{-1} day^{-1}.

Methane emissions from cattle in EnterFe for New Zealand rise from 0.5 to 0.75 Tg CH_4, while the values have remained relatively constant since 2014.
An increase is also evident for dairy cattle, which has plateaued since 2014 (0.25 Tg CH_4 in 1990 to 0.55 Tg in 2024). Non-dairy cattle, on the other hand, show similar values of around 0.5 Tg CH_4 in both 1990 and 2024.
Thus, in 1990, dairy and non-dairy cattle contributed equally to methane emissions from the cattle population; however, this share has changed over the years for dairy cows, so that by 2024 they will account for about two-thirds of the emissions. This shift in the ratio is quite similar across the subcategory populations (see fig. A5). Furthermore, the GE values for non-dairy cattle remain constant at 150 MJ head^{-1} day^{-1}, while the value for dairy cattle rises from 160 to 225 MJ head^{-1} day^{-1}.

In the USA, methane emissions from dairy cattle rise from just under 2 Tg CH_4 to 2 Tg, with the slight increase not beginning until around 2005. The other two livestock categories, on the other hand, fluctuate overall, for both non-dairy and dairy cattle, between 4.9 and 5.5 Tg for non-dairy livestock and between 6.5 Tg CH_4 and 7.1 Tg for dairy (see fig. 4.13). Unlike in the other two countries, non-dairy cattle here emit notably more than dairy cattle (2024: 5 Tg CH_4 versus 2 Tg).

As with the other countries, the emission curves correspond fairly well with the livestock numbers from fig. A6, where non-dairy cattle (80 million head) are kept in clear greater numbers than dairy cattle (20 million head). The curve for dairy cattle does not match the livestock numbers exactly, as these remain constant. However, the GE for dairy cattle rises steadily from 200 to 260 MJ head^{-1} day^{-1} between 1990 and 2022, while the Ym value decreases from 6.5% to 6.1% over the same period.

When comparing cattle emissions across countries in fig. 4.13, it is striking that the U.S. produced more than three times as much in 2022 as the other two countries combined (6.8 Tg CH_4 compared to less than 2 Tg combined). This is also reflected in the number of animals: in 2022, the US had approximately 100 million cattle, while DEU and NZL each had around 10 million (see fig. A4:A6).

Germany and New Zealand produce similar amounts of methane from cattle farming in 2024 (approx. 0.8 Tg CH_4), in 1990, DEU had produced twice as much, but also kept twice as many cattle (approx. 20 million animals in 1990 compared to 10 million in 2024).

Share of CH_4 emissions from EnterFe of cattle

The share of CH_4 produced through EnterFe processes in cattle on entire amount of emitted methane for the agriculture sector of a country declines slightly for Germany (approx. from 78% 1990 to 76% 2024, see fig. 4.14) and USA (from 76% 1990 to 74% 2022, see ibid.). For the share on total methane emissions, both curves increase over time quite linear, but is the increase fur the US mild from 20% (1990) to 25% (2024), is it for DEU quit intense from 1990 also a value of around 20-25% to 45% in 2024.

New Zealand time series for the development of the share of enteric fermentation cattle CH_4 emission looks a bit different to those described (see fig. 4.14). Both curves for the share on total and agriculture methane emission show a similar form, they just have slightly different values. The share on agriculture on the one hand grew from 1990 to 2024 from 45% to 70%, on the other hand the share for total emission from 40% to a bit less than 65%.

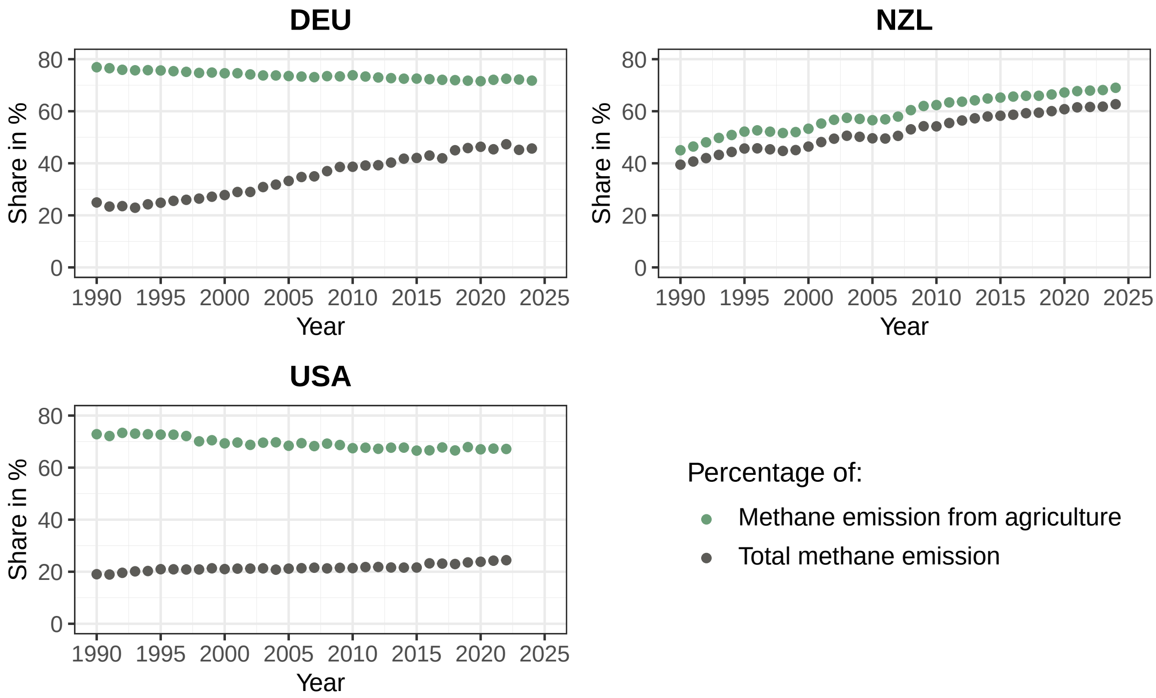

Figure 4.14: Time series of the share of methane emissions resulting from enteric fermentation of cattle on methane produced in the entire agriculture sector and the total amount of CH_4 per country and year. The calculation of the share are based on the information of the reported CRT.

4.2.3 Reductionpotential in enteric fermentation processes of cattle

Hier muss definitiv in der Disskusion ein Vergleich mit der atmospheric growth rate von 20.1-21.7 Tg CH_4 / year dür diese dekade kommen -> in sangios et al/ siehe präsi + kann hier auch darüber sprechen, dass linearität von Ym eindeutig den von DE überwiegt

Figure 4.15 shows the average reduction in CH_4 emissions during the decade from 2010 to 2019 in the countries NZL, DEU, and USA for various possible scenarios in which the shift in the rumen microbiome was successful to varying degrees. The scenarios are 10, 20, 30, 40, and 50, where the numbers refer to the percentage by which the microbiome is shifted toward non-methane-producing microbes. Table table B9 in the appendix lists the precisely calculated amounts of saved CO_2 equivalent (CO_2eq.) and CH_4.

For the various scenarios, the CH_4 diagram shows that, under identical scenarios, by far the largest amount of CH_4 can be saved in the United States, and that Germany and New Zealand make roughly equal contributions over the ten-year period examined, with Germany contributing slightly more. Overall, a 10% shift would result in annual emissions of approximately 0.9 to 1 Tg CH_4 less over the decade 2010-2019, with about 0.75 Tg attributable to the United States alone.

At 30%, the total reduction in methane emissions would be approximately 2.8, and at 50%, it would be almost exactly 4.5 Tg. The rise between the different reduction scenarios seems to be linear in fig. 4.15, although table B9 shows a slightly non-linear rise.

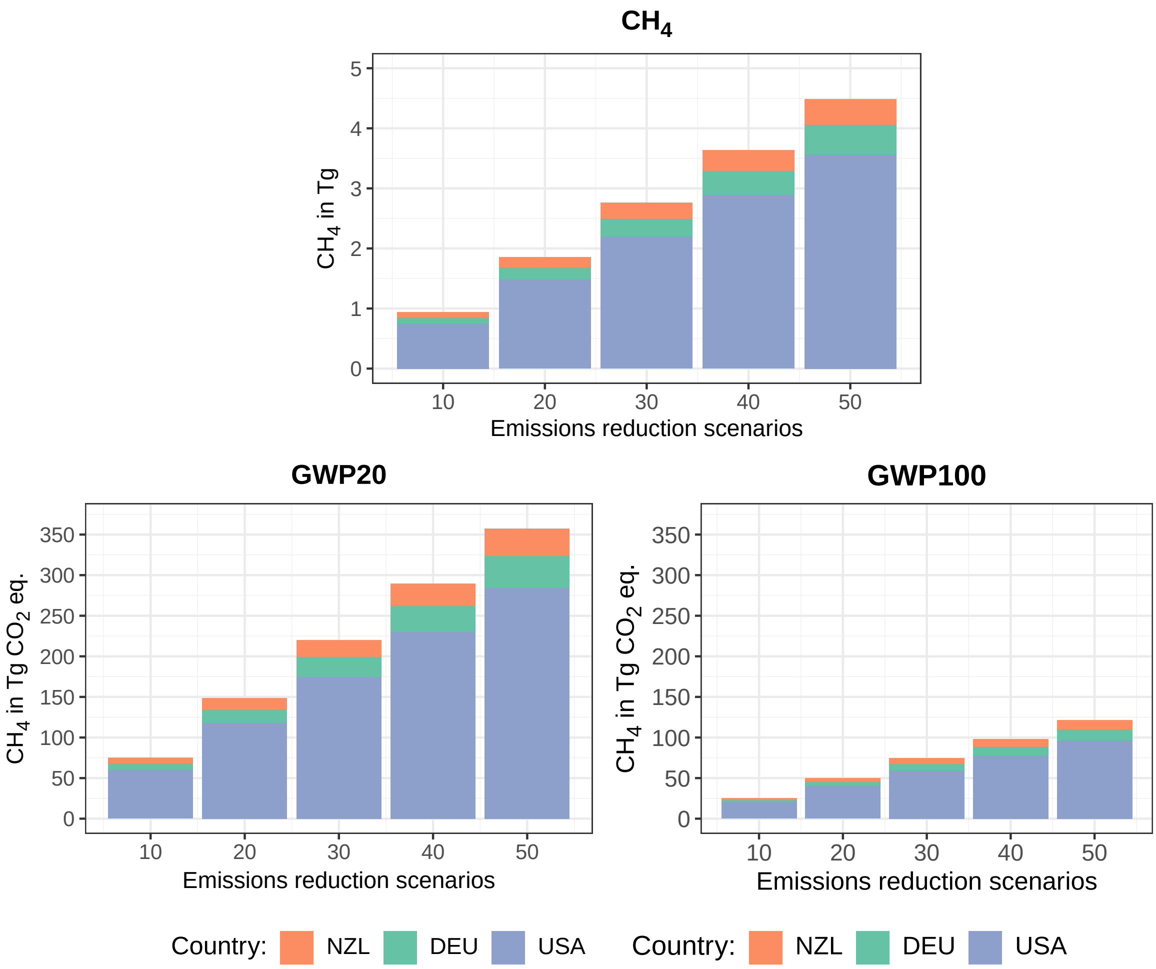

Figure 4.15: Saved CH_4 through different scenarios (10, 20, 30, 40 and 50%) of a successful shift of rumen microbiome towards less methanogenic microbes. The saved methane is plotted as Tg CH_4 against the these five "Emission reduction scenarios" (CH_4). The values for the global warming potential (GWP) for 20 years (79.7) and 100 years (27) are plotted as well in Tg CO_2eq. (Intergovernmental Panel On Climate Change (Ipcc), 2023). The saved CH_4 is calculated as an average for the decade 2010-2019 for each country (Germany (DEU), New Zealand (NZL) and United States of America (USA))

For global warming potential (GWP), clear quantitative differences can be observed depending on the amount of gas emitted. For example, in a “scenario with a 30% emission reduction” over a 20-year time horizon, approximately 220 Tg less eq. CO_2 would be emitted across all countries combined. Over a 100-year time horizon, this figure would be only 75 Tg. The relative dominance of the United States remains evident in these figures.

4.2.4 Prediction for 2030

Figure 4.16 (DEU), 4.17 (NZL) and 4.18 (USA) show the total methane emission for each country as a time series from 1990 to the last published year in the CRT and beyond until 2030 as a forecast. The forecast is shown for three different linear models (using 5, 10, or 15 years as the reference period for the fit) and for a scenario in which no further intervention was made, as well as for three scenarios with varying degrees of success in cattle immunization (10, 30, and 50% shifts in the rumen microbiome). The equations as well as the R^2 are given in the annex in table B10 (DEU), B11 (NZL) and B12 (USA).
Most of the R^2 of the linear models are above 0.5, but there are 5, which have values below 0.1 that are not due to mathematical reasons, because there was no slope predicted or the values did not change over time.

A similar trend is in all three figures visible for the three linear fits used. All models show a negative development, although the max. confidence intervall (CI) of the 5 year model has a positive trend.
Besides, the forecast for the fit with 10 or 15 years as time reference show very similar prediction while the 5-year linear model differs from them. But in DEU the 5-year fit predicts higher values for CH_4 production for 2030 than the other two models, while for NZL and USA the fits with the longer time sample as reference higher values predicts.
Two additional points stand out when comparing Germany with the other two countries (fig. 4.16 to fig. 4.17 and 4.18).
First, in Germany, the predicted range of emission values at least partially meets the two emission targets set by the global methane pledge (GMP), whereas in the other two countries, the emission values exceed the targets.
Second, the value for the 60% of 2010 emissions target is higher than the 70% of 2020 emissions target for DEU. For the other two countries, it is the other way around. Even so, the two values are in a similar range for all of them.

A closer look at the values in fig. 4.16 for Germany shows that the five-year model for predicting the emissions trend, assuming unchanged mitigation efforts, forecasts a nearly constant trend with a slight decline (from 1.9 Tg CH_4 to 1.8 Tg), although the confidence intervals here are quite wide and do show even a possible increase of methane emission.
The 10- and 15-year models, on the other hand, show a clear downward trend with a value of approximately 1.5 Tg CH_4 in 2030, which corresponds almost exactly to the 60% target of the

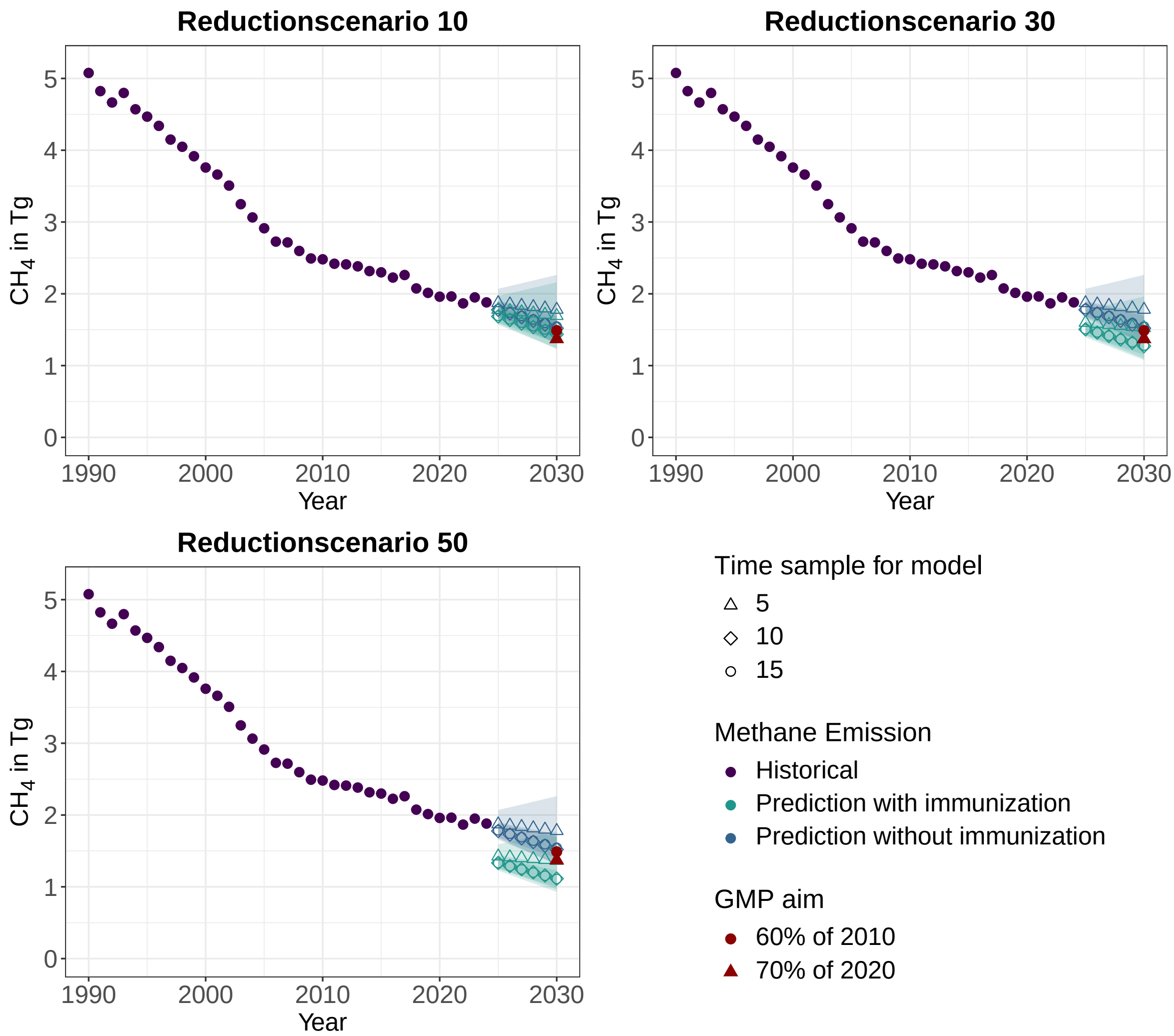

Figure 4.16: Historical values for total methane emissions in Tg for **Germany** from 1990 to 2024 and the forecast of methane emissions through 2030. The projection was generated using three different linear models with varying time reference periods (the last 5, 10, or 15 years), which are listed in table B10 along with the corresponding equation and R^2. Total emissions were not extrapolated directly. Instead, they were determined using a prediction of total emissions excluding enteric fermentation in cattle, as well as the four indicators (DE, GE, NE_x, and herd size), to calculate the missing emissions from cattle production. In addition to predicting total CH_4 emissions without additional mitigation measures (blue), the calculation was also performed for three reduction scenarios (10, 30, and 50% shift in the rumen microbiome) (green). For each linear extrapolation, the confidence intervall are displayed as a colored area in the corresponding color (confidence level of 95% was used). The two global methane pledge (GMP) methane emission targets are also shown in red: 60% of 2010 methane emissions as a circle and 70% of 2020 methane emissions as a triangle.

GMP. The range defined by the CI is also significantly narrower here, ranging from about 1.4 to 1.75 Tg, with the 70%-target remaining just above the lower CI range.

The cumulative range of all three confidence intervall values points to a range of approximately 1.4 Tg to 2.25 Tg CH_4.

The projected values for the reduction scenario 10 hardly differs from the scenario without mitigation measures and deviates by only about 0.1 Tg (see fig. 4.16).

A greater difference can be observed for Scenario 30. Here, the projected emissions for the 5-year adjustment would be roughly at the level of the GMP target of 60% of 2010 emissions and thus at values comparable to those in the 10- and 15-year models under "Projection without vaccination." For this scenario, the values for these two models under "with immunization" are just below the target of 70% of 2020 emissions, namely at approximately 1.3 Tg CH_4. The range of cumulative CI for the year 2030 extends from 1.1 Tg to just under 2 Tg of emitted CH_4. This means, for these models assuming a 30% change in the rumen microbiome, no further increase in methane levels would be expected until 2030.

The scenario 50 with an immunisation of cattle that leads to a 50% shift of microbiome towards less methan production, had an combined CI range from slightly below 1 Tg to 1.75 Tg CH_4 emitted in 2030. Where the models and their CIs would be completly below both GMP emission targets for 2030 (see fig. 4.16, approx. slightly below 1 Tg to 1.3 Tg CH_4).
In addition, the prediction for the 5-year linear fit would be 2030 at nearly exact the 70% emissions of 2020 (around 1.4 Tg CH_4), although the max. CI value is with 1.75 Tg clearly above both emission aims.

Figure 4.17 shows the prediction for NZL. The prediction without immunization has a combined CI range from slightly above 1 Tg to 1.35 Tg emitted methane for 2030. Here, the CI is as before for DEU notable wider for the 5-year model as the other two. Neither the values directly predicted through the models (1.2 Tg CH_4 for the 5 year fit and 1.25 to 1.3 Tg for the 10- and 15-year model) nor the CI range achieves one of the two GMP emission targets. The 60% of 2010 emissions is at 0.8 Tg and 70% of 2020 at slightly low 1 Tg CH_4.

The difference between no immunization and immunization for the reduction scenario 10 is like for Germany in fig. 4.16 round about 0.1 Tg, but this is enough that the minimum of the CI to be low enough to achieve just the 70% target (see fig. 4.17).
At the reduction scenario 30, this aim lays between the prediction of th 5-year (approx. 0.9 Tg CH_4) and the etrapolation with the 10- and 15-year model (1 Tg). Besides, the lower limit for the confidence intervall reaches the other GMP aim.
For the last assumed scenario (50% shift of microbiome) the 10- and 15-year linear model predict methane emission values that are nearly exact the 60% target. So, most of the predicted reange for the methane emission achieves this GMP aim with an range from 0.7 Tg to 0.85 Tg emitted methane for the year 2030.

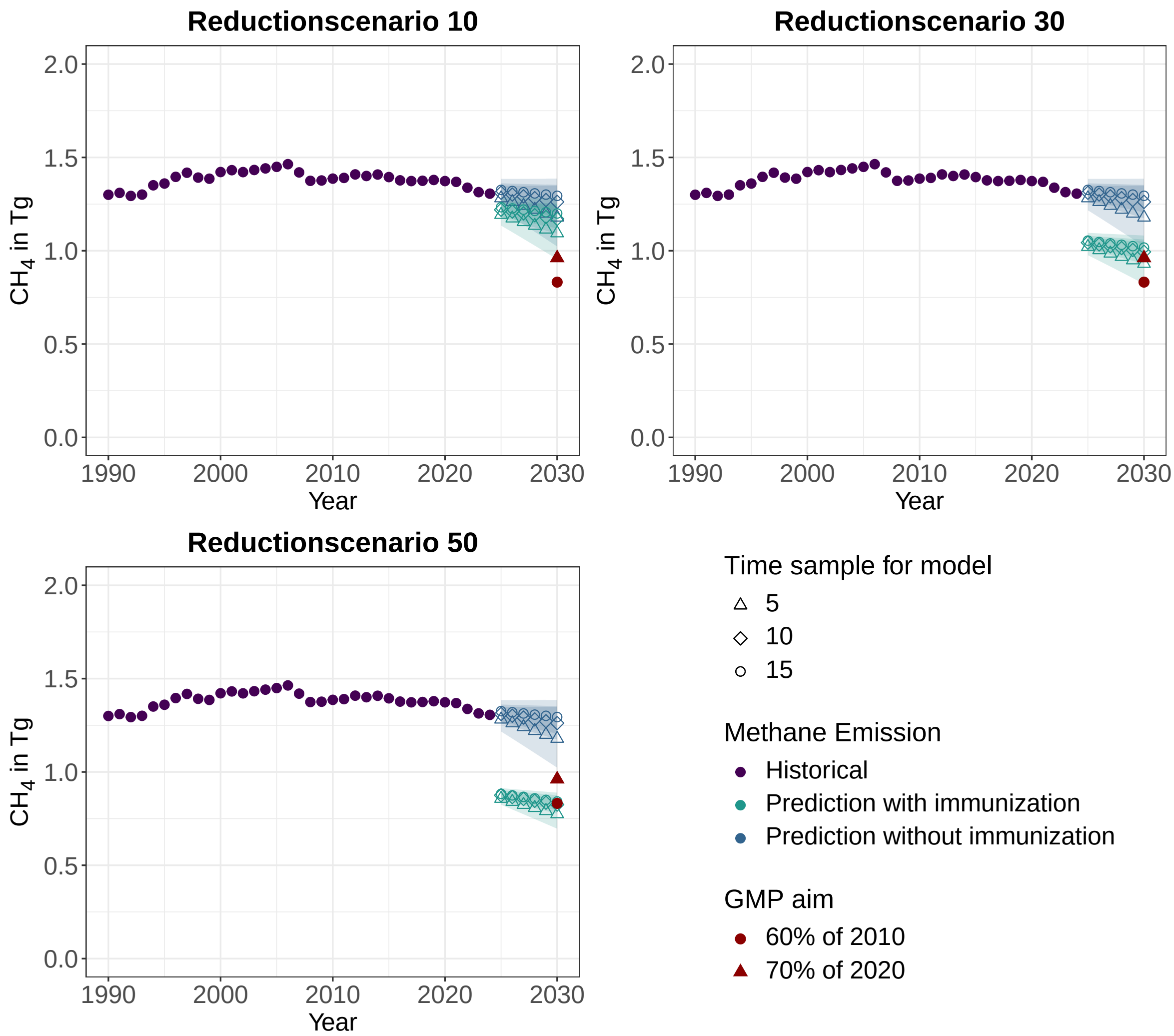

Figure 4.17: Historical values for total methane emissions in Tg for **New Zealand** from 1990 to 2024 and the forecast of methane emissions through 2030. The projection was generated using three different linear models with varying time reference periods (the last 5, 10, or 15 years), which are listed in table B11 along with the corresponding equation and R^2. Total emissions were not extrapolated directly. Instead, they were determined using a prediction of total emissions excluding enteric fermentation in cattle, as well as the four indicators (DE, GE, NE_x, and herd size), to calculate the missing emissions from cattle production. In addition to predicting total CH_4 emissions without additional mitigation measures (blue), the calculation was also performed for three reduction scenarios (10, 30, and 50% shift in the rumen microbiome) (green). For each linear extrapolation, the confidence intervall are displayed as a colored area in the corresponding color (confidence level of 95% was used). The two global methane pledge (GMP) methane emission targets are also shown in red: 60% of 2010 methane emissions as a circle and 70% of 2020 methane emissions as a triangle.

For the U.S., the 10- and 15-year models for "prediction without immunization" in fig. 4.18 predict an emissions value for 2030 that corresponds almost exactly to the 2022 value (approximately 27 Tg CH_4). The 5-year fit, on the other hand, shows a significant decrease by 2030—a reduction of approximately 3 Tg to 24 Tg of methane emitted in 2030. Even with the combined CI range (22 to 28 Tg CH_4), the lowest value is still significantly higher than the nearest GMP emissions target (20 Tg CH_4 for the 70% target).

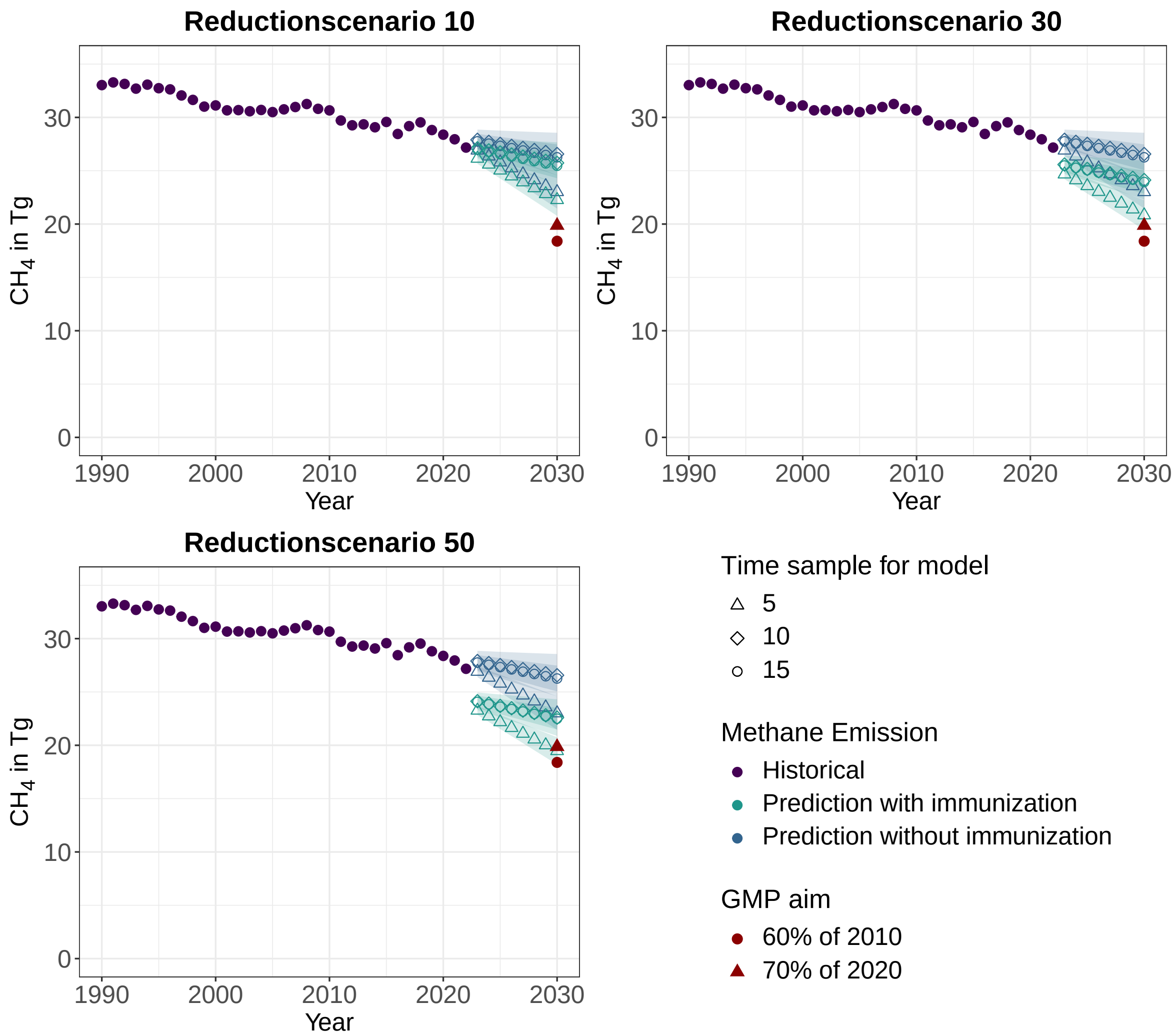

Figure 4.18: Historical values for total methane emissions in Tg for the **USA** from 1990 to 2022 and the forecast of methane emissions through 2030. The projection was generated using three different linear models with varying time reference periods (the last 5, 10, or 15 years), which are listed in table B12 along with the corresponding equation and R^2. Total emissions were not extrapolated directly. Instead, they were determined using a prediction of total emissions excluding enteric fermentation in cattle, as well as the four indicators (DE, GE, NE_x, and herd size), to calculate the missing emissions from cattle production. In addition to predicting total CH_4 emissions without additional mitigation measures (blue), the calculation was also performed for three reduction scenarios (10, 30, and 50% shift in the rumen microbiome) (green). For each linear extrapolation, the confidence intervall are displayed as a colored area in the corresponding color (confidence level of 95% was used). The two global methane pledge (GMP) methane emission targets are also shown in red: 60% of 2010 methane emissions as a circle and 70% of 2020 methane emissions as a triangle.

Even the confidence intervall range of reduction scenario 10, which differs by approximately 1 Tg from the scenario with no additional mitigation efforts, does not meet this target with its lowest value of approximately 21 Tg.

With reduction scenario 30, the prediction for the 5-year model is still slightly above the target at 21 Tg CH_4, but the CI range falls just below 20 Tg and thus below the 70% methane emissions level of 2020. The maximum value for the CI range is approximately 26 Tg, meaning that under this scenario, no values predicted for the year 2030 are above the 2022 value (see fig. 4.18).

In the 50% reduction scenario, the 5-year model prediction, with a value of approximately 19.5 TgCH_4, is slightly below the 70% target but still above the GMP's 60% target relative to 2010 emissions (approximately 18 Tg CH_4). However, the CI range of the 5-year fit (18 to just under 21 Tg CH_4) meets this target.
The other fits, as well as their CI range (21 to 24 Tg CH_4), do not meet either of the two GMP targets even in reduction scenario 50 (see fig. 4.18).

5 DISCUSSION

5.1 Biochemistry

5.1.1 Transformation via conjugation

Promising initial results have been achieved for implementing for the first time a genetic tool for *M. boviskoreani* and *M. millerae*, even though a final assessment is still pending due to the pending whole-genome sequencing(see section 4.1.7).

So in 3 of 28 preparations lines growth could be detected after selection with 20 µg/ml puromycin (Puro) (table 4.3). Within these three preparation lines, six cultures exhibited uniform growth of the corresponding *Methanobrevibacter* species (verified via 16S-sequencing)three each for *M. boviskoreani* and *M. millerae*, although the three cultures for*M. boviskoreani* were all from a single preparation line.

For these cultures, at least some evidence was provided that they had successfully taken up the plasmid, even though whole-genome sequencing is still pending. The hint is based on a performed PCR with the isolated gDNA, where a 838 bp region of the two used plasmids (pMb_0231_hpt and pMm_0231_hpt, see fig. 4.10) was amplified. The gel image of thisPCR shows a clear band at 750 bp in all six cultures (fig. 4.11). Although the expected size for the amplified plasmid region is 838 bp, it is assumed that the 750 bp band is the targeted region, because the used primers were checked for off targets in the gDNA of the possible organisms in the culture. The check with "Primer Blast" on NCBI did not show any possible hits with an PCR product below 1000 bp. Of course, because no negative control was performed, it could be due to unrealistic contamination of the primer set with the plasmid.
Besides this contamination, the signal in the gel image could be explained to two other possibilities. On the one hand, *E. coli* could still be present in the culture, and replicate the plasmid, which has the ori for *E. coli* on it. However, the bacterial 16S-Sequencing did not detect a signal for *E. coli*.
On the other hand, the plasmid could still be present in the culture medium and be extracted along with the gDNA during isolation. What argues against this, however, is that the band at 750 bp exhibits a visually very similar intensity for the three *M. boviskoreani* cultures (see lanes 2, 5, and 6 under “Plasmid” in fig. 4.11), although these lanes represent different dilutions or subculturing stages relative to one another; thus, for lane 5 (as the second transfer),

only 1% of the recovery culture from lane 6 should remain in the culture.
For the three *M. millerae* lanes (see 1, 3, and 4 in fig. 4.11) is it the same, although compared to the *M. boviskoreani* lanes the 750 bp has a lower intensity. In addition, all three *M. millerae* show a weak band at just under 500 bp.

Derived of the compare for these cultures, some more or less influencing factors for the conjugation can be assumed, although here a rather small sample was investigated and to verify this further investigation is definitely needed.
First, growth status of *Methanobrevibacter* seems to be relevant. So all cultures with growth are in the 16 preparation lines with freshly grown cultures (first visible sign of growth on the day the conjugation was performed). This aligns with the literature, where cultures from late exponential or early stationary phase were used for successful transformation (Capovilla et al., 2025; Fink et al., 2021). In which actual growth phase the used initial cultures were is not dectable here, because of missing OD600 values for the *M.* cultures. This should be changed in future investigation.
Second, all three growing preparation lines had as donator strain for the interdomain conjugation *E. coli* WM6026. Future transformation of *M. boviskoreani* and *M. millerae* should focus on that. The methylation pattern of *E. coli* WM6026 therefor does not impact the *M. boviskoreani* and *M. millerae* (Schwalen et al., 2018, SI).
Third, the recovery time is not so critical for the success (growing preparation line no. 6.6A had a recovery time of 4.5 h while 5.6B of 19.5 h, see table 4.3) of the transformation as described for *Methanothermobacter thermautotrophicus* ΔH in Fink et al. (2021).
Fourth, for the OD600 of the *E. coli* WM6026 no direct trend is detectable from this preparation set, though the range of the OD600 was 1.24 to 1.79 that seems to work.

A problem is, that also leads to the missing whole genome sequncing, PCR of the bacterial 16S-rRNA produces clearly a signal at 1000 bp (see fig. 4.11), which was sequenced as derived from *S. epidermidis*. Because of that, the next step would be to purify these cultures through plating on selective media and picking a single colonie for the following analyses.

If whole-genome sequencing confirms that the plasmid has been successfully integrated, the next step would be to test the counter-selection implanted in the plasmid design using 8-Azahypoxanthin. Also, it would be interesting to test if other transformation methods as electroporation could work.

To improve the conjugation protocol, it would likely be useful to first test whether a specific growth phase of *M.* has an effect by measuring its OD600 beforehand, because a clear trend towards fresh grown cultures is visible for this conjugation experiment.
To determine whether the OD600 of *E. coli* WM6026 influences success—or whether the ratio of *E. coli* cells to archaea cells does, as described, for example, as a critical factor in Kopp et al. (2004) for the conjugation of *Sorangium cellulosum*, further test series must be conducted.

5.1.2 Plasmid constructs

The most interesting finding for the plasmid construction, is the discovery of aminoacids sequence of the *pac* gene, which was on the parent plasmid and originally planned to be on the new constructed ones for the conjugation and is also several times used in the literature Max du hattest das in der literatur doch auch gennant gesehen oder?. So it differs in 5 aminoacids from the aminoacids sequence in the originally *pac* from *Streptomyces alboniger* that leed to an longer β-sheet at the C-terminal end site (see fig. 4.9).

If the changed 3-D-structure really has an influence on the function or the performed conjugation is not clear. To answer this question it needs further investigation.
Just to be on the safe side, for the used plasmids in the conjugation (see fig. 4.10) *pac* gene was replaced with the codon optimized original *pac* from *Streptomyces alboniger*.

In general the constructed plasmids followed the scheme in fig. 1.2, where the selection of a successful conjugation works with a resistance against puromycin via *pac*-gene and a counterselection with a sensitivity against 8-Azahypoxanthin (8-Aza).
If the whole-genome sequencing proofs the success of the transformation, a next step for the plasmid experiments would to test the other planed plasmids with a different promoter (*PglnA* instead of *hmtB*, see table B8).

5.1.3 Sensitivity tests

Puromycin
Since the selection of successfully transformed *Methanobrevibacter* is based on resistance to puromycin (Puro), it was first necessary to determine whether, and at what concentration, *M. boviskoreani* and *M. millerae* are sensitive to the antibiotic.

The growth curves in fig. 4.3 for *M. boviskoreani* showed that it exhibits no real sensitivity up to a concentration of 2 µg/ml Puro. At 5 µg/ml, an initial change in the growth curve can be observed. At 10 µg/ml and 20 µg/ml, the organism appears to be more sensitive, although growth could be detected through spontaneous resistance at 10 µg/ml after 4–6 days and at 20 µg/ml after approximately 8 days.
M. thaueri appears to be more sensitive to Puro (see fig. 4.4). Thus, a change in the growth curve can already be observed at a concentration of 2 µg/ml, and at concentrations of 10 µg/ml and 20 µg/ml, there is no growth at all within 10 days.
Should I also mention here that in another series of experiments with millerae conducted by the research group, it was already determined that this strain also shows no growth at 20 µg/mL within 11 days?

Selection using Puro therefore works for *M. boviskoreani* and *M. thaueri*, although they are a bit less sensitive than described in the literature for other archaea (minimum inhibitory concentration (MIC) 2 and 5 µg/ml for *Methanococcus voltae* and *Methanosarcina mazei* in Mondorf et al. (2012) and Possot et al. (1988)). For the upcoming conjugation experiments, a concentration of 20 µg/ml puromycin was chosen for selection. Should growth occur within 8 days, it can be assumed that this was mediated by the introduced resistance.

8-Azahypoxanthin hier muss ich definitiv noch einkürzen

As for the second and relevant function of the plasmid in the genetic modification of the *Methanobrevibacter* genome, namely, the counter-selection with 8-Azahypoxanthin (8-Aza), it is interesting to know how sensitive the organism is to the base analog. Although the *hpt* gene on the plasmid already provides a backup mechanism that induces artificial sensitivity to 8-Aza. Concentrations of 100, 250, 500, 750, and 1000 µg/ml 8-Aza were tested, hereby the 8-Aza dose not dissolve in H_2O, therefore DMSO was used as solvent.

As can be seen from fig. 4.5, 4.7, and 4.6, this series of measurements for *M. boviskoreani*, *M. millerae*, and *M. thaueri* has little to no predictive value regarding sensitivity to 8-Azahypoxanthin. However, the solvent control revealed a problem: DMSO itself has an inhibitory effect on the growth of the tested *Methanobrevibacter* cultures and is therefore not suitable as a solvent for the base analogs in this context.

In the case of *M. boviskoreani*, a slight effect of 8-Aza can still be observed (see fig. 4.5). For

example, the triplicates for the sample with a final concentration of 100 µg/ml 8-Aza show a curve very similar to that of the DMSO sample, even though only one-tenth of the DMSO volume was added to the 22.8 ml culture (0.1 ml to 1 ml). The maximum measured OD600 value, at approximately 0.1, is slightly below that of the positive control (0.2), and growth was delayed by one to two days. At 250 µg/ml, growth was delayed by an additional 2 days, and at concentrations of 500 µg/ml and higher, no OD600 values above 0.5 were measured within 13 days, with the exception of one replicate for the 1000 µg/ml solution. This indicates—particularly for the last three concentrations, but also between 100 and 250 µg/ml—that the base analog has an effect on the growth of *M. boviskoreani*; however, the extent of this effect cannot be determined, since the solvent DMSO itself inhibits growth by one-third.
For *M. millerae*, DMSO also shows an inhibition of approximately one-third compared to the positive control, even though a delay in growth could not be detected, as the second measurement was not taken until three days later. In addition, the 100 µg/ml preparation again resembles the DMSO control, but no pattern is discernible beyond that. For example, the 1000 µg/ml preparation grows almost exactly like the 100 µg/ml preparation; the 750 µg/ml sample includes one replicate that even reaches maximum OD600 values comparable to the positive control, as well as one replicate that showed no growth at all (see fig. 4.7).
In the case of *M. thaueri*, even less growth is visible, since—apart from an outlier reading on one day (Day 2) for a replicate of the DMSO sample—growth is only visible in the positive control. It should be noted, however, that growth occurred only after 5 days for this one replicate and after 9 and 14 days for the other two replicates. Furthermore, no OD600 value greater than 0.2 was measured.

A plausible explanation for the negative effect of DMSO on the growth of *M.* species is that, as reported in Atcher and Alfonso (2013), DMSO acts as a mild oxidizing agent toward dithiols at room temperature. This would result in the DTT, which is added to the culture as a reducing agent to remove residual oxygen and keep the medium completely anaerobic, reacting with the DMSO to form its cyclic structure with a disulfide bridge, whereas DMSO reacts to form dimethyl sulfide (Sanz et al., 2002). Consequently, no reducing agent is present in the medium, and the cultures are exposed to greater oxidative stress.

This hypothesis is supported by the following:
First, when samples were collected for photometric measurements, the cultures containing DMSO turned red more quickly and also took longer to lose their color than the positive

controls or than had previously been observed in other cultures (section 4.1.5).
Second, in the *M. thaueri* measurement series, a DSM 1523 medium was also prepared as a triplicate; this was not inoculated but only had 1 ml DMSO added. Here, as with the other cultures, the OD600 was measured daily. Within a few days, the culture turned distinctly red and did not fade (see fig. 4.8).
However, this also means that the *Methanobrevibacter* themselves can partially reduce the solution, since in this case the cultures did not turn red as quickly or as distinctly as in the negative control, and decolorization also occurred repeatedly. However, the *Methanobrevibacter* do not appear to be able to fully compensate for the absence of DTT, resulting in inhibited growth.

In the future, an alternative solvent should be sought.

5.1.4 General characterization of Methanobrevibacter

Besides the experiments leading to the first attempt to transform *M. thaueri* and *M. boviskoreani*, general characterization experiments were performed to improve our understanding of the *Methanobrevibacter* species in the rumen of cattle.

Cellcount per ml

An important value to know for different experiments is the cellcount per ml for a specific OD600 (e.g. for plating to calculate the plating efficiency or to calculate the ratio of *E. coli* and *Methanobrevibacter* (*M.*) for the conjugation). For this bachelor thesis first reference values were counted with a cell-chamber and a phase-contrast microscope for single OD600 of *M. thaueri*, *M. boviskoreani*, *M. millerae* and *M. wolinii* (see table 4.1).

Compared to the cellcount per ml for *E. coli* ($8 \cdot 10^8$ cells/ml) at a OD600 given by Merck KGaA (2026) the counted cells/ml for the *M.* cultures are quite high (e.g. see table 4.1 *M. wolinii*: $10.819 \cdot 10^8$ cells/ml at a OD600 of 0.387).
Since only one or two cultures have been counted for these organisms so far, these values apply only to the measured OD600 values. Whether a linear relationship exists—as is generally assumed for OD600 values $\trianglelefteq$ 0.3 (Steinbüchel et al., 2021, pp. 9. 37), remains entirely unclear at this point. Nor can it be conclusively stated here whether there are differences between the *M.* species; one indication of this could be that *M. boviskoreani* has more cells/mL with an OD600 that is 0.1 smaller.

Conducting another cell count at various OD600 values would be helpful in clarifying this and creating a calibration curve.

Two factors contributed to the inaccuracy of the count.
First, the chamber depth of 0.1 mm (GmbH, 2026) was so great that the cells could be located in multiple planes, requiring the count to switch back and forth between them. Here, cells can easily be overlooked because the correct plane was not selected, or they can be counted twice.
Second, at 400x magnification, the *M.* cells are still quite small on a phase-contrast microscope, and they also tend to form chains or clumps, which carry the risk of no longer being able to distinguish cells from one another (compare the example image shown in fig. A2).
To mathematically mitigate the increased risk of miscounting, an attempt was made to count approximately 150 to 250 cells per large square instead of fewer than 150. This, of course, leads to a significantly increased workload.

To further address this issue in the future, one could utilize the intrinsic fluorescence of methanogenic archaea due to the cofactor F_{420} (Lambrecht et al., 2017) and perform the counting using fluorescence microscopy. This would make it easier to identify the small cells at 400x magnification. A 1,000x magnification is difficult to achieve during specimen preparation due to the height of the counting chamber and the required distance from it.

Cultivation

In this bachelor thesis the *M.* species could be cultivated successfully in DSM 1523 and 119, which do not contain any hard to get rumenfluid, as is usually the case (Lee et al., 2013; Li et al., 2023; Miller & Lin, 2002). Because there was no visible difference between the cultivated cultures in these two media, most of the experiments were performed in DSM 1523, which is easier to handle due it does not contain the toxic fatty acid mix and the sludge fluid.

The positive controls of the sensitivity tests towards puromycin (fig. 4.3 and 4.4 in section 4.1.4) and 8-Azahypoxanthin (fig. 4.5, 4.7 and 4.6 in section 4.1.5) are indirect growth-curve for *M. thaueri*, *M. boviskoreani* and *M. millerae* in DSM 1523. In general, it can be said, based on these growth-curves, that the used *M.* reach their growth maximum within 2 to 4 days. This maximum is usually around a OD600 of 0.3 and does not exceed 0.4. For calculating general growth parameter as geration time or growth rate furteher growth experiements

are neede with a higher time resolution, espcially in the exponential growth phase.
Compared with growth of methanogenic archaea in the literature (Hoedt et al., 2016; Li et al., 2023), are they growing as fast as the reference organism or a bit faster, but in the litrature OD600 values of 1.2 could be achieved for some cultures. This difference in growth density could be due to the missing rumen fluid, but needs further investigation.

If we look at the cultivation of *M. boviskoreani* and *M. thaueri* a bit more closely, it can be said for *M. boviskoreani* that it grows quit reliably in the described manner and can also be easily spread plated (see fig. 4.1). The only unusual aspect is the formation of precipitates (see fig. A1), which makes photometric measurements somewhat more difficult. In this case, it is important to homogenize the culture as thoroughly as possible beforehand, even if the precipitates cannot be completely dissolved.
The *M. thaueri* on the other hand grows much more unreliably, as you can see in the comparism of the positiv control for Puro and 8-Aza (fig. 4.4 and fig. 4.6), where the triplicates for 8-Aza started to grow after 5 to 14 days while for Puro within 3 days the max. OD600 was reached. In addition, this *M.* species does not grow as dense as the other in this bacherlor thesis and lyses pretty quick afterwards (e.g. within 4 days all triplicates of the positiv control in fig. 4.4 are below 0.1). This growth problem is also showing in plating. Here the spread plating on DSM 1523 (1.5% agar) and pour plating with DSM 119 (1% agar) were not successful (fig. 4.1 and A3). The roll plating was successfull but *M. thaueri* bottle only 15 colonies while the others had above 2000 (see table 4.2).
This growing behavior of *M. thaueri* makes it more difficult to analyses it, and also to plan its grows for the transformation, where it is necessary that the culture grows within two or three days.

For the growing experiments would it be helpful to check if the *M. boviskoreani* and *M. thaueri* can use a liquid alternative substrate to H_2, so a plate reader can be used for the experiments. In contrast to the literature (Li et al. (2023) said *M. boviskoreani* JH1^T could use EtOH) *M. thaueri* and *M. boviskoreani* could not use EtOH as an alternative.
Another substrate that some *Methanobrevibacter* can use is formate, so that should be tested next (Hoedt et al., 2016; Li et al., 2023).

Roll plating

The advantage of roll plating over spread or pour plating is that it can be performed in a

standard laboratory and does not require an anaerobic atmosphere (see **??**). Only the picking of the colonies must be done inside a glove-box. In general, it can be said that the technique works well, and for *M. boviskoreani* and *M. wolinii*, less starting material will need to be used next time, since there are too many colonies, which are consequently too small for selective picking (see fig. 4.2 and table 4.2). This also leads to a disclaimer, it is not proven through sequencing that the right *Methanobrevibacter* grew, but because of the constant consumption of the H_2/CO_2 atmosphere there was definitely anaerobic growth.
Still this plating technic has the potential to be a time saving alternative to the spread plating. But it needs improvement for a fitting quantity, so less but bigger colonies grow, that can easily be picked.

An important aspect of the roll-out technique is that the agar must still be completely liquid when the culture is added and rolled. In addition, the bottle must be spun into the same horizontal position as quickly as possible while rotating in the ice bath. Otherwise, the agar becomes chunky, as is easily recognizable in the case of *M. wolinii* (see fig. 4.2), and this hinders the identification of growing colonies a lot. It is also important to regularly (every 2 to 4 days) replenish the gas mixture of H_2 and CO_2 to enable growth.

The plating efficiency shown in table 4.2 are very low, with values in the lower two-digit ppm range, and even below 1 ppm for *M. thaueri*. But these values should be interpreted with caution, since the OD600 of the initial solution was not measured. Instead, it was assumed retrospectively that the cultures have a similar OD to those obtained by cell counting.
For future determinations of plating efficiency, it would be important to measure the OD600 in advance and to plate only at an OD600 corresponding to those in table 4.1.

5.2 Geography

The geographical analyses is focused on the three organisation for economic co-operation and development (OECD) countries: Germany (DEU), New Zealand (NZL) and United States of America (USA). For further information see section 2.2.1

5.2.1 Evaluation of immunization of cattle" as a mitigation strategy

As outlined in section 2.2.3 two reference data are used. On the global scale the growth rate of methane (20.9 Tg/year) for the decade 2010 to 2019 (Saunois et al., 2025). On the national scale, the emission targets of the global methane pledge (GMP) to reduce the national emission about 30% compared to 2020 methane emissions and 40% to 2010 (Appelhans et al., 2025). DEU, NZL and USA signed the GMP.

National scale

For the evaluation, if mitigation of methane prduction in the enteric fermentation processes of cattle can be a important missing piece to achieve the signed emission targets, the CH_4 had to be predicted to the year 2030. As described in section 4.2.4 the total emission were predicted without any extra mitigation and for diffrent success scenarios of shifting the microbiome of cattle towards less methanogenic archaea. For this purpose, the prediction was performed via three linear models that had different time sample as reference - last 5-, 10- and 15-years. Figure 4.16 (DEU), 4.17 (NZL) and 4.18 (USA) show, the results of the 10- and 15-year model are quite identical while the 5-year model usually differs a bit more. This probably due to the smaller data sample for the 5-year model so it is probably more susceptible for change in the last five years. By examining the trends in national methane emissions for individual sectors (see fig. 4.12), the linear models for each country can be evaluated.
For DEU (see fig. 4.16), it can even be concluded that the 5-year model, with a Prediction of a lower methane emission reduction than the other two models (23.5 Tg to 26.5 Tg CH_4), is likely the most plausible. This can be explained by the fact that in DEU, the energy and waste sectors have remained unchanged for the past six years of the data set (2018-2024), and since emissions are already at such a low level, there is little room for further reduction (see fig. 4.12). From 1990 to 2018, CH_4 emissions from this two sectors decreased continuous. This is still factored into the 10-year and 15-year models, leading to a prediction with a stronger decrease.
For NZL (fig. 4.17) and USA (fig. 4.18), such a clear attribution cannot be made. However, the 5-year model there can be described as optimistic, as it assumes that the sharper reduction in methane levels seen over the past few years will continue at this rate. The 10- and 15-year models, on the other hand, are somewhat more pessimistic. They reflect the fact that there have been periodic periods of several years with stronger reductions in the past, but these were followed by years of stagnant methane emissions or even slight increases.

For the three tested reductionscenarios in fig. 4.16, 4.17 and 4.18 (10, 30 and 50% microbiom shift in the rumen towards less methanogenic archaea) it can be said, the 10% scenario has a to little impact for all three countries that it could help to reach the goals, if they not already reached it. This last case is only true for the 10- and 15-year model of DEU.

The 30% reductionscenario ensures that NZL and DEU would reach with all models at least one reduction scenario. For NZL the 70% and for DEU the 60%, this is due to constant total emission values for NZL and a stronger decrease of methane emissions between 2010 and 2020 for DEU. The 50% reductionscenario would even be enough for these two countries, that all models and their CI would be below both emission goals (see fig. 4.16 and 4.17).
The situation is somewhat different for the USA. In none of the scenarios do the pessimistic models (10- and 15-year) meet either of the two emissions targets of the GMP (see fig. 4.18). The optimistic model (5-year) would come at least close to the 70% target with a 30% reduction and fall short of it in the 50% reduction scenario, but would still meet the 60% target relative to 2010.

This difference between the countries can be attributed to the different composition of sectors of the national emission of these states (see fig. 4.12). Though is for DEU and NZL the agriculture in the year 2024 the main CH_4 source while in the USA energy and agriculture contribute the same and waste less but has still a substantial impact.
So two things get clear. First, if the agriculture sector and therefor the EnterFe are not the dominant methane emission source for the country, a change there alone will not be enough to achieve the GMP aims. As example for the USA only if the reduction over tha past 5 years for the energy and waste sector is continued similarly, then emission reduction from cattle can be te missing peace. However, if efforts to reduce methane emissions from the energy and waste sectors were intensified, the need for mitigation measures for cattle might be limited at first.
Second, if agriculture is the dominant sector, it seems there is a need for new mitigation options to reduce methane production in a relevant way, because the emissions of agriculture did not change much in all three investigated countries (see fig. 4.12). The CH_4 emission derived from enteric fermentation in cattle seems to be a valid target for that, if reduction is about 30% or more.

Global scale

On the global scale several things gets quit obvious, if you compare the averaged yearly saved methane emission for the different reduction scenarios for the decade 2010-2019 (see fig. 4.15) with the yearly growth rate (20.9 Tg CH_4) calculated in Saunois et al. (2025) for the same decade.

First of all, no reduction scenario (10, 20, 30, 40, or 50% shift in the rumen microbiome) would have been sufficient for the cumulative reduction in CH_4 emissions to offset the annual growth rate. Thus, neither changes in a few countries nor emission reductions in individual sectors are sufficient to stop methane accumulation in the atmosphere or even to enable a decrease in global methane levels. Nevertheless, a reduction scenario of 30 in the three countries (approx. 2.75 Tg CH_4) alone could have offset over 10% of the global growth rate. Under reduction scenario 50, this figure would even be just over 20% at 4.5 Tg CH_4.
Considering that climate change and rising temperatures are causing natural CH_4 sources to become increasingly active, a prominent example being the thawing of permafrost (Saunois et al., 2025; Schädel et al., 2026; Turetsky et al., 2020). Therefore, efforts to reduce methane emissions from cattle should aim for a reduction of at least around 30% in order to make a significant contribution to mitigate climate change.

The second part of the fig. 4.15 underlines again with the calculation of CO_2eq., that methane reduction really can be a short-term solution to soften climate change impacts and allow so to adapt/ mitigate further. So would the saved methane emission for the reduction scenario 30 (2.75 Tg) on a time horizon of 20 years correspond to 220 Tg CO_2, for a 100 year time horizon 75 Tg CO_2. The CO_2eq. for the global warming potential (GWP)20 would be roughly the same as the total CO_2 emission (no other GHG were counted) of Spain (220.08 Tg CO_2) for the year 2024 (European Commission. Joint Research Centre., 2025).

Finally, it is noticeable in fig. 4.15 that the amount of methane saved increases linearly with the scenarios. In the scenarios, two values are varied: first, methane conversionfactor (Ym), which has a linear relationship with methane emissions (see eq. (1)), and second, digestibility of food (DE), which has a nonlinear mathematical relationship with methane emissions (see section 3.2.4) . Although the theoretically assumed increase in dietary efficiency leads to further reductions in CH_4 emissions, this component is almost negligible because of the non-visual impact, so it has virtually no relevance to the significance of the method as a mitigation strategy.

Comparison with other mitigation strategies

Another mitigation strategy is the use of food additives, which also affect methanogenic archaea; for example, halogenated methane compounds such as bromoform inhibit the key enzyme (methyl-coenzyme M reductase) involved in methanogenesis (Boadi et al., 2004;

Garnett & Eckard, 2024; Malyugina et al., 2025). In this context, the addition of red algae such as *Asparagopsis taxiformis* in in vivo experiments with beef cattle resulted in up to an 80% reduction in methane production (Malyugina et al., 2025, Table 4). However, even aside from this best-case scenario, reductions of over 30% were regularly achieved in vivo. To be competitive, the immunization strategy should also aim to achieve at least a 30% reduction.
An advantage of immunization over feed additives is that it does not require daily administration to the animals and can be easily used in grazing animals, whereas feed additives are more difficult to implement in this context because the animals are not fed continuously or there is no controlled feed composition (Baca-González et al., 2020; Malyugina et al., 2025).

5.2.2 Characterization of national emission

The composition of the methane emission from the diffrent sectors for the countries DEU and USA fits the described global pattern in the section 1 and literature (Appelhans et al., 2025; Mundra & Lockley, 2024) quit well (fig. 4.12). So in the USA the three main methane emitting sectors are energy, waste and agriculture, whereas agriculture emission increased slightly, energy and waste decreased. In Germany, these three sectors were also dominant in 1990, and while emissions from agriculture have remained virtually unchanged, emissions from waste and energy have decreased so significantly that they now emit virtually no CH_4.
DEU clearly demonstrates here—as is already evident in the USA—that methane emissions from the energy and waste sectors can already be reduced very effectively using current methods. Although it must be noted here as a disclaimer that, in part, the reduction in the energy sector was achieved by outsourcing energy production or coal extraction to other countries—which reduces emissions nationally but not globally (Appelhans et al., 2025). At the same time, as shown in fig. 4.12, the challenge of reducing emissions from the agriculture sector remains particularly striking (Edelenbosch et al., 2024; Intergovernmental Panel On Climate Change (IPCC), 2022).
New Zealand also shows only minor changes in the agricultural sector; however, over the entire period under consideration from 1990 to 2024, there is no other relevant CH_4 emitter besides agriculture. This clearly demonstrates that there can be significant national differences, which must be taken into account when assessing the significance and application of methods. Thus, for the United States, it is likely initially easier and more relevant to reduce emissions from energy and waste.

These national differences are also evident in the share of EnterFe CH_4 emissions from cattle relative to total national emissions in fig. 4.14. Thus, CH_4 from enteric fermentation processes in cattle is a major and relevant single source in all three countries; however, in the U.S. in 2022, it accounted for only 22% of total emissions, whereas in DEU it accounted for just under 50% and in NZL as much as 60%. It can, however, be noted that EnterFe from cattle is a good single methane emission target for the three countries examined.

When examining cattle emissions from enteric fermentation (see fig. 4.13) for the three countries, it is striking that the globally described increase in cattle emissions due to rising demand for animal products (Appelhans et al., 2025; FAO, 2009) is only slightly observable, or that there is even a decrease in emissions (see Germany). This is due to the fact that only OECD countries were considered. The described increase in demand and emissions is primarily attributable to non-ANNEX-I countries.

Furthermore, a comparison with the trends in cattle population numbers in fig. A4, A5, and A6 shows that the amount of methane emissions from cattle correlates very strongly with changes in population size. In addition, DEU shows that while increased production through higher-performing animals has led to an increase in GE, the simultaneous reduction in the number of animals has nevertheless resulted in a reduction in methane emissions (Appelhans et al., 2025; Niebuhr et al., 2026) . However, this method of reducing population size and methane emissions cannot be continued indefinitely, and there is also the question of to what extent animal welfare is still ensured in these high-performance animals (Appelhans et al., 2025). It can therefore be concluded that population size is the decisive factor in methane production in cattle farming.

5.2.3 Future steps

For the geography site next steps to improve the calculation and rating would be to check if shifting the microbiome of the rumen can have an influence on other emission sources. One realistic option here is the manure management. From the calculation described in IPCC (2006b) one value already seems promising, the B_0. B_0 is needed for the calculation of the EF for manure management. The value represents thereby maximum methane production from the manure produced by an animal category, this could be influenced by a different ratio of methanogen to non-methanogen anaerobic microbes. But this values has to be measured, so this hypothesis, therefore, can only be tested once the first successful applications of immunization have been made.

In addition, right now, just the gross effect was investigated off methane and its reduction in cattle. So could a counter effect be that the cow and shifted rumen microbiome produce less CH_4 but more CO_2, so the shown saved carbon dioxide in fig. 4.15 could be less. If the immunization strategy is in the in-vivo state, this definitely needs to be checked.

Furthermore, it would be interesting to extend the calculations to other countries in order to better assess the significance of such a strategy on a global scale. Of particular interest here would be countries that are not ANNEX-I states, in contrast to the three examined so far, although differences in emission patterns were already evident in those countries as well (see section 5.2.2)

6 CONCLUSION

The interdisciplinary approach of this bachelor thesis to unit the small scaled investigation of the biochemistry with the greater view of the geography allowed a more holistic view towards the problem of hard-to-abate methane emissions from the agriculture sector.

The analyses of the national methane emissions for the three countries of interest (Germany (DEU), New Zealand (NZL) and the United States of America (USA)) showed that the agriculture sector, and therefor the emissions resulting from enteric fermentation processes mainly in cattle, are a big potential source for reducing methane emissions. For Germany and New Zealand is agriculture the only sector left for potential relevant methane reduction (see fig. 4.12).

The general geodata analyses of the national methane emissions from DEU, NZL and USA proofed the assumption derived from global methane emission pattern there is a need for fitting mitigation strategy for CH_4 emission from enteric fermentation processes in cattle (Appelhans et al., 2025; Intergovernmental Panel On Climate Change (Ipcc), 2023; Saunois et al., 2025). In addition, this single emission source contributes that much to the national emission (from 25% in the USA 2022 to 45% in DEU and 60% in NZL, see fig. 4.14) it will have a relevant impact on mitigate climate change. Especially the calculation of the GWP20 and GWP100 in fig. 4.15 showed clearly, methane reduction could be really helpful short-term to stay below the 2 °C aim of the UNFCCC (2015). For this purpose as already pointed out in section 5.2.1 a shift of the rumen microbiome towards non-methanogenic microbes needs to be at least at 30%.
If the immunization of cattle can be helpful for this short-term mitigation, depends on the stage of biochemist development and investigation of a potent and specific vaccine against methanogenic archaea.

The biochemistry part of this bachelor thesis could collect several basic information, based on the investigition of mainly two strains *Methanobrevibacter thaueri* CW^T (*M. thaueri*) and *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*), on the poorly understood genus *Methanobrevibacter* (*M.*), which is the dominant methanogenic archaea in the rumen of cattle (Baca-González et al., 2020, p. 3).
A huge step for further investigation would it be, if the promising results of the attempt for the

transformation via interdomain conjugation are proofed true through the still pending whole genome sequencing (see section 4.1.7 and 5.1.1). This would allow for the first time to analyses the *M.* further based on genetic engineering. Besides this, it could be shown that *M. thaueri* and *M. boviskoreani* can be cultured and plated on media (DSM 1523 and 119) without rumenfluid. In addition, first cellcounts per ml were determined (fig. A2) allowing first calculation of plating efficiency.
One problem that has been identified and must be resolved for future experiments is that the solvent DMSO has a clear inhibitory effect on the growth of *M.* (see section 4.1.5) due to its oxidative effect (Atcher & Alfonso, 2013). For chemicals that are poorly soluble in water, such as 8-Aza, which is needed for counterselection (see section 1.1), an alternative solvent must be used.

As gets obvious here, the investigation on a possible vaccine for cattle against its methanogenic mircobiome is still in the beginning and not in the stage of in-vivo test, so some time ist still needed until a possible vaccine could be available on the market. This means the mitigation strategy will not help to mitigate climate change in the next decade.
The best approach to reduce methane notably over the next years is to use the reduction potential, as for the USA is quit obvious with approx. (fig. 4.12), in the energy and waste sectors, which are, as shown for DEU in fig. 4.12, easier to modify with today's technics (Appelhans et al., 2025; IPCC, 2022).

Nevertheless, this immunization strategy can be very useful in the long term for achieving the goal of net-zero (UNFCCC, 2015) within the agricultural sector, at least to some extent. This will even be more relevant, if taken into account, that the methane emission from natural sources such as permafrost will increase in the future due to the temperature rise (Schädel et al., 2026). Nevertheless, it must be noted here that even with a mitigation strategy that reduces emissions per head of cattle, it will still be necessary in the future to reduce the population to a sustainable size (Edelenbosch et al., 2024), which will be challenging with a growing demand for animal products (Appelhans et al., 2025; FAO, 2009). But the immunization or other strategies to reduce CH_4 production in cattle can allow a population size that can be socially accepted and politically feasibly.
Furthermore, the basic research conducted here on methane production by methanogenic archaea may be important not only for the development of a vaccine, but also for other mitigation strategies, such as food additives.

BIOCHEMICAL TERMS

β-sheet	A common secondary structure of proteins. It consists of β-strands arranged in parallel or antiparallel orientations. β-strands are illustrated as arrows. Another typical secondary structure is the α-helix.
16S-Sequencing	16S sequencing targets the 16s genes, which code for the 16S ribosomal RNA (rRNA). The 16S rRNA is a part of the procaryotic ribosome. These 16-s base sequenz is specific for the organism.
AlphaFold	Is a KI based software for prediction of 3-D structure of proteins. For the prediction it needs an aminoacid sequence.
archaea	Archaea constitute one of the three major domains of life (bacteria, archaea, eukaryotes), and, like bacteria, they belong to the prokaryotes (Fritsche, 2024, p. 3).
auxotrophic	Auxotrophic organism can not produce relevant bio-molecules for growth and surviving. Therefor, they must obtain these from their environment (Fritsche, 2024, p. 65).
C-terminal end	The end of the amino acid sequence of a protein, where the carboxy group of the terminal amino acid has no covalent connection to another amino acid.
colony	In a microbiological context, a colony is defined as a clearly distinct cluster of cells that can be traced back to a single parent cell through cell division.

complex media	Complex media contain natrual, organic moleculas, that are not precisely defined chemically(e.g. yeast extract or pepton) So a lot of diffrent microbes can grow on them (Fritsche, 2024, p. 359).
conjugation	Conjugation is the process of horizontal gen-transfer between a donor-cell and an acceptor-cell. The connection or "mating" between the cells is made by a pilus (also called "sex-pilus"), which allows the transport of the DNA. The conjugation can also happen between different species or domains. Needed for the conjugation is a plasmid that contains the gene for conjugation (the *tra*-region) and connection with a pilus (Fritsche, 2024, p. 207).
glove-box	A sealed, airtight chamber system with built-in gloves that allow you to work from the outside while the internal atmosphere remains controlled (e.g. a anaerobic atmosphere).
in-vitro	Experiments are not performed in/ at the living organism, but in an artificial setting.
ORI	Origin of replication, is the region of the DNA, where the machinery responsible for replicating the entire genomic DNA first binds and begins replication.

PCR	The polymerase chain reaction is a in-vitro method to amplify a specific sequence of DNA. For this purpose, specific fitting primer has to be designed, that flank this DNA-sequence on both complementary DNA-strings..
phase-contrast microscope	Phase-contrast microscopy is based on the fact, that objects or medium not only changes the amplitude of light but also the phase of it. The phase-contrast microscope makes this phenomena visual for the human eye (Pawlizak, 2009).
plasmid	A small ring of double-stranded DNA. Occurs naturally in archaea and bacteria, but is also used for scientific purposes. Here often as a vector for a transformation.
Plating	Cultivate microorganism on a solid medium (contains a gelling agent like agar). THerby is the plating the process how the organism get distributed on the plate / solid media. Different technics are possible like spreading or pouring (Fritsche, 2024, p. 364).
pour plating	A plating technique in which the solution containing microorganisms is mixed with the agar-containing culture medium while it is still liquid (at about 45 ℃). This mixture is then poured into a Petri dish, where it solidifies(Fritsche, 2024, p. 364).
primer	Short nucleotide sequence that serves as a starting point for the polymerase to replicate the complementary DNA strand. For this purpose, the primer hybridizes to a specific sequence on the singlestranded-DNA (ss-DNA) template (Fritsche, 2024).

promotor	Region on DNA, which allows binding of relevant enzyms for the transcription of genes. So it is a essential region for gene expression.
restricitionsite	Restrictionsites are a specific sequence of approx. 4 to 10 bp that are recognized by a specific restriction-enzyme. This cuts the double-stranded DNA at this site in a specific behavior.
ribosome	Ribosomes are a mix out ribosomal RNA (rRNA) and proteins. They play a central role in protein biosynthesis by catalyzing the translation of mRNA into an amino acid sequence.
spread plating	A plating technique in which the solution containing microorganisms is put on the solid agar-containing culture medium. The solution is then spread across the agar plate using a Dragalski spatula (Fritsche, 2024, p. 364).
suicide plasmid	A plasmid that has no ORI and therefore can not be replicated in the cell except it gets integrated in the gDNA.
transformation	The process of getting a extern DNA into the target organism and its expression. If the target organism is a animal eucaryotic cell it is called transfection.

GEOGRAPHICAL TERMS

ANNEX-I	ANNEX-I and Non-ANNEX-I countries is a classification of the UNFCCC. ANNEX-I are the OECD countries or with an economy in transition.
compound events	When two wether extremes occur at the same time a shortly after each other, so so that they reinforce each other, as example a heatwave and droughts (IPCC, 2023).
CRT	Common reporting table is a template for countries to report its emission of GHG fitting IPCC standards. The UNCFF has oblige the ANNEX-I countries to publish this every year since 1990.
gross	In the context of climate change, the gross effect refers to the direct, physical influence of a greenhouse gas (such as methane) on the Earth's energy balance, considered in isolation. The net effect, by contrast, accounts for masking effects. For example,e caused by co-emitted cooling agents, such as aerosols released during methane extraction, which can partially offset the gross warming contribution.

GWP	Global warming potential is a concept used to represent the relative impact of various GHGs on a uniform scale over a specific time period. It takes into account their effectiveness in causing radiative forcing as well as their atmospheric lifetime. CO_2 serves as a reference, so that the effect of 1 g of greenhouse gas is converted to x g of CO_2 eq.. The relevant time period is indicated as the number of years following the GWP designation (e. g., GWP20 refers to a 20-year time horizon).
planetary boundaries	Is a concept that uses the variability of values of the Holocene as a reference to evaluate thresholds for different processes to maintain a stable system on earth, where human mankind can definitely thrive. The nine main processes are "Climate change", "Biosphere integrity", "Land-system change", "Freshwater use", "Biogeochemical flows", "Ocean acidification", "Stratospheric ozone depletion", "Atmospheric aerosol loading" and "Novel entities" (Planetary Boundaries Science (PBScience), 2025).
radiative forcing	Is a concept that quantifies the change of the energy balance of the hole planet. The unit is W/m^2. It depends on many factors as e.g. aeorosol conten of the atmosphere, planetary albedo and the concentration of GHG (Irfan et al., 2024).

zone of an increasing risk	"Refers to the stage when human activities push Earth beyond the Planetary Boundaries, entering a “Zone of Increasing Risk.” In this zone, the further boundaries are exceeded, the greater the chance of causing serious damage, destabilizing key Earth system processes, and disrupting life-support functions" (Planetary Boundaries Science (PBScience), 2025).

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APPENDIX

A Figures

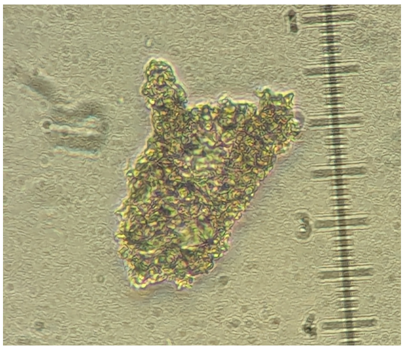

Figure A1: Phase-contrast microscope picture of *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*) culture precipitate at 1000 times magnification. *M. boviskoreani* was cultivated in DSM 1523.

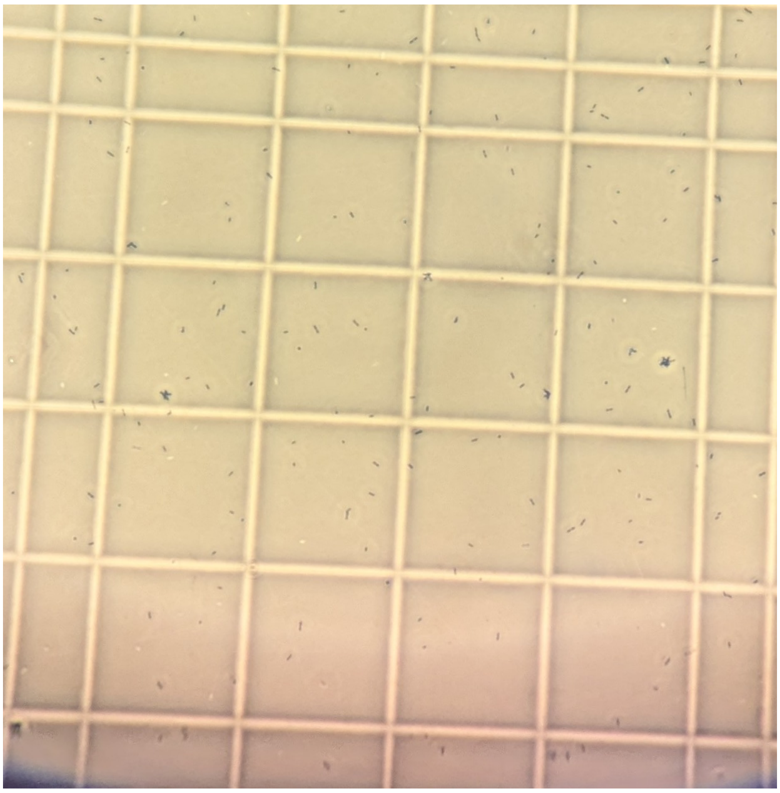

Figure A2: Phase-contrast microscope picture of *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*) at 400 times magnification. One group square with its sixteen small squares of a Neubauer improve counting-chamber is shown.

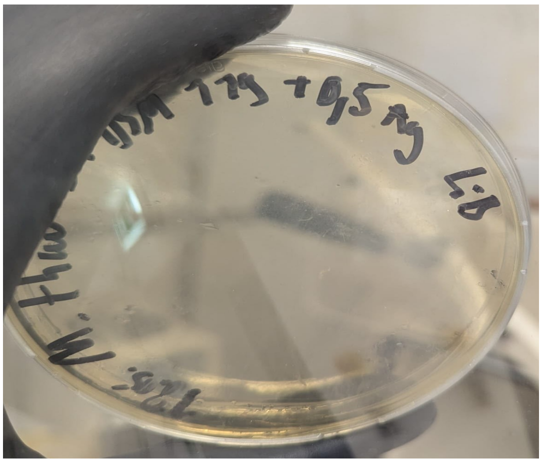

Figure A3: Pour plating result of *Methanobrevibacter thaueri* CW^T (*M. thaueri*) DSM 119 (1% agar). 100 µl of a full-grown culture have been plated. Cultivated in a jar with H_2/CO_2 gas phase. No growth detected.

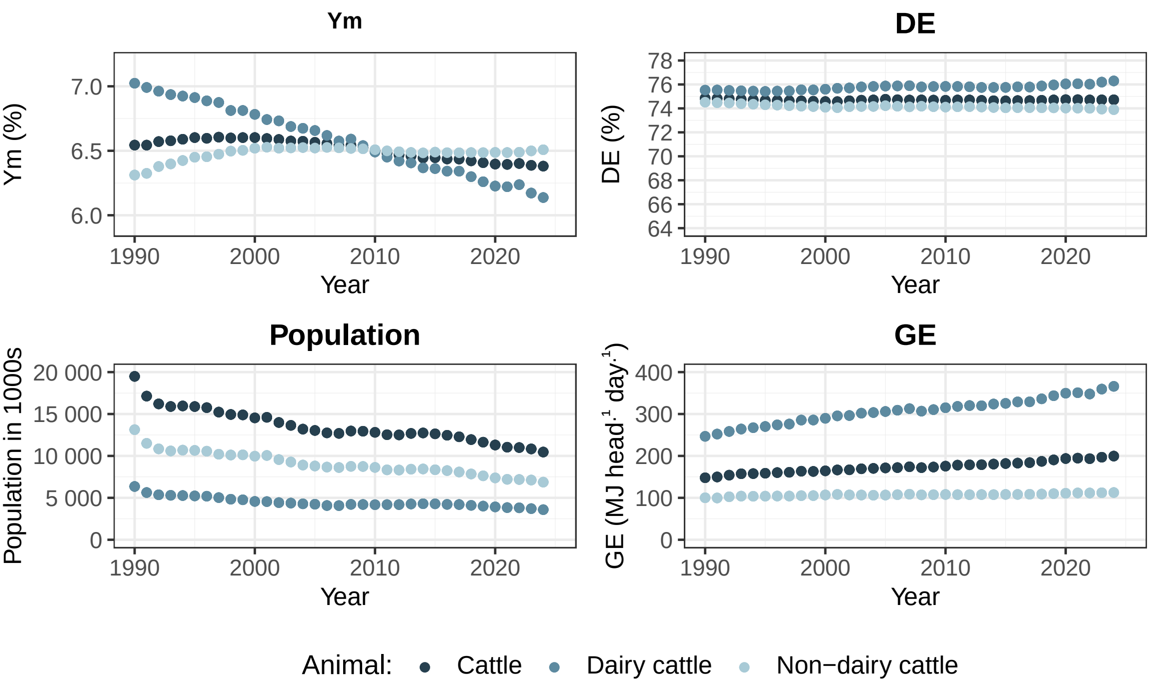

Figure A4: Time series of methane conversionfactor (Ym), digestibility of food (DE), population-size and gross energy intake (GE) of cattle and its subcategories dairy and non-dairy for Germany. The data for dairy and non-dairy cattle is from the published CRT. The values for cattle is calculated as weighted mean of its subcategories, where the population-size is used as weight.

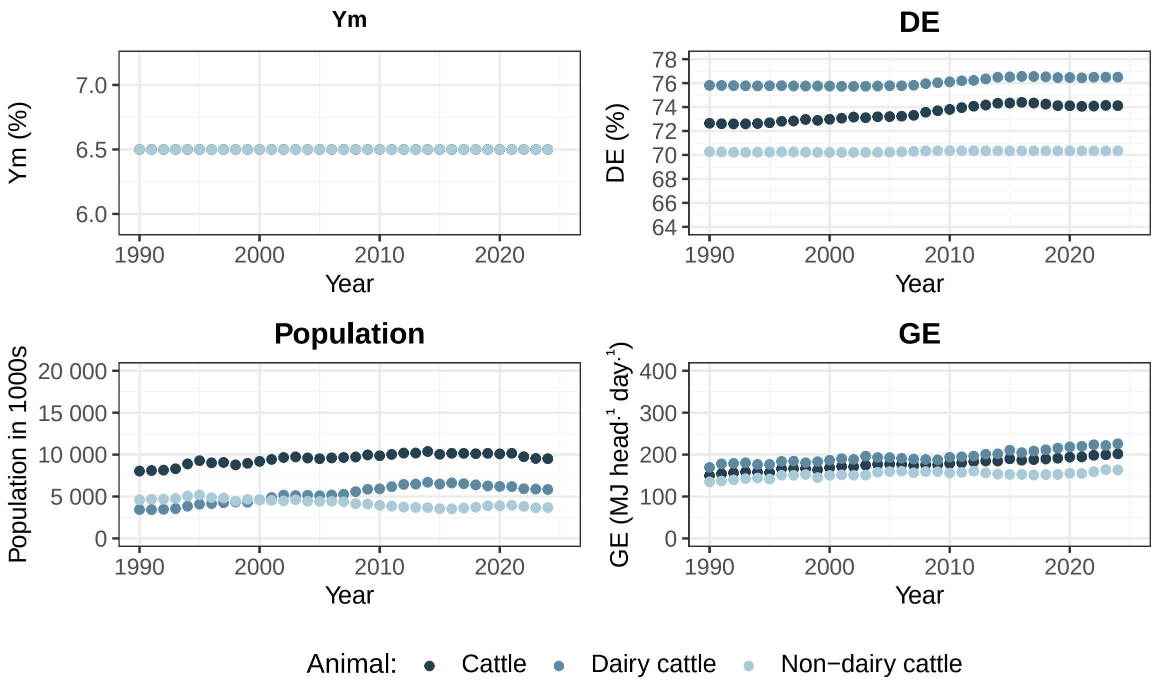

Figure A5: Time series of methane conversionfactor (Ym), digestibility of food (DE), population-size and gross energy intake (GE) of cattle and its subcategories dairy and non-dairy for New Zealand. The data for dairy and non-dairy cattle is from the published CRT. The values for cattle is calculated as weighted mean of its subcategories, where the population-size is used as weight.

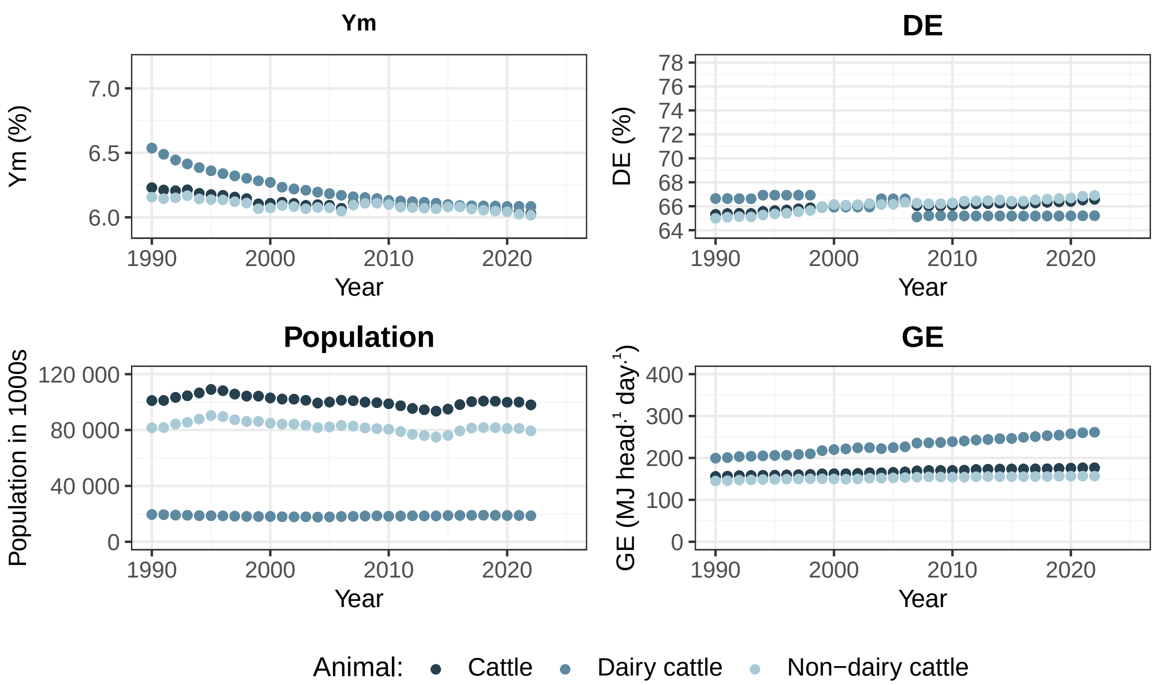

Figure A6: Time series of methane conversionfactor (Ym), digestibility of food (DE), population-size and gross energy intake (GE) of cattle and its subcategories dairy and non-dairy for United States of America. The data for dairy and non-dairy cattle is from the published CRT. The values for cattle is calculated as weighted mean of its subcategories, where the population-size is used as weight.

B Tables

Table B1: Counted cells for *Methanobrevibacter wolinii* SH^T (*M. wolinii*) with an OD (600 nm) of 0.387. Cells were counted with a phase-contrast microscope and a Buerk counting chamber at 400x magnification. Per chamber filling four square (Sqr.) with a volume of $4 \cdot 10^{-6}$ ml were counted. The arithmetic mean (ø cellnumber) per chamber filling was calculated. The culture was diluted 1:20 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	ø cellnumber
1	138	159	226	233	189
2	260	271	293	265	272.25
3	180	197	216	188	195.25
4	207	191	211	227	209

Table B2: Counted cells for *Methanobrevibacter wolinii* SH^T (*M. wolinii*) with an OD (600 nm) of 0.396. Cells were counted with a phase-contrast microscope and a Neubauer improved counting chamber at 400x magnification. Per chamber filling four large-square (Sqr.) with a volume of $4 \cdot 10^{-6}$ ml were counted. The arithmetic mean (ø cellnumber) per chamber filling was calculated. The culture was diluted 1:20 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	ø cellnumber
1	177	239	263	254	233.25
2	246	241	267	208	240.5
3	175	176	161	221	183.25
4	217	185	205	180	196.75

Table B3: Counted cells for *Methanobrevibacter thaueri* CW^T (*M. thaueri*) with an OD (600 nm) of 0.235. Cells were counted with a phase-contrast microscope and a Neubauer improved counting chamber at 400x magnification. Per chamber filling four large-square (Sqr.) with a volume of $4 \cdot 10^{-6}$ ml were counted. The arithmetic mean (ø cellnumber) per chamber filling was calculated. The culture was diluted 1:10 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	ø cellnumber
1	189	232	250	265	234
2	237	209	224	263	233.25
3	186	253	230	237	226.5
4	146	187	216	262	202.75

Table B4: Counted cells for *Methanobrevibacter thaueri* CW^T (*M. thaueri*) with an OD (600 nm) of 0.116. Cells were counted with a phase-contrast microscope and a Neubauer improved counting chamber at 400x magnification. Per chamber filling four large-square (Sqr.) with a volume of $4 \cdot 10^{-6}$ ml were counted. The arithmetic mean (ø cellnumber) per chamber filling was calculated. The culture was diluted 1:10 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	ø cellnumber
1	255	181	189	171	199
2	127	92	131	193	135.75
3	110	136	153	109	127
4	127	93	109	156	121.25

Table B5: Counted cells for *Methanobrevibacter millerae* ZA-10^T (*M. millerae*) with an OD (600 nm) of 0.378. Cells were counted with a phase-contrast microscope and a Neubauer improved counting chamber at 400x magnification. Per chamber filling four large-square (Sqr.) with a volume of $4 \cdot 10^{-6}$ ml were counted. The arithmetic mean (ø cellnumber) per chamber filling was calculated. The culture was diluted 1:20 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	Quadrat_5	ø cellnumber
1	186	175	173	156	220	182
2	118	204				161
3	211	178	232	211	184	203.2
4	128	128	201	204	160	164.2

Table B6: Counted cells for *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*) with an OD(600 nm) of 0.287. Cells were counted with a phase-contrast microscope and a Neubauer improved counting chamber at 400x magnification. Per chamber filling four large-square (Sqr.) with a volume of $4 \cdot 10^{-6}$ ml were counted. The arithmetic mean (ø cellnumber) per chamber filling was calculated. The culture was diluted 1:20 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	ø cellnumber
1	123	162	166	170	155.25
2	162	228	205	242	209.25
3	170	128	137	213	162
4	187	174	263	237	215.25

Table B7: The total average of counted cells (ø cellnumber) for each *Methanobrevibacter* (*M.*) culture with corresponding OD at 600 nm (OD600). Counting was done with a phase-contrast microscope and a Buerk or Neubauer improved counting chamber at 400x magnification. ø cellnumber was calculated as arithmetic mean of shown cellcuonts in tab. B1 through B6. The dilution factor takes into account the dilution of the culture being counted.

culture	OD600	Dilution factor	ø cellnumber
M. wolinii	0.387	20	216.375
M. wolinii	0.396	20	213.438
M. thaueri	0.235	10	224.125
M. thaueri	0.116	10	145.750
M. millerae	0.378	20	177.600
M. boviskoreani	0.287	20	185.438

Table B8: Suicide plasmid constructs for transformation of *Methanobrevibacter thaueri* CW^T (*M. thaueri*) and *Methanobrevibacter boviskoreani* AbM4 (*M. boviskoreani*) via conjugation.The listed plasmids follow the general structure shown in fig. 1.2, with a cassette for *E. coli* containing the conjugation gene and homologous arms for recombination. The name of each plasmid is derived from the organism for which it possesses the appropriate homology arms (pMt for *M. thaueri* and pMb for *M. boviskoreani*) and the four named columns, where ORI refers to *M.* in this case and the promotor for the *pac*. *pac_org* is the *pac* with the original sequence derived from *Streptomyces alboniger*, *pac_opt* is the codon optimized version for the affected organism. The size of the constructed is given in bp as well.

name	ORI	promoter	pac	extra feature	size (bp)
pMt_0231_hpt	/ (0)	hmtB (2)	pac_opt (3)	hpt (1)	6145
pMt_0331_hpt	/ (0)	PglnA (3)	pac_opt (3)	hpt (1)	6254
pMt_0241_hpt	/ (0)	hmtB (2)	pac_org (4)	hpt (1)	6145
pMt_0341_hpt	/ (0)	PglnA (3)	pac_org (4)	hpt (1)	6254
pMb_0231_hpt	/ (0)	hmtB (2)	pac_opt (3)	hpt (1)	6214
pMb_0331_hpt	/ (0)	PglnA (3)	pac_opt (3)	hpt (1)	6323

Table B9: Lists saved CH_4 for five scenarios of different success in shifting the microbiom from methanogen towards non-methanogen (10, 20, 30, 40 and 50%). It was calculated an average for the decade 2010 to 2019 for the three countries Germany (DEU), New Zealand (NZL), United States of America (USA). The CO_2 eq. was calculated for the GWP20 and GWP100 with factors taken from Intergovernmental Panel On Climate Change (Ipcc) (2023, Table 7.15, Chapter 7) (79.7 for GWP20 and 27 for GWP100)

country	red.-scenario	CH_4 [Tg]	GWP20 [Tg CO_2 eq.]	GWP100 [Tg CO_2 eq.]
DEU	10	0.104	8.321	2.819
DEU	20	0.207	16.460	5.576
DEU	30	0.306	24.424	8.274
DEU	40	0.404	32.220	10.915
DEU	50	0.500	39.853	13.501
NZL	10	0.088	7.032	2.382
NZL	20	0.174	13.906	4.711
NZL	30	0.259	20.628	6.988
NZL	40	0.341	27.204	9.216
NZL	50	0.422	33.639	11.396
USA	10	0.750	59.739	20.238
USA	20	1.480	117.953	39.959
USA	30	2.192	174.707	59.186
USA	40	2.887	230.061	77.938
USA	50	3.564	284.071	96.235

Table B10: Listed linear models for predicting relevant values for calculating 2030 CH_4 emission of **Germany**. The relevant indicators are total methane emission without cattle enteric fermentation (EnterFe) and Ym (methane conversion factor), DE (digestibility), NE_x (net energy intake), population size for dairy cattle (DC) and non-dairy cattle (NDC). For fitting the linear models three different time periods were used (the last 5, 10 or 15 years of the data set). The fitted value is thereby depended on the time. Besides the equation, the R^2 is given as well. R^2 for the 15-year model of "NDC YM [%]" turns zero because no change is predicted.

time sample	fitted value	equation	R^2
5	total CH_4 without cattle EnterFe [kt]	$y = -6.25 \cdot x + 13677.282$	0.070
5	DC population size [1000s]	$y = -78.386 \cdot x + 162269.508$	0.956
5	DC Ym [%]	$y = -0.023 \cdot x + 52.062$	0.712
5	DC DE [%]	$y = 0.063 \cdot x - 51.879$	0.735
5	DC NE_x [MJ head^{-1} day^{-1}]	$y = 1.868 \cdot x - 3630.669$	0.720
5	NDC population size [1000s]	$y = -110.076 \cdot x + 229727.445$	0.893
5	NDC Ym [%]	$y = 0.005 \cdot x - 4.499$	0.821
5	NDC DE [%]	$y = -0.035 \cdot x + 144.304$	0.891
5	NDC NE_x [MJ head^{-1} day^{-1}]	$y = 0.129 \cdot x - 215.365$	0.888
10	total CH_4 without cattle EnterFe [kt]	$y = -36.946 \cdot x + 75747.569$	0.792
10	DC population size [1000s]	$y = -76.559 \cdot x + 158578.922$	0.988
10	DC Ym [%]	$y = -0.024 \cdot x + 55.219$	0.946
10	DC DE [%]	$y = 0.057 \cdot x - 40.026$	0.932
10	DC NE_x [MJ head^{-1} day^{-1}]	$y = 1.941 \cdot x - 3778.256$	0.946
10	NDC population size [1000s]	$y = -168.698 \cdot x + 348278.384$	0.974
10	NDC Ym [%]	$y = 0.002 \cdot x + 2.939$	0.483
10	NDC DE [%]	$y = -0.016 \cdot x + 106.264$	0.749
10	NDC NE_x [MJ head^{-1} day^{-1}]	$y = 0.209 \cdot x - 377.797$	0.957
15	total CH_4 without cattle EnterFe [kt]	$y = -40.092 \cdot x + 82103.312$	0.934
15	DC population size [1000s]	$y = -44.459 \cdot x + 93727.349$	0.767
15	DC Ym [%]	$y = -0.023 \cdot x + 52.942$	0.977
15	DC DE [%]	$y = 0.032 \cdot x + 10.375$	0.716
15	DC NE_x [MJ head^{-1} day^{-1}]	$y = 1.532 \cdot x - 2951.872$	0.940
15	NDC population size [1000s]	$y = -125.537 \cdot x + 261079.881$	0.917
15	NDC Ym [%]	$y = 0 \cdot x + 6.57$	0.000
15	NDC DE [%]	$y = -0.014 \cdot x + 102.431$	0.852
15	NDC NE_x [MJ head^{-1} day^{-1}]	$y = 0.144 \cdot x - 246.555$	0.863

Table B11: Listed linear models for predicting relevant values for calculating 2030 CH_4 emission of **New Zealand**. The relevant indicators are total methane emission without cattle enteric fermentation (EnterFe) and Ym (methane conversion factor), DE (digestibility), NE_x (net energy intake), population size for dairy cattle (DC) and non-dairy cattle (NDC). For fitting the linear models three different time periods were used (the last 5, 10 or 15 years of the data set). The fitted value is thereby depended on the time. Besides the equation, the R^2 is given as well. R^2 is NA for indicators where the variance of the values is zero. In addition there are two zero values because of rounding, first the slope of the 15-year model for "NDC DE [%]" and the R^2 for the 15-year model of "NDC population size [1000s]".

time sample	fitted value	equation	R^2
5	total CH_4 without cattle EnterFe [kt]	$y = -12.753 \cdot x + 26300.235$	0.997
5	DC population size [1000s]	$y = -102.756 \cdot x + 213780.671$	0.887
5	DC Ym [%]	$y = 0 \cdot x + 6.5$	NA
5	DC DE [%]	$y = 0.011 \cdot x + 54.263$	0.570
5	DC NE_x [MJ head^{-1} day^{-1}]	$y = 0.715 \cdot x - 1352.955$	0.797
5	NDC population size [1000s]	$y = -71.703 \cdot x + 148784.07$	0.732
5	NDC Ym [%]	$y = 0 \cdot x + 6.5$	NA
5	NDC DE [%]	$y = 0 \cdot x + 70.346$	NA
5	NDC NE_x [MJ head^{-1} day^{-1}]	$y = 0.937 \cdot x - 1835.457$	0.852
10	total CH_4 without cattle EnterFe [kt]	$y = -10.058 \cdot x + 20851.018$	0.978
10	DC population size [1000s]	$y = -88.716 \cdot x + 185394.664$	0.930
10	DC Ym [%]	$y = 0 \cdot x + 6.5$	NA
10	DC DE [%]	$y = -0.007 \cdot x + 90.206$	0.305
10	DC NE_x [MJ head^{-1} day^{-1}]	$y = 0.906 \cdot x - 1740.452$	0.895
10	NDC population size [1000s]	$y = 22.947 \cdot x - 42611.147$	0.211
10	NDC Ym [%]	$y = 0 \cdot x + 6.5$	NA
10	NDC DE [%]	$y = 0 \cdot x + 70.346$	NA
10	NDC NE_x [MJ head^{-1} day^{-1}]	$y = 0.508 \cdot x - 968.768$	0.773
15	total CH_4 without cattle EnterFe [kt]	$y = -9.839 \cdot x + 20408.364$	0.992
15	DC population size [1000s]	$y = -29.407 \cdot x + 65583.652$	0.219
15	DC Ym [%]	$y = 0 \cdot x + 6.5$	NA
15	DC DE [%]	$y = 0.021 \cdot x + 33.254$	0.469
15	DC NE_x [MJ head^{-1} day^{-1}]	$y = 1.019 \cdot x - 1969.163$	0.966
15	NDC population size [1000s]	$y = -0.097 \cdot x + 3942.39$	0.000
15	NDC Ym [%]	$y = 0 \cdot x + 6.5$	NA
15	NDC DE [%]	$y = 0 \cdot x + 71.161$	0.481
15	NDC NE_x [MJ head^{-1} day^{-1}]	$y = 0.105 \cdot x - 153.685$	0.096

Table B12: Listed linear models for predicting relevant values for calculating 2030 CH_4 emission of the **USA**. The relevant indicators are total methane emission without cattle enteric fermentation (EnterFe) and Ym (methane conversion factor), DE (digestibility), NE_x (net energy intake), population size for dairy cattle (DC) and non-dairy cattle (NDC). For fitting the linear models three different time periods were used (the last 5, 10 or 15 years of the data set). The fitted value is thereby depended on the time. Besides the equation, the R^2 is given as well. Slope of the 15-year model for "DC DE [%]" is zero due to rounding issues.

time sample	fitted value	equation	R^2
5	total CH_4 without cattle EnterFe [kt]	$y = -529.671 \cdot x + 1091556.297$	0.989
5	DC population size [1000s]	$y = -78.221 \cdot x + 176838.967$	0.789
5	DC Ym [%]	$y = -0.001 \cdot x + 8.017$	0.704
5	DC DE [%]	$y = 0.006 \cdot x + 53.297$	0.845
5	DC NE_x [MJ head^{-1} day^{-1}]	$y = 0.736 \cdot x - 1400.748$	0.977
5	NDC population size [1000s]	$y = -526.204 \cdot x + 1143925.596$	0.765
5	NDC Ym [%]	$y = -0.01 \cdot x + 27.096$	0.960
5	NDC DE [%]	$y = 0.081 \cdot x - 96.049$	0.952
5	NDC NE_x [MJ head^{-1} day^{-1}]	$y = 0.173 \cdot x - 295.476$	0.921
10	total CH_4 without cattle EnterFe [kt]	$y = -245.886 \cdot x + 518212.392$	0.753
10	DC population size [1000s]	$y = 21.479 \cdot x - 24555.96$	0.144
10	DC Ym [%]	$y = -0.003 \cdot x + 12.927$	0.802
10	DC DE [%]	$y = 0.002 \cdot x + 62.028$	0.293
10	DC NE_x [MJ head^{-1} day^{-1}]	$y = 0.671 \cdot x - 1269.793$	0.990
10	NDC population size [1000s]	$y = 638.685 \cdot x - 1209266.975$	0.544
10	NDC Ym [%]	$y = -0.006 \cdot x + 19.106$	0.787
10	NDC DE [%]	$y = 0.049 \cdot x - 32.878$	0.820
10	NDC NE_x [MJ head^{-1} day^{-1}]	$y = 0.113 \cdot x - 173.444$	0.838
15	total CH_4 without cattle EnterFe [kt]	$y = -236.704 \cdot x + 499649.489$	0.876
15	DC population size [1000s]	$y = 35.307 \cdot x - 52471.673$	0.536
15	DC Ym [%]	$y = -0.005 \cdot x + 16.261$	0.916
15	DC DE [%]	$y = 0 \cdot x + 65.618$	0.011
15	DC NE_x [MJ head^{-1} day^{-1}]	$y = 0.617 \cdot x - 1159.782$	0.993
15	NDC population size [1000s]	$y = 104.563 \cdot x - 131253.734$	0.040
15	NDC Ym [%]	$y = -0.006 \cdot x + 17.665$	0.876
15	NDC DE [%]	$y = 0.043 \cdot x - 19.257$	0.890
15	NDC NE_x [MJ head^{-1} day^{-1}]	$y = 0.106 \cdot x - 158.969$	0.915

DECLARATION

I herewith declare that I have composed the present thesis myself and without use of any other than the cited sources and aids. Sentences or parts of sentences quoted literally are marked as such; other references with regard to the statement and scope are indicated by full details of the publications concerned. The thesis in the same or similar form has not been submitted to any examination body and has not been published. This thesis was not yet, even in part, used in another examination or as a course performance. Furthermore, I declare that the submitted written (bound) copies of the present thesis and the version submitted on a data carrier are consistent with each other in contents.

Place, date	Signature